

Cross Asset Volatility

A Practical Guide to Volatility Trading Across Asset Classes

Volatility has emerged as a distinct asset class, offering **uncorrelated risk-return profiles** beyond traditional asset beta. The evolution of **Quantitative Investment Strategies (QIS)** has enabled investors to access volatility trading across asset classes with increasing efficiency and liquidity. This report provides a practical roadmap for navigating the volatility landscape, focusing on the most liquid and tradable instruments in equities, commodities, FX, and interest rate derivatives.



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Executive Summary

Overview

Volatility has emerged as a distinct asset class, offering **uncorrelated risk-return profiles** beyond traditional asset beta. The evolution of **Quantitative Investment Strategies (QIS)** has enabled investors to access volatility trading across asset classes, supported by deeper listed and OTC derivatives markets, improved analytics, and richer flow and positioning data. This report provides a practical roadmap for navigating the volatility landscape, focusing on the most liquid and tradable instruments across equities, commodities, FX, and interest rate derivatives.

Key Highlights

Market Structure: We review the volatility market structure for the most liquid derivative instruments in equities, commodities, FX, and rates, highlighting the key drivers of supply and demand.

Analytics & Data: Our approach leverages market analytics and flow data to monitor volatility and correlations, helping to identify dislocations in gamma, vega, and delta positions.

We organize the trading framework around four pillars:

- **Tail & Carry:** Combining carry strategies with tail hedges can improve return consistency by collecting steady carry while retaining convex protection against large market moves. This category represents a significant portion of systematic volatility strategies, balancing income generation with protection against extreme market moves.
- **Relative Value:** Relative value opportunities arise when related volatility markets dislocate, such as index versus single-stock volatility, FX pairs, or rates pairs.
- **Beta with Options:** Using options to express directional views or generate yield offers risk-managed exposure and potential outperformance versus delta-one strategies. Options with timing signals provide flexibility in structuring trades to match specific risk tolerances and market outlooks.
- **Event Trading:** Targeting macro and company-specific events allows for the capture of unique trading opportunities driven by implied risk premia. Event-driven volatility trades can be structured around earnings, central bank meetings, geopolitical developments, and other catalysts.

Cross-Asset Opportunities:

A cross-asset portfolio of volatility carry and tail strategies can improve diversification and risk-adjusted returns. Because volatility relationships differ across equities, FX, commodities, and rates, combining strategies across asset classes can reduce reliance on any single market regime. Relative value volatility trades across assets unlock additional alpha, especially by leveraging correlations between fixed income and equities.

Conclusion & Outlook

Cross-asset volatility trading is evolving, offering investors tools for protection, carry, and alpha. These strategies hedge and generate income for linear positions, while also enhancing beta returns and influencing the primary market. Portfolio construction techniques will be explored in future research as we continue to track developments and share insights on emerging opportunities in the volatility space.

1. Overview and Scope

This report aims to provide a **practical guide** to the most liquid instruments tradable in QIS format in **equities, commodities, FX, and interest rate** derivatives; outline **key analytics** for market insight; and present a **framework for identifying trading opportunities**. In terms of payoff complexity and implementation flexibility, equity derivatives offer the broadest range of options. We anchor the framework in the equity volatility space and compare opportunities in other asset classes. We also explore opportunities in relative value across assets and the use of correlation. Finally, we lay out points to consider for portfolio construction, which will be discussed in more detail in follow-up reports.

Volatility Market Structure: Size, Liquidity & Flows

Across asset classes, implementation is primarily driven by liquidity, standardization, tenor availability, and the balance between hedging demand and yield-seeking supply.

Equity Derivatives

For QIS strategies, equity index options are by far the most liquid, with the US market dominating by a large margin. Listed markets reflect a significant volume of the flow, as OTC and exotic products can be hedged through them at low cost. Estimates from the open interest indicate around \$18 trillion notional in SPX options, compared to \$2.2 trillion in SX5E, the largest in Europe.

Figure 1: Notional open interest in listed SPX options by maturity and strike

Billions of \$, as of 9/8

Moneyness	<1W	1W-1M	1M-3M	3M-1Y	1Y-2Y	2Y-5Y
<50%	12	255	206	900	84	32
50-60%	1	71	40	258	100	86
60-70%	9	174	95	420	123	104
70-80%	135	473	209	813	188	131
80-90%	87	812	444	833	63	22
90-95%	113	956	543	1117	193	89
95-100%	301	1254	342	670	62	12
100-105%	372	1312	527	907	167	51
105-110%	32	226	185	719	33	9
110-120%	2	148	175	305	34	11
120-130%	0	12	11	110	24	6
130-140%	0	12	2	58	15	10
140-150%	0	1	4	12	10	8
>150%	0	3	1	19	7	16

Source: J.P. Morgan, Bloomberg Finance L.P.

Figure 2: Key players and locations for equity volatility supply and demand

Blue - supply, red - demand (see details in [US Equity Derivatives Strategy report](#))

Moneyness	<1W	1W-1M	1M-3M	3M-1Y	1Y-2Y	2Y-5Y
<50%						
50-60%						
60-70%						
70-80%						
80-90%						
90-95%						
95-100%						
100-105%						
105-110%						
110-120%						
120-130%						
130-140%						
140-150%						
>150%						

Source: J.P. Morgan, Bloomberg Finance L.P.

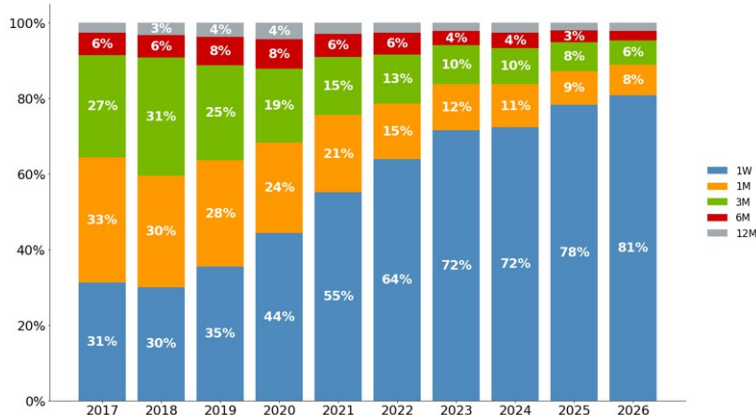
Since the launch of 0DTE in May 2022, the tenor at which volume peaks has kept getting shorter (Figure 3), with liquidity deteriorating slightly post 1Y. The bid-offer spread is around 0.01% on premium for ATM options to 1M tenor, and liquidity is good for up to 1% delta strikes. Single-stock options are mainly used for dispersion trades and intraday momentum strategies, with the SPX Top 75 representing the most popular.

Among the main derivatives players, retail structured products dominate the **long-dated vega supply**, while insurance products provide long-dated bids but at a reduced size ([link](#)). **Portfolio hedging** is popular for tenors around 1Y. The **short-dated gamma**

space has been driven by large **index overwriting** ETFs, particularly since 2023, as well as 0DTE strategies with a carry bias ([link](#)). In single names, dispersion players take the other side of vega supply, but the **single-stock levered ETF short gamma** effect has recently started to overtake the gamma supply ([link](#), [link](#)).

Figure 3: Total SPX volume by time to expiry

% of total volume, as of 9/8



Source: J.P. Morgan, OptionMetrics

In equities, we have the unique product of VIX, with around 14 million listed lots in open interest and bid-offer around 5 cents for 1M ATM. Liquidity is concentrated within 3M and dries up post 6M. Supply comes more from systematic and hedge fund players looking to harvest vol of vol risk premium, facilitating demand for calls and call spreads as a portfolio hedge from asset managers. Volatility carry players also buy put structures on VIX as a conservative way to short index volatility.

Overall, equities remain the deepest listed volatility market, with SPX dominating index liquidity, 0DTE reshaping short-tenor activity, and VIX providing a distinct tradable volatility instrument.

Commodity Derivatives

The most liquid underlyings for QIS options strategies are **gold and oil (Brent and WTI)**. Options on the gold ETF (GLD) provide a liquid alternative to trade gold like a stock, with \$250bn listed notional open interest. This is very sizable compared to options on gold futures, for which there is \$350bn listed notional and a sizable OTC market. Liquidity is mainly below 6M, with 0.5 vol as bid-offer in normal market conditions. Historically, there has been demand for longer-dated calls in gold as a portfolio hedge, but that is diminishing with the recent rally, and producers are more active in buying puts to lock in prices. Retail and PB supply downside vega from accumulator-like products.

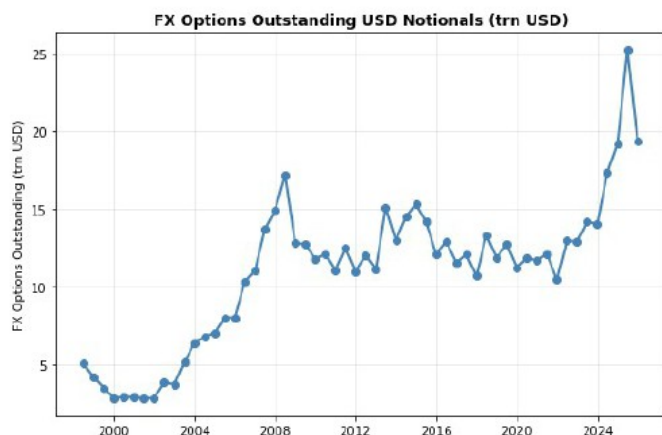
Listed options on Brent comprise around \$400bn notional, with a slightly smaller size in WTI. Liquidity is generally good out to around 12M, with 0.5-1 vol bid-offers under normal market conditions, though spreads can widen in high-volatility regimes. The OTC market is very sizable, with a large portion of the flow hedged through the listed market. While gold and WTI trade on CME exchanges with physical settlement on expiry for futures, Brent has more liquidity on ICE and facilitates cash settlement. The structure flow tends to be producers hedging WTI with longer-dated calendar swaps and consumers such as airlines hedging with shorter-dated calls on Brent.

FX Derivatives

The FX volatility market is among the largest and most liquid globally, with options outstanding notional approaching \$20tn and a deep ecosystem spanning G10 and a number of EM currency pairs (Figure 4). Trading remains predominantly OTC. While **EUR/USD** and **USD/JPY** dominate volumes, EM pairs such as **USD/CNH** and **USD/SGD** as well as some of the non-USD crosses offer meaningful liquidity and opportunities across relative value, correlation, and volatility carry strategies as well (Figure 5).

Beyond EUR/USD and USD/JPY, G10 liquidity is concentrated in AUD/USD, USD/CAD, GBP/USD, and USD/CHF, with tradable depth also available in **NZD/USD**, **USD/SEK**, and **USD/NOK**. Tight bid-offers support efficient execution across vanilla options, with liquidity in risk reversals, butterflies, barriers, digitals, DNTs, FVAs, and correlation swaps facilitating a broad range of directional and relative value strategies.

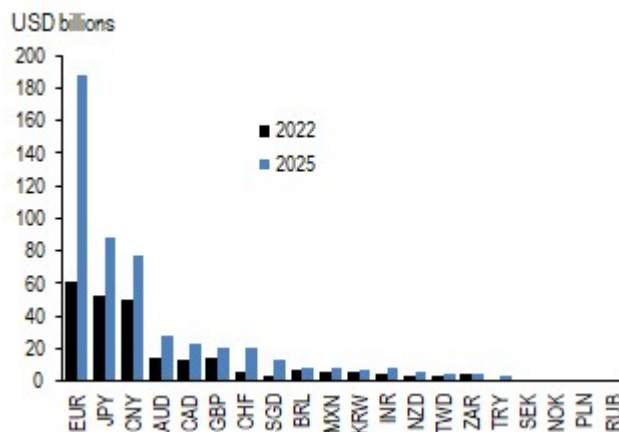
Figure 4: FX options liquidity (in USD) over time



Source: J.P. Morgan, BIS

Figure 5: Average daily FX options turnover versus the USD from the 2022 and 2025 BIS surveys

Details given in the latest [VXY Rebalancing](#) note



Source: J.P. Morgan, BIS

A key feature of FX volatility is the depth of cross-currency options markets. EUR and JPY crosses, including EUR/GBP, EUR/JPY, AUD/JPY, and KRW/JPY, are actively traded and often reflect regional macro themes, global risk sentiment, and central bank divergence. CHF and JPY crosses provide safe-haven exposure, while commodity currencies such as AUD, NOK, CAD, and ZAR are widely used for macro and carry strategies. EM FX volatility is concentrated in USD/CNH and USD/SGD, where liquidity rivals smaller G10 pairs. Persistent corporate demand for USD calls against local currencies creates structural flow demand, while pension funds and macro and systematic investors provide volatility supply.

Market structure also differs across currencies. Institutional investors, asset managers, and corporates all play a part in EUR volatility markets, while Japanese structured-product flows remain an important component of JPY volatility. CHF markets are influenced by private-bank activity, while EM volatility reflects a combination of corporate hedging demand and yield-seeking investors, alongside geopolitical narratives.

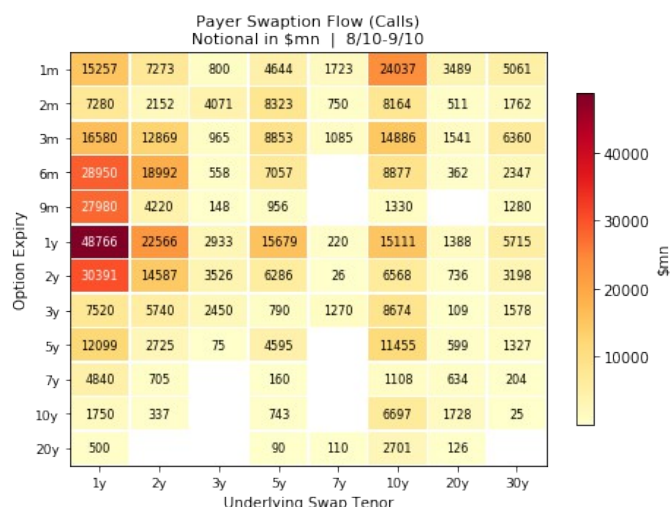
Interest Rate Derivatives

Exchange-traded instruments for trading volatility include options on government bond and interest rate futures, such as those on US Treasuries and SOFR futures, respectively. These represent the most standardized and liquid means of trading interest rate volatility. Daily volumes for UST options across tenors exceed 1mn contracts (\$100-200bn notional). **Swaptions** (options on interest rate swaps) are the primary OTC instrument for trading volatility in swap rates. Swaptions offer greater flexibility in terms of underlying tenor, expiry, and strike. The annual volume in the USD swaption market alone is on track to exceed \$16 trillion in notional. Market activity is concentrated in G10 currencies and benchmark swap tenors. Caps and floors are another class of OTC options that provide insurance against rates moving above or below specified levels, respectively. Liquidity is good for standard expiry x tail combinations (Figure 6): the “benchmark” expiries are 1m, 3m, 6m, 1y, 2y, 3y, 5y, 10y, and the benchmark tails are 1y, 2y, 5y, 10y, and 30y (though other points, such as 10yx20y and 20yx10y, can be relatively liquid as well).

The **supply** of USD rate volatility typically comes from dealers warehousing risk, fast-money relative value accounts, programmatic supply (from both fast and real money), callable bond issuance, synthetic duration trades, and institutions monetizing convexity or selling protection when they view implied vol as rich. The **demand** comes from corporate hedging using caps and floors, MBS hedging, ALM hedging (including bank ALM desks), synthetic duration expressions, and systematic long vol strategies in the longer expiries. Figure 7 summarizes the primary supply and demand sources in different regions of the vol surface.

Figure 6: Swaption liquidity is concentrated in standard expiries and tails

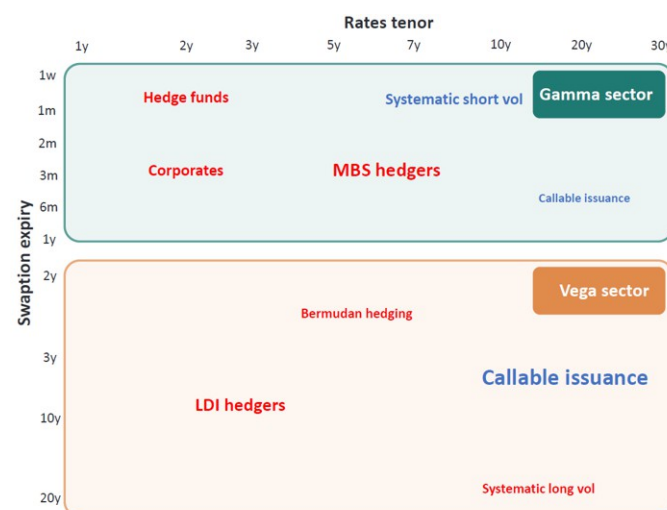
Amount of notional transacted in swaption payers across expiry x tenor buckets, 8/10-9/10; \$mm



Source: J.P. Morgan, DTCC, Bloomberg Finance L.P.

Figure 7: Primary supply and demand sources across the rates vol surface

Red (blue) indicates sources of demand (supply)



Source: J.P. Morgan

2. Analytics and Data

To identify market dislocations, we use **risk parameter analytics** and **flow data analytics** to provide a more comprehensive picture of market dynamics. We group these diagnostics into three categories: **volatility** surface parameters, **correlation** parameters, and **flow-based** measures of gamma, vega, and delta positioning. These indicators map directly into the strategy pillars discussed later: surface parameters help frame carry and tail trades, correlation parameters support relative value and cross-asset trades, and flow data helps identify positioning-driven opportunities.

2.1 Market Analytics — Volatility Surface Parameters

Insights from options market parameters reside in the **correlation and volatility** spaces, both within each asset class and across asset classes. Unlike trends, both volatility and correlation exhibit **mean reversion** while being subject to macro regime shifts.

For parameters from the **volatility surface**, we monitor the absolute values and percentiles versus history for the implied and realized volatility, skew, convexity, and term structure using root time anchor points. In practice, we focus less on any single level and more on whether a parameter is stretched versus its own recent history, and whether that stretch is consistent across tenor, strike, and asset class.

Figure 8: Cross-asset volatility parameters

As of 9/8

		Implied ATM			Realized			Downside Convexity (10dP-50d)			Skew (25dP-25dC)			Upside Skew (50d-25dC)			Term Spread	
		1M	3M	1Y	1M	3M	1Y	1M	3M	1Y	1M	3M	1Y	1M	3M	1Y	3M-1M	1Y-3M
Equity	SPX Index	11.8%	13.5%	15.8%	8.2%	13.3%	12.8%	6.3%	8.8%	9.7%	3.8%	4.9%	6.2%	1.3%	1.5%	2.1%	1.8%	2.3%
	NDX Index	17.2%	19.7%	22.1%	12.7%	24.7%	19.6%	7.3%	8.2%	8.4%	4.6%	4.8%	5.0%	1.6%	1.6%	1.6%	2.5%	2.4%
	SX5E Index	12.9%	14.2%	16.4%	7.7%	12.1%	15.4%	5.1%	6.6%	8.1%	3.0%	3.8%	4.8%	1.1%	1.3%	1.7%	1.4%	2.1%
	DAX Index	13.1%	14.7%	16.3%	8.4%	12.7%	15.8%	4.7%	6.5%	8.5%	2.9%	3.8%	5.0%	1.0%	1.3%	1.6%	1.6%	1.6%
	UKX Index	10.3%	11.4%	13.1%	5.6%	9.1%	11.3%	4.2%	5.1%	6.3%	2.0%	2.6%	3.7%	0.6%	0.9%	1.3%	1.0%	1.7%
	SMI Index	12.1%	12.4%	13.7%	9.6%	10.0%	12.0%	4.7%	5.4%	6.0%	2.4%	3.0%	3.7%	0.8%	1.1%	1.4%	0.3%	1.3%
	HSI Index	15.7%	17.1%	19.5%	14.0%	17.4%	18.8%	3.5%	3.7%	3.9%	0.9%	1.1%	1.4%	-0.1%	0.0%	0.2%	1.4%	2.4%
FX	NKY Index	29.1%	26.5%	25.0%	22.7%	32.7%	28.6%	7.6%	7.3%	6.9%	3.7%	3.4%	3.1%	1.0%	0.9%	0.7%	-2.5%	-1.5%
	EURUSD	5.0%	5.3%	6.3%	4.3%	4.5%	5.4%	0.7%	0.9%	1.3%	0.2%	0.2%	0.2%	0.0%	-0.1%	-0.2%	0.3%	1.0%
	GBPUSD	5.5%	6.0%	7.3%	3.5%	5.2%	6.2%	0.8%	1.2%	2.1%	0.4%	0.7%	1.0%	0.1%	0.1%	0.2%	0.5%	1.3%
	EURJPY	9.4%	8.7%	8.3%	8.2%	7.5%	6.7%	3.2%	2.8%	1.8%	2.3%	1.9%	0.7%	0.8%	0.6%	0.0%	-0.7%	-0.4%
	USDJPY	10.5%	9.5%	8.7%	9.7%	8.5%	8.3%	3.0%	2.5%	1.5%	2.3%	1.8%	0.3%	0.9%	0.6%	-0.1%	-1.0%	-0.8%
	AUDUSD	7.2%	7.5%	8.5%	4.9%	6.1%	7.8%	1.0%	1.3%	2.1%	0.5%	0.7%	1.0%	0.1%	0.1%	0.2%	0.3%	0.9%
	CADUSD	4.6%	4.6%	5.1%	4.8%	4.2%	4.3%	0.4%	0.5%	0.8%	0.0%	0.0%	0.0%	-0.2%	-0.2%	-0.2%	0.0%	0.5%
Commod	XAU	24.0%	23.6%	23.1%	25.5%	26.4%	28.4%	0.6%	1.1%	1.6%	-1.8%	-1.8%	-1.8%	-1.5%	-1.6%	-1.8%	-0.4%	-0.5%
	WTI	42.6%	43.5%	30.7%	34.6%	50.0%	54.6%	5.4%	0.8%	7.0%	-0.1%	-5.2%	2.2%	-1.3%	-4.3%	-0.3%	0.9%	-12.8%
Rate	USD_2Y	86.4	91.1	95.6	65.9	74.5	78.1	13.7	13.0	13.3	-0.8	-0.5	-1.3	-4.8	-3.9	-4.2	4.7	4.6
	USD_5Y	79.9	83.9	88.2	59.5	68.6	68.0	9.5	6.5	6.0	-1.6	-3.4	-4.3	-4.3	-4.4	-4.7	4.0	4.3
	USD_10Y	69.1	74.3	81.5	54.2	60.4	58.4	4.0	4.9	3.1	-3.5	-4.0	-5.2	-4.3	-4.4	-4.6	5.2	7.2
	EUR_2Y	69.5	73.9	81.1	51.4	64.1	74.8	0.3	0.8	1.8	-10.7	-10.5	-9.7	-11.0	-7.5	-7.1	4.4	7.1
	EUR_5Y	62.6	67.8	75.2	50.2	56.4	62.3	1.3	-0.1	0.7	-7.1	-7.7	-7.7	-7.9	-5.7	-5.4	5.5	7.5
	EUR_10Y	56.5	62.0	69.5	47.7	50.0	53.7	0.3	0.3	0.4	-9.7	-9.4	-9.6	-9.9	-6.7	-6.5	5.2	7.4
	GBP_10Y	72.1	82.7	91.2	54.8	68.6	75.3	-3.6	-4.2	-2.4	-11.3	-10.7	-10.3	-8.7	-6.6	-6.6	10.6	8.4
	JPY_10Y	65.9	66.4	70.1	57.6	60.3	57.9	0.7	0.3	-7.1	-1.5	-3.7	-10.9	-1.9	-2.9	-5.4	0.4	3.7

*Rates volatilities are from swaptions and quoted in annualized basis points (abp)

Source: J.P. Morgan, Bloomberg Finance L.P.

We find it helpful to consider the **2-year percentile range** of the values, as this provides a better idea of the magnitude of the dislocation within the current volatility regime, rather than relying only on longer histories that may span very different macro and market environments.

The shape of the vol surface is usually driven by moves in the front months. As a result, we find the **term structure** and **skew** to be helpful signals for how to position for points on the surface, particularly **regions for short-dated gamma vs. long-dated vega** ([link](#), [link](#), [link](#)). In commodities, skew can also be helpful for flagging the **direction** of potential future tail events ([link](#)).

Figure 9: Cross-asset volatility parameters 2Y percentiles

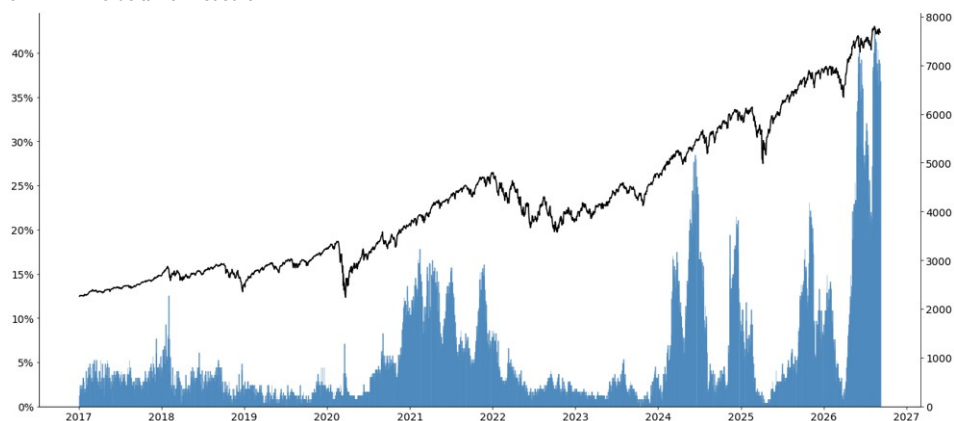
As of 9/8

		Implied ATM			Realized			Downside Convexity (10dP-50d)			Skew (25dP-25dC)			Upside Skew (50d-25dC)			Term Spread	
		1M	3M	1Y	1M	3M	1Y	1M	3M	1Y	1M	3M	1Y	1M	3M	1Y	3M-1M	1Y-3M
Equity	SPX Index	20%	25%	40%	6%	51%	33%	20%	20%	18%	19%	11%	4%	24%	11%	5%	83%	87%
	NDX Index	34%	49%	71%	13%	82%	45%	39%	14%	18%	28%	8%	2%	25%	7%	1%	93%	93%
	SX5E Index	18%	21%	58%	1%	15%	35%	14%	2%	15%	15%	7%	15%	21%	12%	22%	76%	96%
	DAX Index	24%	35%	58%	1%	23%	33%	10%	6%	39%	16%	11%	28%	29%	26%	21%	87%	94%
	UKX Index	47%	53%	63%	4%	36%	43%	26%	11%	41%	28%	22%	49%	30%	35%	59%	70%	80%
	SMI Index	57%	49%	60%	26%	18%	37%	36%	28%	31%	26%	25%	52%	26%	38%	61%	44%	82%
	HSI Index	4%	1%	3%	3%	15%	12%	60%	59%	56%	76%	81%	77%	83%	86%	88%	69%	92%
FX	NKY Index	83%	73%	74%	49%	67%	82%	39%	15%	0%	33%	5%	0%	32%	8%	1%	12%	30%
	EURUSD	4%	6%	18%	14%	6%	0%	62%	51%	57%	57%	56%	58%	61%	60%	59%	82%	99%
	GBPUSD	3%	2%	16%	0%	0%	1%	34%	34%	36%	40%	40%	35%	55%	46%	41%	87%	99%
	EURJPY	65%	54%	21%	65%	60%	25%	97%	89%	25%	97%	92%	31%	97%	91%	43%	5%	17%
	USDJPY	69%	44%	16%	56%	32%	12%	91%	82%	24%	93%	88%	28%	93%	88%	29%	8%	18%
	AUDUSD	9%	5%	4%	1%	2%	0%	5%	2%	9%	13%	14%	20%	22%	24%	36%	54%	79%
	CADUSD	37%	30%	11%	52%	34%	9%	13%	11%	25%	78%	76%	70%	71%	75%	73%	33%	61%
Commod	XAU	86%	87%	88%	73%	83%	100%	51%	74%	78%	6%	25%	77%	4%	13%	44%	33%	30%
	WTI	66%	75%	76%	47%	76%	99%	66%	29%	29%	53%	31%	21%	49%	30%	21%	73%	25%
Rate	USD_2Y	39%	46%	43%	28%	20%	6%	16%	8%	26%	7%	9%	7%	8%	7%	7%	74%	48%
	USD_5Y	30%	35%	41%	17%	20%	3%	13%	8%	2%	8%	3%	0%	8%	5%	0%	72%	76%
	USD_10Y	24%	27%	33%	18%	21%	2%	6%	19%	1%	6%	4%	7%	7%	5%	9%	79%	81%
	EUR_2Y	54%	58%	71%	40%	38%	47%	11%	16%	16%	18%	16%	14%	18%	16%	14%	83%	52%
	EUR_5Y	40%	46%	63%	36%	37%	35%	21%	16%	44%	19%	17%	15%	19%	16%	10%	97%	88%
	EUR_10Y	33%	45%	54%	35%	37%	17%	17%	16%	19%	18%	16%	12%	17%	16%	10%	90%	94%
	GBP_10Y	28%	38%	48%	22%	35%	40%	8%	8%	10%	14%	10%	10%	10%	11%	8%	98%	90%
	JPY_10Y	92%	95%	92%	73%	76%	100%	38%	65%	8%	28%	28%	7%	26%	13%	6%	44%	77%

*Rates volatilities are from swaptions; in rates
Source: J.P. Morgan, Bloomberg Finance L.P.

In equities, the number of stocks that experience **inverted skew** can be used as a gauge of crowding on the upside, where protection or call spread replacement strategies and orthogonal long-vega hedges should be considered ([link](#)).

Figure 10: Percentage of stocks with upside inverted skew in the SPX T250 universe vs. SPX spot 3M ATM - 25-delta vol measure



Source: J.P. Morgan, Bloomberg Finance L.P.

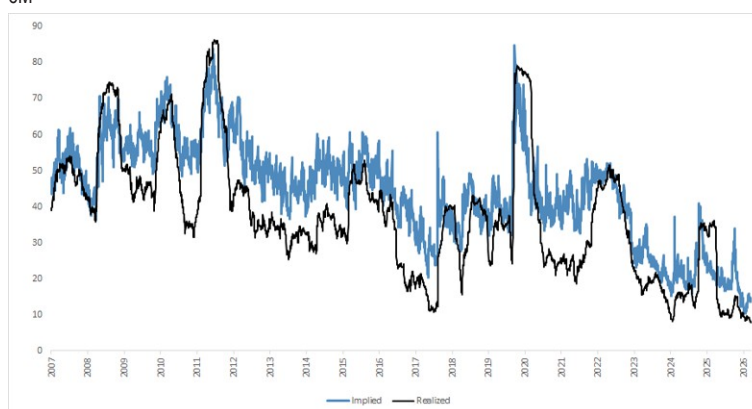
2.2 Market Analytics — Correlation Parameters

Correlations Within Asset Class — Equities

Correlation serves as a **good signal for macro conditions** while also being a liquid tradable product, particularly within equities, with FX correlation trading being quite prevalent as well. One can explore dislocations in **spot dispersion, volatility dispersion, and pairwise correlation** with equity derivative payoffs. The mean-reverting nature provides trading opportunities in distressed markets ([link](#)), while at depressed levels, it can be used to cheapen the risk premium for macro hedges.

Figure 11: Implied and realized correlation of SPXT75

6M



Source: J.P. Morgan, Bloomberg Finance L.P.

We monitor correlations within equities with the **Dispersion Chartbook** ([link](#)), which is the backbone for the most popular volatility dispersion trade, with PnL starting to resemble that from the dot-com era ([link](#), [link](#)) (see 3.2 Relative Value). Since the AI boom started in late 2022, equities have entered a low-correlation regime, as single-stock volatility has kept delivering. Dislocations can occur between **country indices and stocks** as well as between **sector indices and stocks**, especially post correlation events ([link](#)).

Figure 12: Correlation parameters for global indices

As of 9/9

Index		3M Implied vs Realized Correlation						
		Implied	1W Change	2Y %ile (Imp Corr)	Realized	1w Change	Spread (RC - IC)	2Y %ile (Spread)
AMER	SPX	12.5	0.7	12.6	6.6	0.9	-5.9	24.0
	NDX	20.3	0.4	14.4	18.6	1.0	-1.7	57.7
EMEA	SX5E	23.2	0.9	16.8	14.7	0.7	-8.5	19.8
	DAX	10.5	1.0	15.0	3.9	0.8	-6.7	16.4
	SMI	21.4	6.0	20.9	3.3	-0.7	-18.1	8.4
APAC	HSCEI	33.5	1.2	3.3	26.1	-0.0	-7.4	51.2

Source: J.P. Morgan, Bloomberg Finance L.P.

Figure 13: Correlation parameters for US sector indices

As of 9/9

	Name	Index	Top Names	3M Implied vs Realized Correlation					
				Implied	1W Change	2Y %ile (Imp Corr)	Realized	Spread (RC - IC)	2Y %ile (Spread)
AMERICAS	Semi	SMH	25	51.8	0.3	34.1	70.8	18.9	100.0
	Tech	XLK	25	27.0	-4.9	38.3	31.8	4.8	76.0
	Financials	XLF	25	38.2	2.0	14.8	25.6	-12.6	2.6
	Healthcare	XLV	25	22.5	0.6	48.7	34.1	11.5	94.8
	Energy	XLE	20	57.3	-0.5	35.9	51.1	-6.2	51.3
	Telecom	XLK	15	16.3	-0.8	11.0	19.5	3.2	86.2
	Materials	XLB	10	21.9	-1.3	1.0	23.5	1.7	73.1
	Discretionary	XLY	30	20.5	0.5	25.1	15.2	-5.4	37.7
	Industrials	XLI	30	28.1	4.8	63.7	22.1	-6.0	40.7
	Utilities	XLU	20	37.3	-3.8	43.1	41.4	4.2	86.4
	Staples	XLP	20	32.6	1.2	76.4	38.2	5.5	89.6
	Real Estate	XLRE	20	24.1	-6.0	2.4	28.5	4.4	83.8
	Software	IGV	30	33.2	-1.7	60.1	29.7	-3.5	34.3

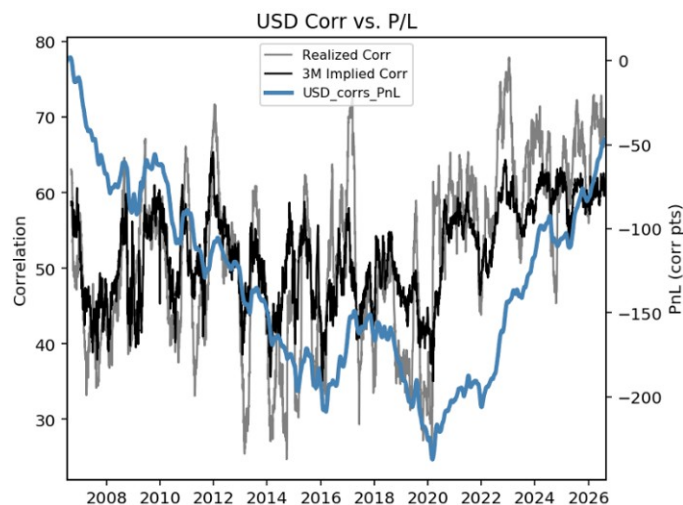
Source: J.P. Morgan, Bloomberg Finance L.P.

Correlations Within Asset Class — FX

FX is a particularly attractive asset class for correlation trading because correlations can be directly extracted from liquid volatility triangles. Within G10, and many G10-EM combinations, all three legs of the triangle trade actively, making correlation both observable and tradeable in a way that is largely unique among major asset classes. USD-centric correlations typically rise during periods of global macro uncertainty, broad USD trends, and synchronized central bank cycles, while stable growth and carry-friendly environments generally favor dispersion and lower correlations. That said, there has been a clear regime change since 2020, with USD correlations remaining elevated and continuing to perform as US exceptionalism, Fed policy, and global dollar funding conditions have become dominant market drivers (Figure 14).

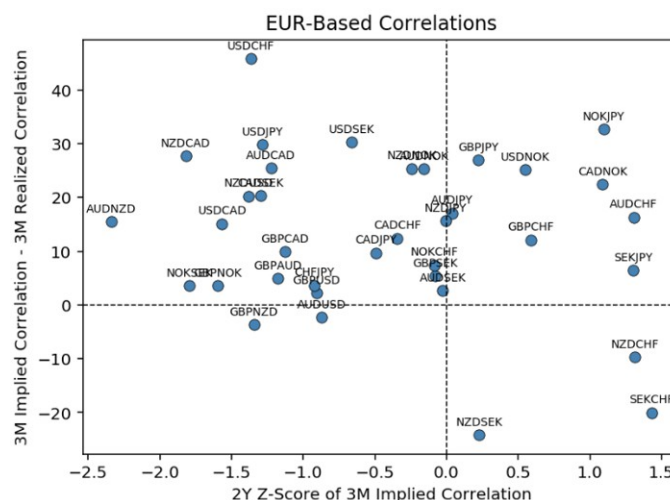
The most direct instruments for trading correlations are correlation swaps and basket options, although exposure can also be gained through RV vol trades. In practice, we combine model valuation, carry, and relative richness measures such as z-scores and implied-versus-subsequently-realized correlation scatterplots. Given structurally elevated USD correlations since 2021, we increasingly prefer to express bullish USD correlation views through the cross-correlation space, where greater common USD-driven moves should push cross-correlations lower (Figure 15).

Figure 14: USD corrs went through a regime change in 2020, exhibiting robust performance since then



Source: J.P. Morgan

Figure 15: With the USD corrs already elevated, selling x-corrs has been a better value than buying the 60pts USD corrs



Source: J.P. Morgan

FX correlation trends tend to be persistent, creating opportunities to harvest correlation risk premia and making correlations a distinct source of return alongside vols, skew, and term structure. The breadth of liquid G10, EM, and cross-currency markets makes FX one of the few asset classes where investors can efficiently trade not only vols but also the relationships between vols, correlations, and currency returns. To help identify opportunities, we developed a parametric correlation fair value model ([link](#)) that can be used to screen and trade correlation swaps.

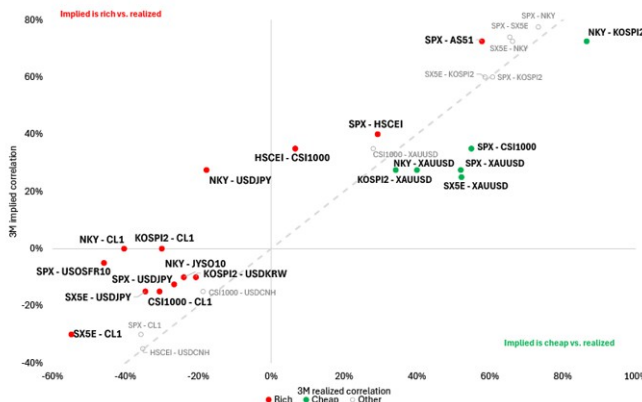
For cross-asset correlations, we focus on **fixed income/equities pairs**, as investors like to use the fixed income leg as a way to **cheapen directional bets on equities with the discount from correlation** to achieve leveraged payouts (see 4.3 Bringing Cross-Asset Correlations to Equities). We can also observe from the lookback window how the dollar story has brought gold, rates, and USD FX to the same trade over the year.

As of 9/8

		Correlation with SPX					Correlation with SPX 2Y Ptile				
		2Y	1Y	6M	3M	1M	2Y	1Y	6M	3M	1M
Equity	NDX Index	95.1%	92.7%	91.7%	91.1%	84.3%	30%	0%	0%	2%	2%
	SX5E Index	37.4%	58.1%	58.3%	57.1%	30.1%	56%	88%	77%	57%	16%
	DAX Index	35.0%	54.6%	54.5%	49.6%	53.4%	51%	93%	76%	41%	55%
	IXX Index	23.7%	36.7%	32.7%	2.5%	16.1%	44%	84%	35%	3%	25%
	SMI Index	22.4%	34.2%	32.4%	5.2%	24.9%	19%	64%	38%	10%	31%
	HSI Index	18.6%	8.8%	10.2%	11.6%	1.8%	32%	6%	36%	52%	37%
	NKY Index	13.6%	19.0%	24.2%	29.8%	-13.5%	62%	60%	67%	70%	14%
FX	EURUSD	0.2%	26.5%	48.9%	41.4%	28.5%	42%	91%	99%	82%	66%
	GBPUSD	19.4%	32.5%	44.8%	33.2%	19.9%	24%	78%	95%	71%	39%
	EURJPY	16.7%	3.3%	-3.1%	-4.1%	-33.8%	4%	1%	1%	4%	8%
	USDJPY	13.4%	-15.0%	-35.7%	-26.4%	-41.0%	30%	4%	1%	13%	18%
	AUDUSD	46.7%	61.4%	67.4%	70.4%	58.6%	75%	100%	98%	87%	56%
	CADUSD	19.5%	19.9%	23.8%	11.2%	39.4%	0%	37%	64%	14%	74%
	DXY INDEX	-2.2%	-27.8%	-49.6%	-36.5%	-39.5%	56%	6%	1%	1%	19%
Commod	XAU	15.6%	32.7%	50.5%	49.0%	51.7%	82%	100%	100%	88%	88%
	WTI	-7.3%	-33.7%	-49.5%	-36.4%	-40.5%	0%	1%	1%	18%	22%
Rate	USD_2Y	20.3%	-20.1%	-42.5%	-37.6%	-47.1%	65%	0%	0%	13%	13%
	USD_5Y	16.1%	-21.9%	-45.8%	-38.4%	-50.8%	72%	1%	0%	13%	13%
	USD_10Y	14.4%	-20.2%	-44.0%	-31.9%	-43.4%	89%	1%	0%	14%	15%
	EUR_2Y	-12.7%	-37.6%	-53.0%	-39.2%	-26.9%	0%	1%	0%	20%	33%
	EUR_5Y	-12.2%	-36.2%	-52.2%	-34.4%	-20.4%	3%	2%	0%	22%	44%
	EUR_10Y	-12.5%	-34.9%	-49.7%	-25.8%	-17.5%	5%	6%	0%	24%	51%
	GBP_10Y	-3.7%	-31.8%	-48.2%	-24.6%	-38.1%	31%	2%	0%	24%	24%
	JPY_10Y	12.5%	-9.2%	-10.5%	4.1%	4.0%	39%	8%	20%	43%	51%

For implied levels, the bid-offer can be less liquid. Nevertheless, in the **hybrid radar** ([link](#)), we also monitor the implied levels from dealers and compare them to realized values, as well as the moves of the pairs.

Data as of 8/7



13

2.3 Flow Data — Gamma, Vega, and Delta

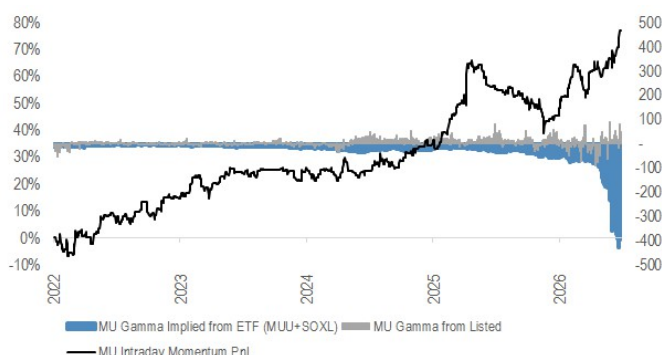
Open interest is a useful starting point for measuring market positioning, but without directional and trade-level context it can misstate the true gamma, vega or delta exposure in the market.

Flow from Listed Options Markets — Gamma/Vega Data

For listed options, we can track flows using the *Lee-Ready algorithm* to determine directions for most trades. Subsequently, we are able to estimate the **dealer gamma imbalance**, which can be very impactful for **short-dated volatility**. More recently, effective short gamma from levered equity ETFs has become very sizable and has a significant impact on the volatility and hence PnL of **systematic intraday momentum strategies** ([link](#)). We now monitor both to better understand the net effect ([link](#)).

Figure 18: Dealer gamma from listed options compared to short gamma from top ETFs in MU

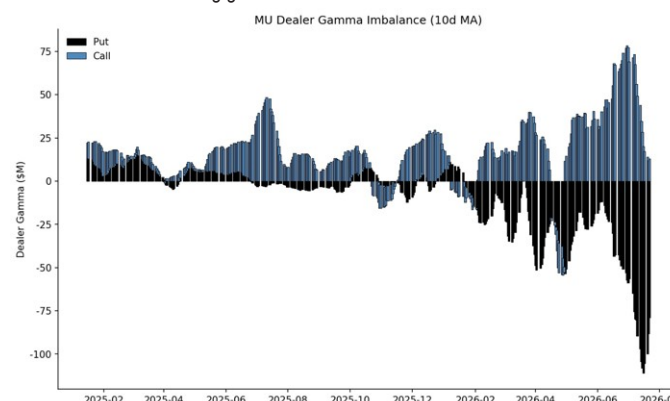
Left - % rtn intraday strategy, right - gamma in \$mm



*As of 7/13, near peak short gamma from ETF and record PnL from momentum strategy
Source: J.P. Morgan, Bloomberg Finance L.P.

Figure 19: Dealers get shorter gamma on puts from listed options only on MU

Positive means dealer long gamma

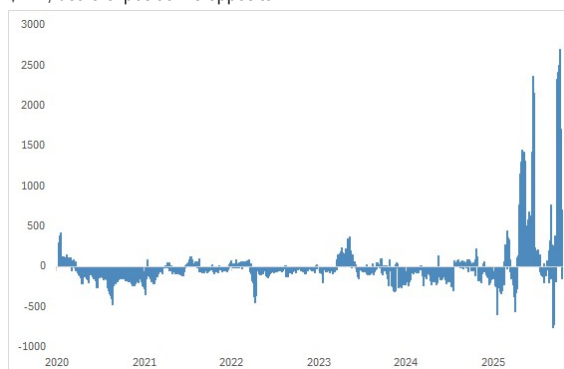


*As of 7/24, realized volatility accelerated during the correction
Source: J.P. Morgan, Bloomberg Finance L.P.

With the same framework, we extend the flow data monitor to **ETFs and options on futures**. This is particularly helpful for commodities such as **gold**, where we captured the gamma squeeze-induced correction ([link](#)).

Figure 20: Net gamma imbalance for GLD options from investors' perspective

\$mm, dealers' position is opposite

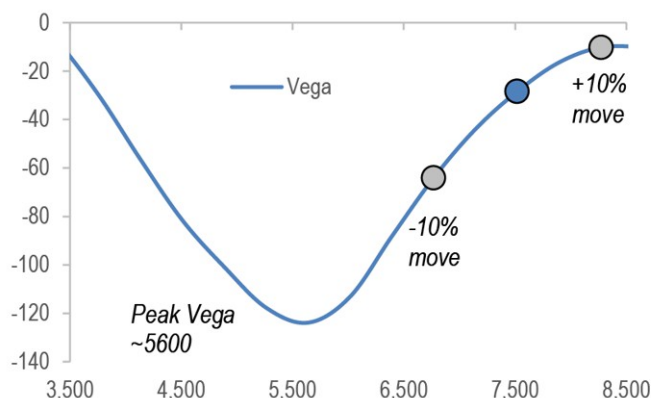


*As of 10/17/2025, expiry prior the October crash
Source: J.P. Morgan, Bloomberg Finance L.P.

The positioning imbalance can also enable us to estimate dealers' **vega position**. However, the number should be considered alongside the **structured product dynamics**, as not all are hedgeable through the listed market ([link](#)), and the profile drives different vega rebalancing needs with movement in spot.

Figure 21: S&P 500 autocallable vega profile

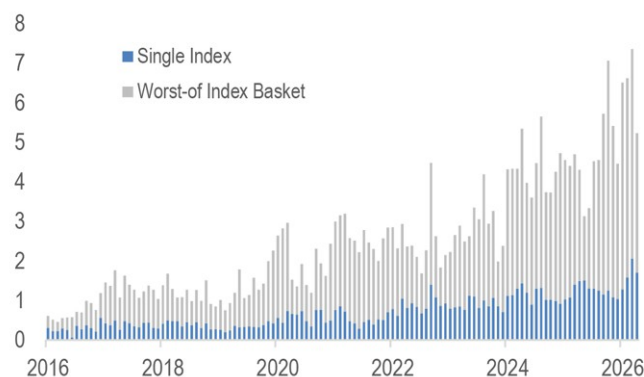
S&P 500 vega vs. spot profile of outstanding products (\$mm)



Source: J.P. Morgan

Figure 22: US index-linked autocallable issuance

Monthly US equity index-linked callable structured product issuance (\$bn)



Source: J.P. Morgan, SRP

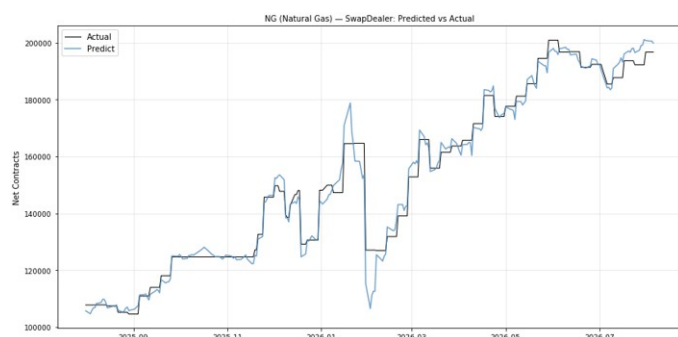
Futures Flow and CFTC Prediction - Delta Data

Using a similar framework, we are able to estimate the futures imbalance from trade-level data. While this gives an idea of the **bias in delta**, it can also be used to **predict CFTC COT data**. The COT is a well-studied signal, especially in commodities. However, the drawback is a 3-day delay due to the Tuesday collection date and the absence of data during government shutdowns. Using the daily futures imbalance, along with 4 other inputs, we are able to **predict the COT daily**, which allows early trading and continuation even during shutdowns ([link](#), [link](#)).

One direct application is for commodity QIS strategies that use COT data as an input, such as rolling pressure risk premium from CTA positions. By replacing the lagged weekly COT release with **daily predictions**, one can **enter positions earlier** ahead of anticipated rolls and maintain **uninterrupted signal generation during government shutdowns**, thereby generating enhanced alpha (Figure 24).

Figure 23: Swap dealer for natural gas futures — predicted vs. actual

Since September 2025, predicted data during government shutdowns has been robust



Source: J.P. Morgan, Commodity Futures Trading Commission

Figure 24: Using daily predicted data gives extra alpha compared to using weekly COT data



Source: J.P. Morgan

3. Opportunities Within Asset Classes

When the volatility risk parameters and/or flow data indicate dislocations, trading opportunities can arise. In this section, we lay out a framework to consider for PnL delivery within each asset class, grouped by four key pillars for alpha: **Tail and Carry**, **Relative Value**, **Beta with Options**, and **Event Trading**.

- **Tail and Carry**

Carry and tail strategies go hand in hand and represent the largest portion of systematic strategies in volatility. Carry strategies can generate **high hit ratios**, but a tail event can **wipe out the accumulated PnL**, while tail strategies can deliver very convex PnL but **bleed significantly** before tail risks are realized. Therefore, the holy grail is “**carry with low drawdown and tail with low bleed**.” While one can implement a carry or tail strategy alone, we prefer solutions that combine both carry and tail by **overweighting the leg that matches the primary investment mandate**.

- **Relative Value**

Successful relative value trades are typically anchored around two time series with a strong relationship. **Volatilities themselves are highly correlated**, giving rise to opportunities for RV trades. Dispersion trades are a popular type of RV trade between indices and stock volatility. In FX and rates derivatives, fair value models can be used to flag dislocations, providing a source of **uncorrelated returns for portfolios**.

- **Beta with Options**

Playing beta with options is an area with large demand in terms of **notional value**, as it is more comparable to delta-one positions. Such strategies offer many advantages, from **providing extra income** with buy-write-like strategies to riding **directional views with low drawdowns** via option replacement strategies. Techniques for **timing, sizing, and payoff choice** are essential to outperform equivalent delta-one strategies.

- **Event Trading**

Not all dates in the time series carry the same weight for price discovery. **Macro news events** such as FOMC meetings and **company news events** such as earnings should be treated differently than “regular days” if possible, as unique trading opportunities could arise during these periods. One can take positions depending on the implied risk premium, and special treatment should be considered in systematic strategies covering such events.

In this section, we outline some interesting strategies within each bucket and provide case studies. We will explore this topic in more detail accordingly in follow-up reports.

3.1 Tail and Carry

3.1.1 Equity Tail Strategy with Delta-Hedged Put Ratios

Long convexity funded by short skew provides tail hedging with low bleed. Improvement can be made with **long gamma overlay**.

We have conducted a full-surface backtest on delta-hedged SPX options to help identify optimal areas for carry and tail and the underlying reasons ([link](#), [link](#)). The two regions that stand out as **good choices for tail hedging** are **3-6M downside wings** and **long-dated ATM-to-slightly-upside strikes** (3.1.2 Equity Tail Strategy with Long-Dated Upside Vega). Both have higher implied volatility compared to the best choice for carry, which is shorter-tenor upside (3.1.6 Equity Volatility Carry with SPX Options).

Figure 25: Sharpe ratio for short options with delta hedging at different tenors and delta strikes

2017-2025

	5dp	10dp	25dp	50dc	25dc	10dc	5dc
10D	0.26	0.27	0.23	0.18	0.46	0.62	0.91
1M	0.05	0.22	0.25	0.23	0.45	0.50	0.46
3M	-0.15	-0.03	0.28	0.31	0.25	0.18	0.16
6M	-0.07	-0.03	-0.02	0.03	-0.03	0.08	0.03
1Y	-0.07	-0.06	-0.13	-0.33	-0.43	-0.29	-0.17

Source: J.P. Morgan, OPRA, Bloomberg Finance L.P.

Figure 26: Average strike implied volatility on trade date

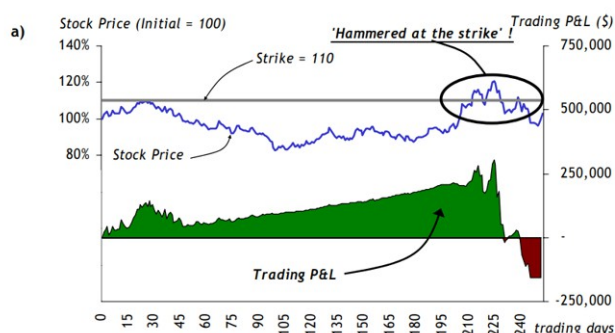
2017-2025

	5dp	10dp	25dp	50dc	25dc	10dc	5dc
10D	25.0%	21.8%	18.2%	15.4%	13.6%	12.8%	12.7%
1M	27.8%	23.8%	19.3%	15.7%	13.5%	12.6%	12.6%
3M	30.3%	25.9%	20.5%	16.3%	13.7%	12.7%	12.6%
6M	31.6%	27.1%	21.5%	16.9%	14.0%	12.8%	12.6%
1Y	31.4%	27.3%	22.0%	17.3%	14.2%	13.0%	12.9%

Source: J.P. Morgan, OPRA, Bloomberg Finance L.P.

The reason lies in the **path dependency** inherent in delta hedging and the **gamma covariance effect** for vanilla options, as highlighted by the PnL attribution work done by JPM Quant strategists ([link](#)). For 1Y options, it could outperform, as **short gamma ramps up as expiry approaches when spot hovers near the strike, coinciding with an uptick in realized volatility** (Figure 27). Whereas for 3-month 5-delta puts, the 5-day realized vol was a hefty 54% when the path sat within $\pm 2\%$ of the strike, significantly overshooting the already elevated implied levels of around 30 (Figure 28, Figure 26).

Figure 27: Example of path dependency of an option's P&L



Just what you need to know about variance swaps (Bossu, Strasser, & Guichard, 2005)
Source: J.P. Morgan

Figure 28: 5-day realized volatility when spot is near the strike

+/-0.5% for 10D 5DC, +/- 2% for 3M 5DP, 2017-2025

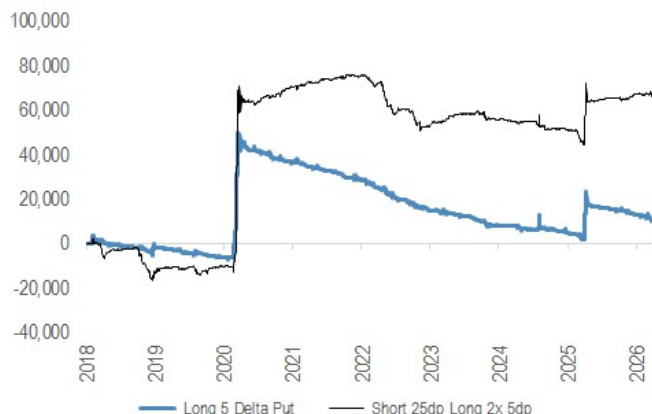
Day	10Day 5Delta Call	3Mth 5Delta Put
	5Delta Put	
1	10.2%	56.6%
2	9.9%	55.7%
3	9.8%	54.8%
4	9.9%	52.8%
5	10.0%	50.4%
Avg	10.0%	54.0%
% Time Near Strike	22.4%	7.0%

Source: J.P. Morgan

If we use hedged 3M 5-delta puts as the base material for a tail strategy, to reduce bleed, we can fund with 3M 25-delta puts. With a 3M 25/5dp 1x2 put ratio, the position is approximately theta-neutral and delivers well in high-sigma crashes (Figure 29).

Figure 29: 3M short 25 dp long 2x 5 dp vs. long 5-delta put only

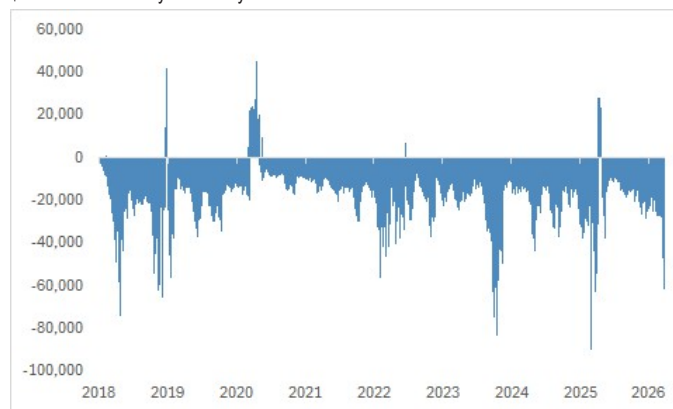
\$1mm notional daily scaled by tenor



Source: J.P. Morgan, Bloomberg Finance L.P.

Figure 30: Net gamma exposure for 3M 25-5 delta 1x2 put ratio

\$1mm notional daily scaled by tenor



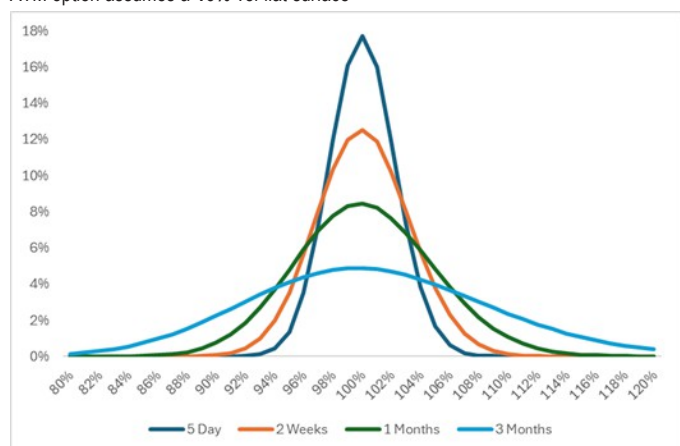
Source: J.P. Morgan, Bloomberg Finance L.P.

The slight drawdown of the theta-neutral strategy leaves the portfolio with a small net short gamma position as a steady state (Figure 30). As a result, the structure suffers during “**slow death**” markets (e.g. late 2018, 2022), in which net short gamma dominates before the convexity effect of the tail leg kicks in.

As the put ratio is **long convexity and short skew**, to protect against mild corrections without bleeding too much, one can add an overlay that is **long gamma funded by skew** with a shorter tenor. Gamma is **maximal near the strike and expiry** (Figure 31), and **shorting skew is an effective way to reduce the carry** (Figure 32), as long as there is protection from another part of the curve or positions from previous days.

Figure 31: Gamma exposure for different maturities and spot levels

ATM option assumes a 16% vol flat surface



Source: J.P. Morgan, Bloomberg Finance L.P.

Figure 32: Short skew funds long gamma at a sizable carry discount

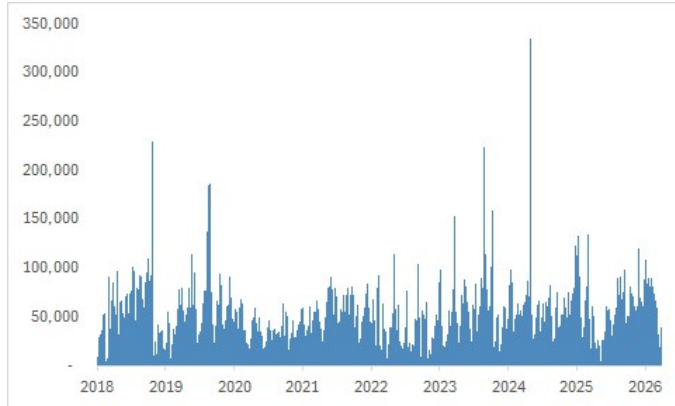
	Gamma	Theta		Gamma	Theta
1M ATM	67,108	- 456	2W ATM	92,729	- 730
1M 25d	44,755	- 454	2W 25d	62,275	- 723
1M 10d	20,340	- 333	2W 10d	28,541	- 466
	Notional	Theta		Notional	Theta
ATM funded by 25d	3.00	- 6	ATM funded by 25d	3.04	- 21
ATM funded by 10d	1.43	- 176	ATM funded by 10d	1.44	- 381
25d funded by 10d	1.83	- 222	25d funded by 10d	1.85	- 474

Source: J.P. Morgan, Bloomberg Finance L.P.

Adding a **long gamma, short skew** position already makes the portfolio **net long gamma on most days** without excess bleed. We add a third leg with ultra-short-dated options for days when gamma is still net negative to construct a **portfolio that is always long gamma locally** (Figure 33).

Figure 33: Final portfolio is always long gamma

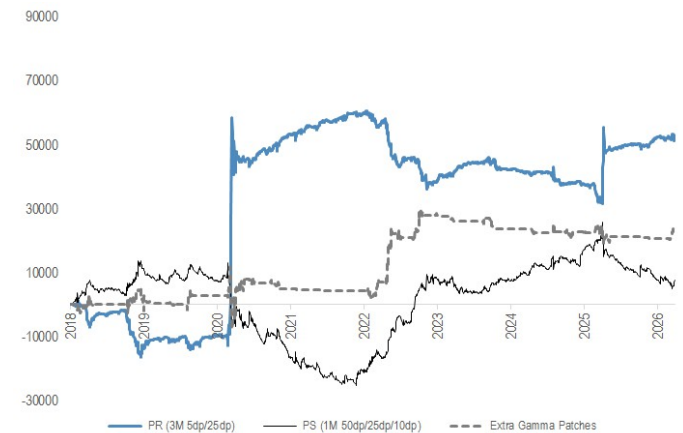
\$1mm notional daily scaled by tenor



Source: J.P. Morgan, ORPA, Bloomberg Finance L.P.

Figure 34: PnL split of the three legs

\$1mm notional daily scaled by tenor



Source: J.P. Morgan, ORPA, Bloomberg Finance L.P.

The final tail portfolio delivered robust returns orthogonal to SPX in nearly all corrections since 2018 with minimal bleed (Figure 35). We tested other indices and for SPX since 2008, finding similar results. The periods in which the tail portfolio bled slightly tended to overlap with strong bull markets in SPX (2021, 2025).

Figure 35: Total “always long gamma” tail portfolio PnL vs. SPX

Left - tail PnL, right - SPX spot



* Assumes no transaction costs

Source: J.P. Morgan, ORPA, Bloomberg Finance L.P.

If operationally allowed, shorting a **shorter-tenor skew** could generate more yield to **fund the gamma and convexity**, but the sizing should be adjusted accordingly to ensure drawdown scenarios are well covered.

In periods in which gamma simply fails to deliver, allocation to “safer” carry trades can help collect theta (see the carry strategies in this section).

Own long-dated, ATM-to-slightly-upside, delta-hedged options to benefit from index skew and path dependency in a rally followed by a crash.

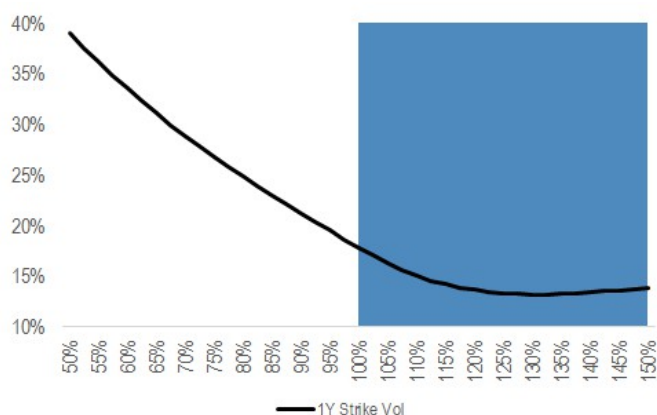
3.1.2 Equity Tail Strategy with Long-Dated Upside Vega

At the longer end of the curve, long **delta-hedged 1Y ATM-to-25-delta** strikes also deliver good tail-like performance (Figure 25). This is a similar behavior to the **long-dated UpVar trade**, which has been a very popular and successful trade, particularly since 2025 ([link](#)), when entry levels were at historical lows.

Index skew makes the downside implied volatility much **higher than the upside implied vol** (Figure 36). Variance swaps also **weight the downside strikes more heavily** in static replication, with $1/(K^2)$ making varswap levels very expensive (Figure 37).

Figure 36: SPX volatility in the 1Y tenor for different strikes

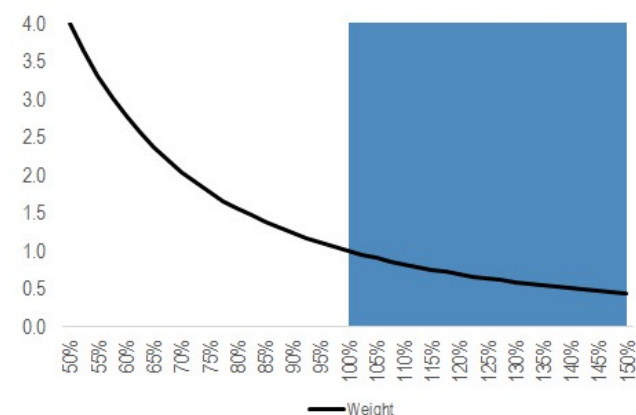
Highlighted area indicates strikes used for ATM UpVar



Source: J.P. Morgan

Figure 37: SPX variance swap weight as a function of strike

Highlighted area indicates strikes used for ATM UpVar



Source: J.P. Morgan

However, by using ATM or slightly higher strikes for UpVar, we **chop off all the expensive areas** above, making the entry level an **almost 9 vol discount to varswap**.

Figure 38: Indicative offers for UpVar on the SPX index for different UpVar barriers and tenors

As of 8/28

Barrier	3M	6M	12M	2Y	3Y	4Y	Spot Scenario
Varswap	17.8%	20.4%	22.3%	23.4%	23.7%	23.8%	
80%			17.7%	18.1%	18.3%	18.5%	125%
85%		16.1%	16.9%	17.4%	17.6%	17.8%	118%
90%	14.3%	15.0%	15.9%	16.5%	16.8%	17.1%	111%
95%	13.0%	13.7%	14.7%	15.5%	16.0%	16.3%	105%
100%	11.3%	12.4%	13.6%	14.7%	15.3%	15.6%	100%
105%	11.8%	12.4%	13.5%	14.4%	14.9%	15.3%	95%
110%		12.9%	13.5%	14.2%	14.6%	14.9%	91%

Source: J.P. Morgan

Figure 39: Expected time in the range for the SPX index for different UpVar barriers and tenors

As of 8/28

Barrier	3M	6M	12M	2Y	3Y	4Y	Spot Scenario
80%			95.1%	91.1%	88.3%	86.2%	125.0%
85%			96.4%	92.8%	88.3%	85.5%	117.6%
90%	96.7%	93.6%	89.2%	84.4%	81.8%	80.0%	111.1%
95%	91.1%	86.9%	82.5%	78.5%	76.6%	75.3%	105.3%
100%	61.3%	63.3%	65.4%	66.8%	67.3%	67.6%	100.0%
105%	12.2%	24.2%	37.7%	48.7%	53.6%	56.4%	95.2%
110%		7.6%	19.8%	34.4%	42.1%	46.8%	90.9%

Source: J.P. Morgan

Figure 40: Median and mean percentage of time in the range for a given tenor

Until 2025

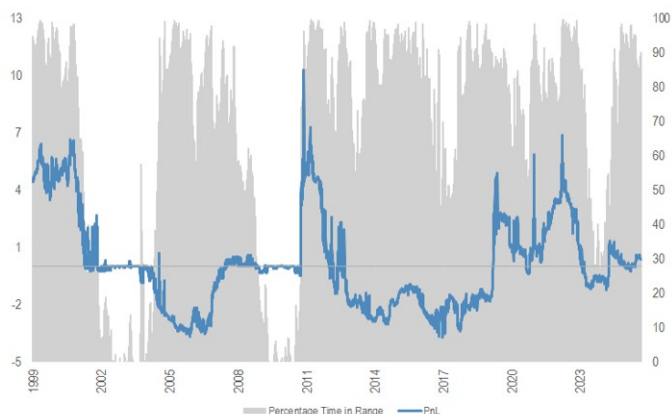
	Data Since 1995				Data Since 2011			
	SXSE		SPX		SXSE		SPX	
	Median	Mean	Median	Mean	Median	Mean	Median	Mean
3M	59	54	70	60	56	52	75	64
12M	72	61	83	70	67	59	86	75
3Y	73	62	91	75	69	65	83	88
5Y	70	62	92	75	77	72	96	94

Source: J.P. Morgan, Bloomberg Finance L.P.

If the tenor is long enough such that most days are above the strike barrier, **UpVar would converge to owning a varswap** (Figure 38). If the market corrects below the strike, the **PnL will be zero** for days below the barrier, and vega will also decrease. The best scenario for the PnL is when the market continues to **rally but is followed by a large correction**. If the tenor is 3Y, the model would price around a 67% time above the strike for SPX and just over 50% for SX5E, whereas on average, the SPX stayed above the strike 88% of the time and 65% for SX5E (Figure 40) over the 3Y tenor.

Figure 41: Backtest of SPX 3Y 105% UpVar at 14.9% entry offer

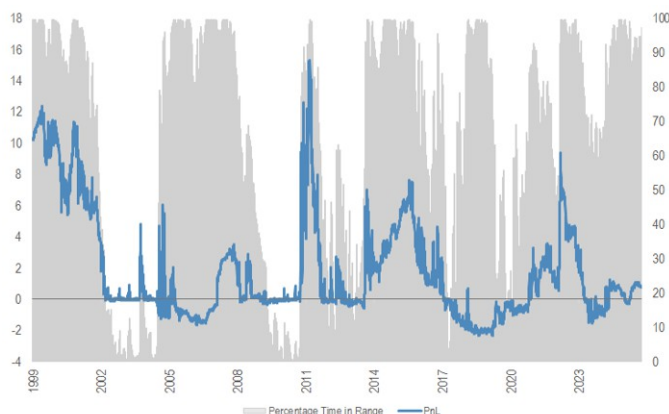
Left - vega PnL, right - % time in range



Source: J.P. Morgan, Bloomberg Finance L.P.

Figure 42: Backtest of SX5E 3Y ATM UpVar at 14.4% entry offer

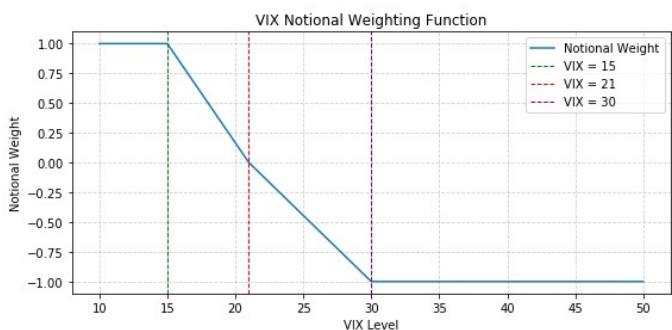
Left - vega PnL, right - % time in range



Source: J.P. Morgan, Bloomberg Finance L.P.

A similar effect can be captured with a single-strike **delta-hedged** option such as a 1Y 25-delta call. The position captures the “**rally followed by crash**” scenario similarly, but better entry timing signals can be used to reduce bleed.

Figure 43: 1Y 25D call notional with VIX signal for direction and sizing



Source: J.P. Morgan

Figure 44: Constant long full notional vs. notional based on VIX level

Daily long 1Y 25-delta call, delta-hedged daily, notional = \$1mn * notional weight



* Assumes no transaction costs
Source: J.P. Morgan

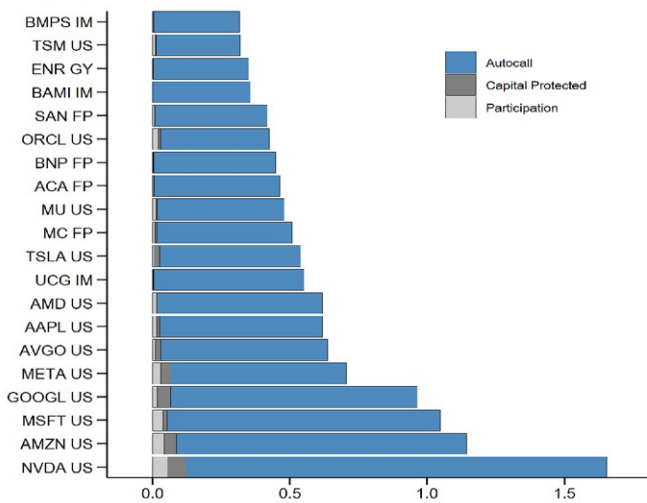
Rather than holding 1Y vega daily, we can size it **using the VIX signal to “buy low, sell high” with VIX regime cutoffs** ([link](#)). Allocate 100% long notional when VIX is below 15, and reduce participation linearly until VIX reaches 21. Above 21, adopt a short bias and increase the short notional linearly, capping at 100% when VIX is above 30. Using estimated data since 2007, the results show a substantial improvement versus maintaining a constant long notional. The pickup has been significant in the post-COVID era, with minimal drawdowns (Figure 44).

3.1.3 Equity Tail Strategy with Mega Cap Downside Vega

Own **long-dated downside vega on mega cap names** to benefit from flat skew, heavy index weight, and attractive entry points due to supply.

We currently see many similarities in terms of dispersion to 1998-1999, when stock vol remained elevated for a long period, as the **evolution of AI provides unprecedented earnings uncertainty**. At the same time, retail structured products continue to pick up momentum, including **selling downside puts** for income, concentrated on mega caps ([link](#)) (Figure 45).

Figure 45: Top shares for notional issuance for multi-share products year-to-date



Source: J.P. Morgan, SPI/WSD

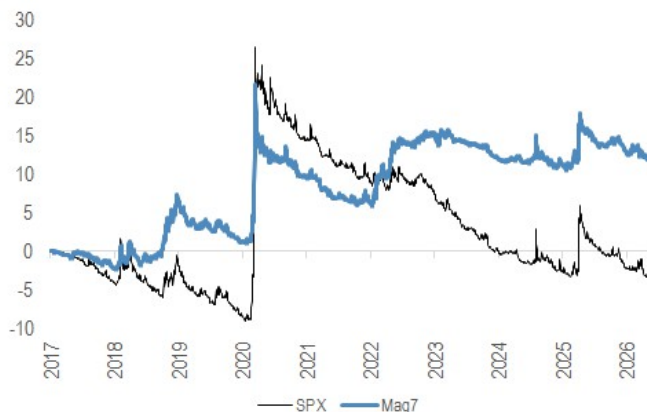
Figure 46: Downside skew on the Mag 7 at record lows compared to SPX



Source: J.P. Morgan, Bloomberg Finance L.P.

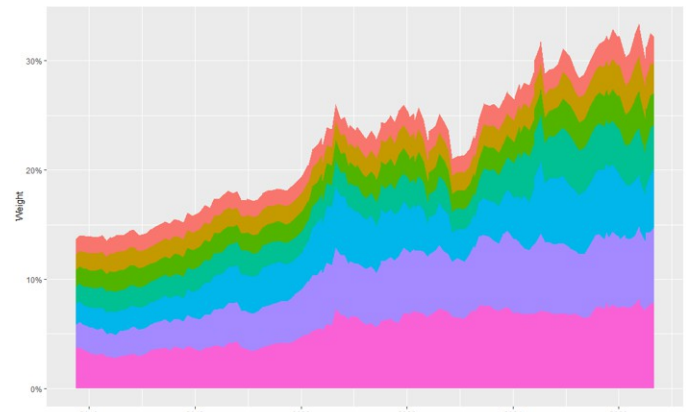
This provides a unique opportunity to own downside vega on mega-cap names at a discount. The entry level for **Mag 7 downside skew is at historical lows** (Figure 46). At the same time, the Mag 7 now represents almost a third of SPX (Figure 48). Owning skew in the names could perform similarly to an SPX hedge but **without paying for the expensive skew and while also enjoying the PnL from idiosyncratic risks**.

Figure 47: PnL of systematic corridor varswaps on the Mag 7 and SPX
1Y 60%-120% corridor, 1/252 * 1 vega notional per day



* Assumes no transaction costs
Source: J.P. Morgan, Bloomberg Finance L.P.

Figure 48: The weight of the top 7 names in SPX now accounts for a third of the index



Source: J.P. Morgan, Bloomberg Finance L.P.

Figure 47 shows the PnL of owning 1Y 60%-120% corridor varswaps on the Mag 7 daily in a systematic manner compared to the same in SPX. The PnL pick up has therefore been **sizable in market drawdowns**, but with a **much lower bleed compared to owning the index downside**. This makes the strategy a nice **tail solution** for long-equity portfolios.

There are a few solutions to improve the bleed. First, we can incorporate a profit-taking mechanism on vega PnL **post market corrections** ([link](#), [link](#)) or adjust the sizing on the new trade accordingly. Second, we can overlay some “safer” carry strategies ([link](#)) or **harvest part of the volatility distribution from the stocks** themselves, as shown below.

Figure 49: PnL of short delta-hedged 1M 25-delta calls on the Mag 7 and improvement with signal

1/252 * 1 vega notional per day, cash PnL



* Assumes no transaction costs
Source: J.P. Morgan, Bloomberg Finance L.P.

Similar to the findings for SPX carry strategies ([link](#), [link](#)), the optimal region for vol carry in stocks is around upside 25 delta in shorter tenors, such as 1M, with the **caveat of earnings treatment**. We applied a signal to **pause new shorts when the term structure of the stock is inverted**. This automatically removed most earnings periods and improved the drawdown significantly (Figure 49).

Figure 50: 1M-3M term structure inversion can filter earnings events for Q2 2026 earnings

ATMF implied vol

	1M	2M	3M	6M	1Y
AAPL					
Prior Earning	27.9%	27.0%	27.0%	27.1%	27.7%
Post Earning	25.3%	25.8%	26.4%	26.9%	27.7%
META					
Prior Earning	48.3%	43.5%	42.9%	42.4%	42.6%
Post Earning	36.1%	36.5%	38.3%	40.2%	41.2%

Source: J.P. Morgan, Bloomberg Finance L.P.

An alternative to overlaying with the upside carry is **to early unwind the trade when the tenor becomes shorter**. This way the vega PnL and mark-to-market will always dominate instead of possible excess bleed on the gamma.

Capture profit post-crash
with a max-loss approach in delta, gamma, and vega positions, which also hedge each other.

3.1.4 Equity Tail Strategy — Trading After Crashes

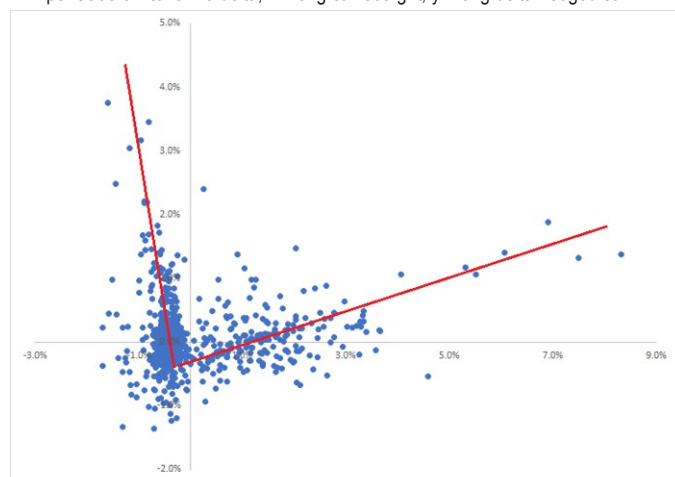
While most research focuses on how to position tail structures with minimal bleed prior to potential corrections, few realize the unique opportunity to trade **AFTER the market has already experienced a crash**. We studied the unique price behavior in detail in the reports *Thinking Outside the Box for Tail Trading* ([link](#), [link](#)).

The tail period is characterized by an inversion in the 1M-3M implied volatility term structure. We implemented three trades with the following setups:

Trades	Implementation	Underlying	Tenor	Sizing
Delta	Long ATM-25Delta Call Spread	SPX	1M	1/22 each time, 3x leverage
Gamma	Long 25Delta Call, Daily Hedge on Close	SPX	5D	1x each time
Vega	Long ATM -25-10 Delta Put Ladder	VIX	1M	Match vega of Gamma Trade (~4bps)

Figure 51: Delta-gamma space relationship

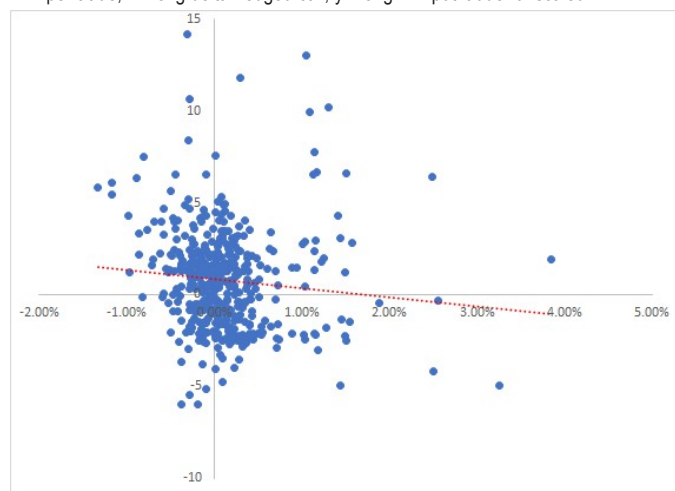
PnL per trade 5D tenor 25 delta, x - long call outright, y - long delta-hedged call



Source: OPRA, J.P. Morgan

Figure 52: Gamma-vega space relationship

PnL per trade, x - long delta-hedged call, y - long VIX put ladder unscaled



Source: OPRA, J.P. Morgan

We highlight the important relationships between the three trades, as they hedge each other to a good extent:

- **Delta - gamma** legs: When the delta trade works, the gamma trade usually performs well due to the “gap up” effect. If the delta doesn’t perform well, it often means that the market has kept tanking, i.e., realized vol is higher and thus gamma PnL is higher.
- **Gamma - vega** legs: When the gamma trade PnL starts to disappoint, it usually means that the market is calming down, with VIX normalizing; hence, there is higher PnL from the vega trade (VIX leg), and vice versa.
- **Delta - vega** legs: They are more related but still diversifying - the delta trade might fail to deliver on the upside (e.g., the market is rangebound/consolidating), but the vega trade will perform in this case as VIX settles down.

Therefore, the **gamma** trade is the most **orthogonal**, which is why we start with the gamma leg and allocate **full notional** upon the signal trigger dates. We then size the vega trade to match the vega notional of the gamma trade. The sizing of

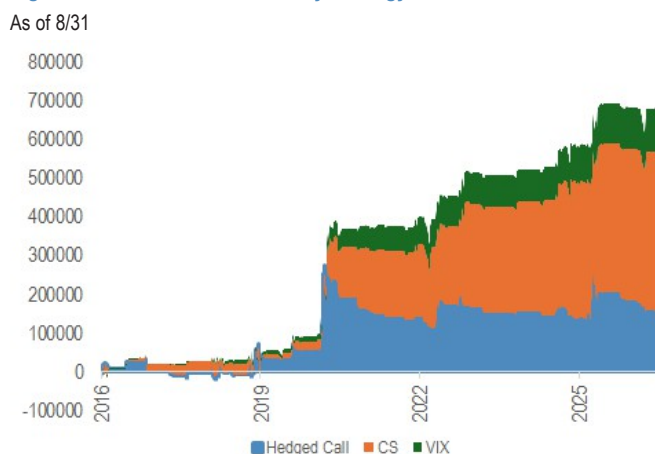
the **delta** trade is one that requires more investigation based on the investor's other positions, but we sized it 3x to roughly match the delta-one return.

Figure 53: Relationships between the three trades



Source: J.P. Morgan

Figure 54: Contribution to PnL by strategy

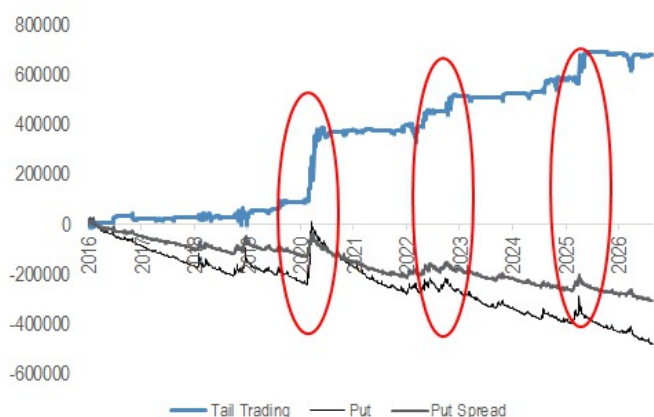


* Assumes no transaction costs
Source: OPRA, J.P. Morgan

A feature we particularly like about the construction of this tail-trading strategy is that the **max loss is almost entirely defined upfront** – the investor is almost always buying options.

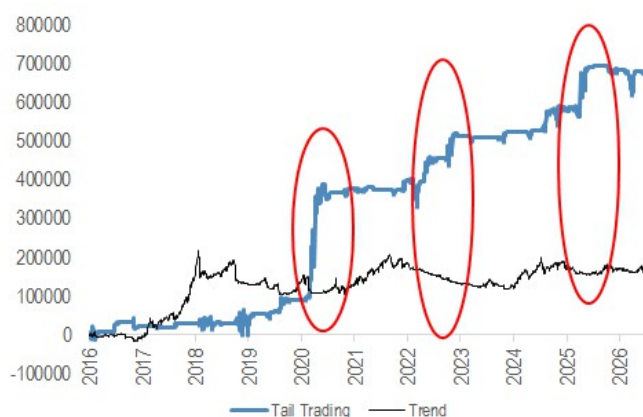
The strategy can be used as a standalone addition to diversify investment portfolios, as an **overlay to systematic hedging strategies** in equities, or as an **overlay to SPX CTA-style trend following**.

Figure 55: SPX put hedging strategies compared with tail trading



Long 3M 30d put or 30d-10d put spread, rolled 1M into the trade
Source: Bloomberg Finance L.P., J.P. Morgan

Figure 56: Tail-trading strategy complements SPX trend strategy



Source: Bloomberg Finance L.P., J.P. Morgan

When this strategy is combined with traditional hedging strategies, such as systematic puts on the SPX (Figure 55), rather than experiencing the typical drag, this “combo hedge” proved to be profitable, boasting a Sharpe ratio above 1 and exhibiting low drawdown:

- The **put spread** provides protection during an **initial market sell-off**.

- The **three-pillar tail trading** act as an effective **profit-taking mechanism from puts** and captures solid returns **in the periods straight after**.

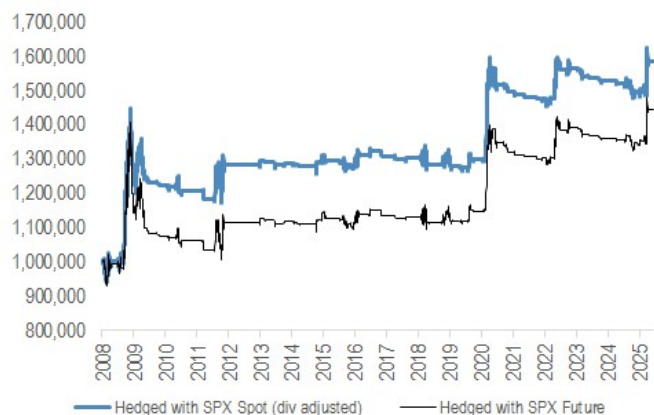
Integrating this with a CTA strategy also demonstrated strong complementary performance (Figure 56):

- When the trend delivery is strong, there is **no interference or drag** (2017, 2021, 1H24).
- When the trend incurs losses due to a “high volatility + mean reversion” market, it **delivers solid PnL** (2020, 2022, 2H24, and 1H25).

We published a follow-up [report](#) that addresses FAQs and extends the backtest to include 2008. We emphasized that the **gamma trade** delivery remained robust, even with the limited data available and slippage when using futures for hedging (Figure 57). The **delta and vega trades** were conducted using a “**fixed premium**” sizing approach to align more realistically with risk limits during tail periods in practice (Figure 58).

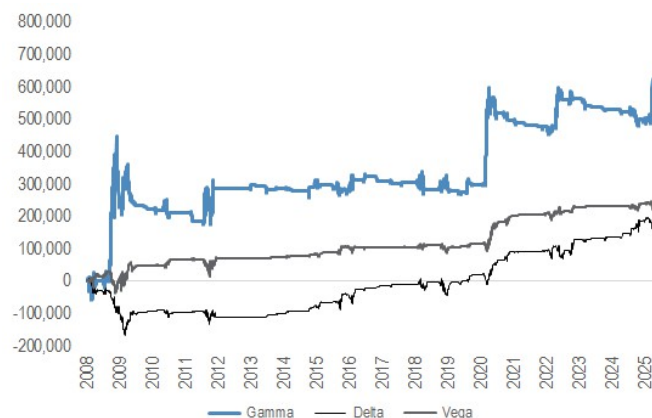
Figure 57: Gamma leg with spot hedge and future hedge

\$1m notional per trade



Source: J.P. Morgan, Bloomberg Finance L.P.

Figure 58: PnL contribution from 3 trades with fixed premium sizing



Source: J.P. Morgan, Bloomberg Finance L.P., OPRA

We explored the PnL impact of delta hedging when **marking the trading book at different volatility levels**. During the 2008 period, marking out-of-the-money calls at **higher volatility** would have yielded superior results due to the accelerated momentum in the “extra shorts,” whereas the **opposite effect** has been observed in recent years.

When expanding the trading signal beyond the signal dates, we found that **gamma trades delivered very poor results outside the signal dates**. However, the **delta and vega trades still provided decent returns even on “normal days,”** although the PnL and hit ratio per trade varied significantly.

Overwriting activities have increased gamma supply in recent years. While the gamma leg delivered well around “Liberation Day,” the performance was less good during the more “orderly sell-off” this year. Therefore it is essential to **have all three legs, as they hedge each other**. The delta leg can deliver even better performance using the buy-the-dip strategy discussed in 3.3.2 Beta in Equities — Long Equities with Options.

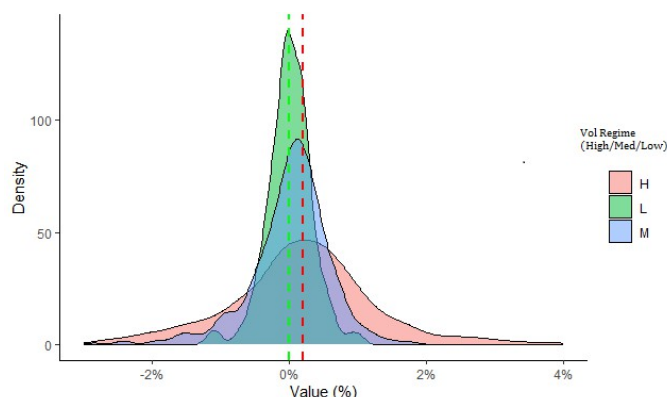
While the strategy is focused on equities, we see some similar reactions in other asset classes, which we will explore in follow-up reports.

3.1.5 Equity Tail-Like PnL with Intraday Momentum Strategy

Generate **robust PnL during high-volatility periods** using an intraday momentum strategy, with careful attention to signal generation and execution details.

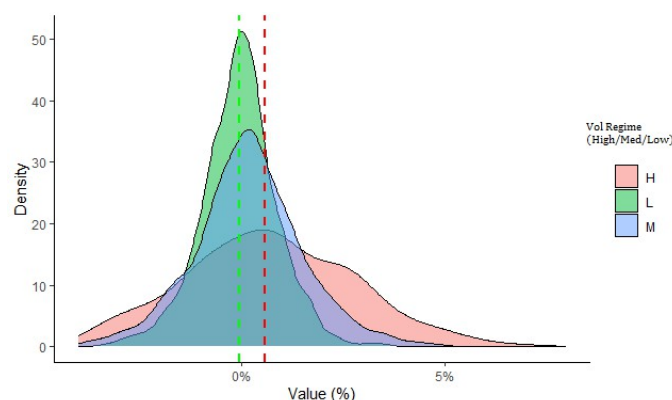
We conducted a deep dive into [intraday momentum \(link\)](#), a defensive strategy that delivers orthogonal returns during market drawdowns. The reason is that intraday momentum **works best on high-volatility days for high-vol underlyings** when close-to-close moves are large due to “momentum into close,” which usually coincides with **equities sell-offs**.

Figure 59: Distribution of SPX intraday momentum strategy PnL in different vol regimes (15/21 boundary)



Source: J.P. Morgan, Bloomberg Finance L.P.

Figure 60: Distribution of SOXX intraday momentum strategy PnL in different vol regimes (25/45 boundary)

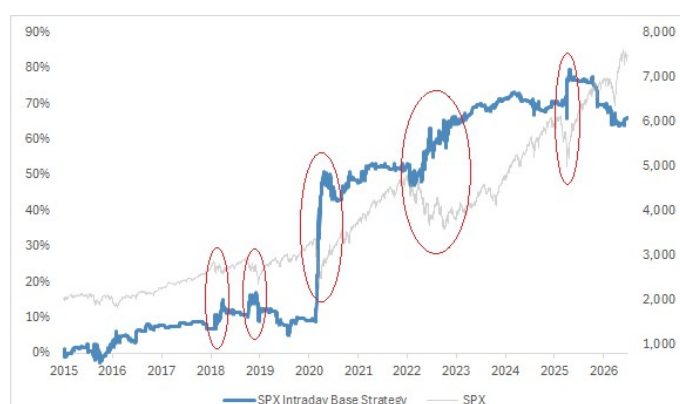


Source: J.P. Morgan, Bloomberg Finance L.P.

A base strategy in SPX has delivered sizable returns historically but has been disappointing in recent years, whereas SOX intraday momentum has delivered better returns, which have been recently amplified by the increase in the levered ETF AUM short-gamma effect.

Figure 61: Intraday momentum base strategy PnL in SPX vs. spot

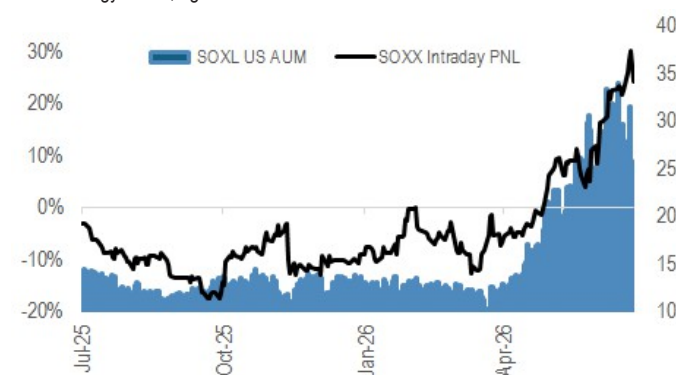
Left - strategy return, right - SPX spot



Source: J.P. Morgan, Bloomberg Finance L.P.

Figure 62: SOXX intraday momentum strategy PnL vs. AUM in SOXL ETF

Left - strategy return, right - ETF AUM

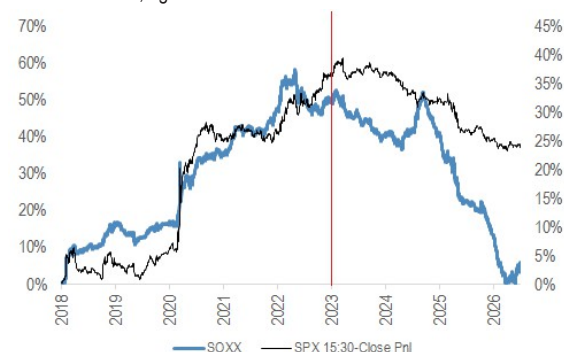


Source: J.P. Morgan, Bloomberg Finance L.P.

The **signal and execution** are crucial to the success of the strategy. For the signal, we examined using fixed percentage moves, as well as using **implied vol breakevens** for stocks. The **higher (more strict) the trigger criteria, the better the result**, despite entering at higher spot levels. However, there is a **trade-off with the number of trades picked up** if we set the signal level too high.

Figure 63: PnL of intraday momentum from 1530 (0-sigma signal)

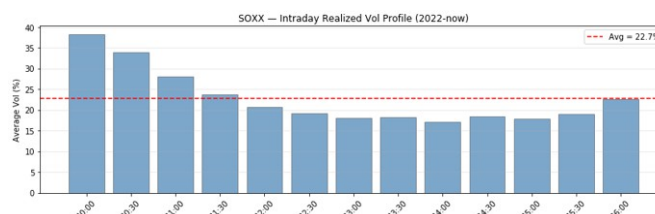
Left - SOXX PnL, right - SPX PnL



Source: J.P. Morgan, Bloomberg Finance L.P.

As for the execution times, we have noticed an increasing effect of **end-of-day reversal since 2023** (Figure 63), possibly contributed by the gamma supply from overwriting strategies, introduction of 0DTEs, and retail activity. As for entry times, the first hour of trading tends to be very volatile, which could induce false alerts on signal activation.

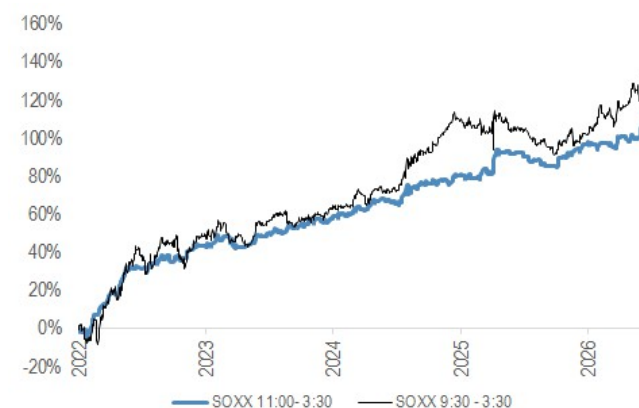
Figure 64: Average SOXX volatility in 30-min windows



Source: J.P. Morgan, Bloomberg Finance L.P.

With modified signal and execution timeframes, we are able to improve the Sharpe ratio of the base strategy from 1.14 to 2.05 and reduce the max DD from -12.5% to -5.6%. Further improvements can be achieved by pairing with a funding leg in option format for lower-volatility days, when intraday reversals could dominate.

Figure 65: SOXX intraday momentum strategy for different trading windows since 2022



*Transaction cost included at 2 cents per trade
Source: J.P. Morgan, Bloomberg Finance L.P.

Harvest volatility carry from the **regions that benefit from the gamma covariance effect**. Behavior varies across different regimes, enabling techniques to minimize drawdowns.

3.1.6 Equity Volatility Carry with SPX Options

In [What to Long, Short and When \(Part I\)](#), we provided an in-depth analysis of the SPX carry trade and showed that **the most efficient place to collect carry is in fact the lower-implied-volatility region of the surface**.

Figure 66: Sharpe ratio for short options with delta hedging at different tenors and delta strikes
2017-2025

	5dp	10dp	25dp	50dc	25dc	10dc	5dc
10D	0.26	0.27	0.23	0.18	0.46	0.62	0.91
1M	0.05	0.22	0.25	0.23	0.45	0.50	0.46
3M	-0.15	-0.03	0.28	0.31	0.25	0.18	0.16
6M	-0.07	-0.03	-0.02	0.03	-0.03	0.08	0.03
1Y	-0.07	-0.06	-0.13	-0.33	-0.43	-0.29	-0.17

Source: J.P. Morgan, OPRA, Bloomberg Finance L.P.

Delta hedging for vanilla options is path-dependent, and to make vol carry safer, it is important to avoid situations where **high short-gamma exposure aligns with large market moves**. Since short gamma peaks near expiry when spot is close to the strike and realized volatility rises, even if that path only materializes about 7% of the time for 3-month 5-delta puts, the **sharp pickup in realized volatility is enough to wipe out carry and trigger large drawdowns** (see 3.1.1 Equity Tail Strategy with Delta-Hedged Put Ratios). In contrast, shorting upside options tends to have realized volatility deliver less than implied by the time the strike comes close, unless entry was too low.

We segmented the time series into distinct market environments to gain more insight. Specifically, we split the data into regimes defined by **implied volatility conditions (VIX regime and term structure)**, trend periods, and earnings windows.

VIX Regime

Grouping results by VIX regimes (cutoffs at 15 and 21) reveals a “**buy low, sell high**” pattern that captures volatility mean reversion, with low-delta short-dated upside trades exhibiting the best Sharpe ratios. However, in high-volatility regimes, shorting front-end gamma is less effective, whereas shorting long-dated vega performs better.

Figure 67: Sharpe ratios for short options in different VIX regimes

Tenor	5dp	10dp	25dp	50dc	25dc	10dc	5dc	Regime
10D	0.20	0.09	-0.02	0.38	0.89	1.11	1.28	Under 15
1M	-0.04	0.06	-0.18	-0.34	-0.01	0.39	0.42	Under 15
3M	-0.54	-0.53	-0.41	-0.38	-0.46	-0.53	-0.50	Under 15
6M	-0.39	-0.46	-0.61	-0.73	-0.71	-0.60	-0.52	Under 15
1Y	-0.30	-0.36	-0.56	-0.73	-0.72	-0.58	-0.61	Under 15
10D	0.86	1.10	0.56	0.29	0.89	1.33	1.57	15-21
1M	0.13	0.46	0.56	0.48	0.71	1.30	1.69	15-21
3M	-0.16	-0.04	0.48	0.55	0.54	0.80	1.11	15-21
6M	0.02	0.07	0.00	0.21	0.08	0.20	0.29	15-21
1Y	-0.06	-0.05	-0.16	-0.46	-0.58	-0.48	-0.39	15-21
10D	-0.03	-0.03	0.13	-0.01	-0.01	0.04	0.26	Above 21
1M	0.05	0.15	0.29	0.34	0.39	0.14	0.02	Above 21
3M	0.58	0.73	0.68	0.57	0.46	0.31	0.26	Above 21
6M	0.79	0.86	0.74	0.60	0.47	0.58	0.45	Above 21
1Y	0.64	0.73	0.71	0.49	0.22	0.26	0.44	Above 21

Source: J.P. Morgan, OPRA, Bloomberg Finance L.P.

Figure 68: Sharpe ratios for short options in different 3M-1M term structure regimes

Peaceful (TS>1), nervous (0<TS<1), distressed (TS<0)

Tenor	5dp	10dp	25dp	50dc	25dc	10dc	5dc	Regime
10D	1.29	1.05	0.57	0.35	0.65	0.81	0.93	Peaceful
1M	0.85	0.83	0.57	0.34	0.57	0.57	0.55	Peaceful
3M	-0.08	0.07	0.35	0.28	0.17	-0.01	-0.09	Peaceful
6M	-0.09	-0.04	-0.07	-0.06	-0.15	-0.08	-0.16	Peaceful
1Y	-0.06	-0.04	-0.12	-0.33	-0.43	-0.32	-0.24	Peaceful
10D	-0.22	0.03	0.07	0.33	0.66	0.85	0.80	Nervous
1M	-0.58	-0.27	-0.02	0.10	0.54	0.76	0.53	Nervous
3M	-0.47	-0.40	-0.09	0.33	0.45	0.69	1.01	Nervous
6M	-0.21	-0.16	-0.12	0.01	0.00	0.19	0.20	Nervous
1Y	-0.17	-0.20	-0.30	-0.50	-0.62	-0.43	-0.31	Nervous
10D	-0.53	-0.55	-0.28	-0.29	-0.17	-0.17	0.53	Distressed
1M	-0.49	-0.40	-0.23	-0.01	0.03	0.12	0.18	Distressed
3M	-0.06	0.04	0.22	0.20	0.17	0.22	0.23	Distressed
6M	0.11	0.14	0.20	0.27	0.29	0.35	0.31	Distressed
1Y	-0.01	0.01	0.01	-0.11	-0.16	-0.02	0.10	Distressed

Source: J.P. Morgan, OPRA, Bloomberg Finance L.P.

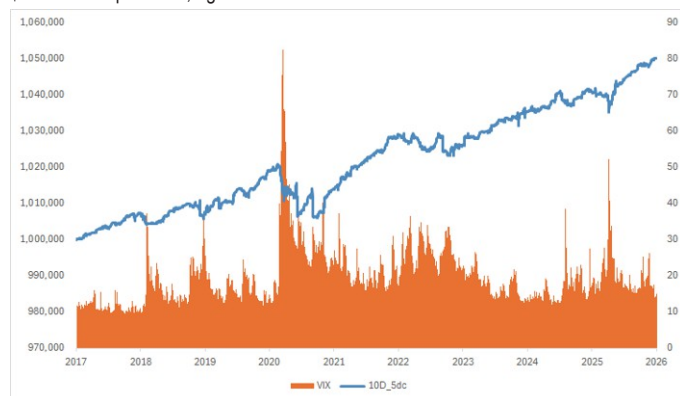
Term Structure

Segmenting by term structure into “peaceful,” “nervous,” and “distressed” regimes shows that short front-end gamma is effective in peaceful, low-volatility markets, but as conditions become nervous, only front-end upside remains reliable, and downside exposure suffers. In distressed markets, shorting front-end wings can still work, but strikes nearer to the money should be paired with long gamma since the risk of sharp upward moves is underpriced ([link](#)).

Shorting 10DTE 5-delta calls is the most effective strategy, with drawdowns occurring during periods of term structure inversion and high VIX. In these cases, pausing trading and owning nearer-the-money gamma are preferred.

Figure 69: Cumulative PnL for daily short 10DTE 5-delta calls delta-hedged at close vs. VIX

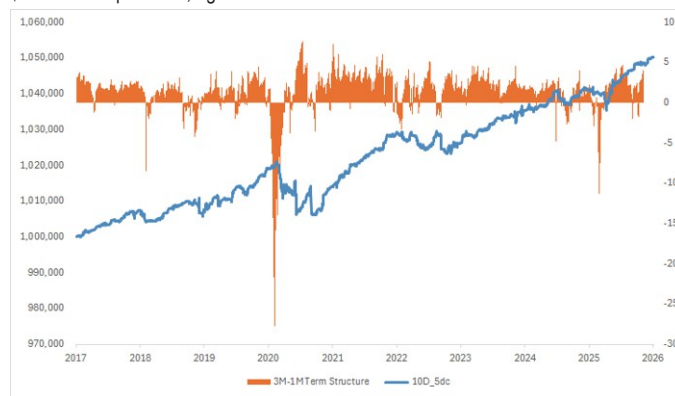
\$1m notional per trade, right axis - VIX



Source: J.P. Morgan

Figure 70: Cumulative PnL for daily short 10DTE 5-delta calls delta-hedged at close vs. term structure

\$1m notional per trade, right axis - 3M/1M TS



Source: J.P. Morgan

Trend Regime

Analyzing results by trend regime using the **JPM Cross Asset Trend Methodology** ([link](#)) shows that short option positions perform poorly during “Extreme Positive” trends, as high trend signals often precede near-term corrections and realized volatility exceeds the sold level. In these periods, implied volatility is typically very low, making short sellers vulnerable to volatility rebounds. While short-dated extreme wings above 5 delta are less impacted, it is advised to reduce short notional exposure during such strong positive trends.

One can also add a long gamma position when the trend signal is extremely positive, such as 1M 10-delta call with delta hedging. The position can cover situations of a spot overshoot followed by a correction and can be very profitable, as entry implied volatility is very low.

Figure 71: Sharpe ratios for short options in different trend regimes

Extreme +ve >0.75, Mild +ve 0.35-0.75, Small +ve 0-0.35, Small -ve -0.35-0, Mild -ve -0.75-0.35, Extreme -ve <-0.75

Tenor	5dp	10dp	25dp	50dc	25dc	10dc	5dc	Regime	Count
10D	0.00	-0.02	-0.27	-0.12	0.14	-0.15	-0.07	Extreme Positive	536
1M	-0.49	-0.23	-0.13	-0.21	-0.08	-0.38	-0.36	Extreme Positive	536
3M	-0.60	-0.60	-0.38	-0.30	-0.17	-0.35	-0.47	Extreme Positive	536
6M	-0.34	-0.37	-0.35	-0.26	-0.27	-0.23	-0.40	Extreme Positive	536
1Y	-0.17	-0.15	-0.13	-0.31	-0.29	-0.13	-0.19	Extreme Positive	536
10D	0.92	0.77	0.84	1.11	1.35	1.54	1.80	Mild Positive	966
1M	0.51	0.74	0.67	0.75	1.03	1.34	1.27	Mild Positive	966
3M	0.09	0.22	0.64	0.53	0.45	0.44	0.49	Mild Positive	966
6M	-0.05	-0.03	-0.00	-0.02	0.03	0.26	0.21	Mild Positive	966
1Y	-0.12	-0.15	-0.34	-0.55	-0.55	-0.28	-0.05	Mild Positive	966
10D	0.46	0.34	0.04	-0.51	-0.52	-0.25	-0.12	Small Positive	487
1M	0.57	0.36	0.12	0.06	-0.07	-0.04	-0.10	Small Positive	487
3M	0.53	0.50	0.32	0.32	0.07	0.21	0.29	Small Positive	487
6M	0.13	0.18	-0.01	0.01	-0.19	-0.23	-0.12	Small Positive	487
1Y	-0.08	-0.06	-0.12	-0.39	-0.62	-0.60	-0.45	Small Positive	487
10D	-0.51	-0.37	-0.18	-0.32	0.06	0.17	0.69	Small Negative	274
1M	-0.33	-0.19	-0.10	-0.22	0.36	0.76	1.21	Small Negative	274
3M	0.57	0.66	0.84	0.74	0.49	0.43	0.55	Small Negative	274
6M	1.03	1.00	0.89	0.79	0.65	0.67	0.61	Small Negative	274
1Y	1.05	1.10	1.14	1.06	0.59	0.21	0.17	Small Negative	274
10D	2.81	2.75	2.65	0.82	0.02	0.34	0.30	Mild Negative	16
1M	2.87	2.62	2.35	0.84	-0.41	0.14	0.40	Mild Negative	16
3M	2.81	2.26	1.97	1.48	0.72	0.58	1.09	Mild Negative	16
6M	2.58	2.13	1.90	1.46	1.22	0.47	0.71	Mild Negative	16
1Y	1.09	1.06	1.17	1.19	0.86	0.11	-0.14	Mild Negative	16

Source: J.P. Morgan, OPRA, Bloomberg Finance L.P.

Earnings Seasonality

Earnings seasonality significantly affects index volatility, especially for short index options on front-month upside wings (10D–1M), with the strongest impact during January and July earnings periods and some effect at year end. “Earnings periods” are defined as trades from the 15th of Jan/Apr/Jul/Oct for five weeks. The highest Sharpe ratios correspond to short-tenor, upside-wing (10- to 5-delta call) strategies during these times.

Figure 72: Sharpe ratios for shorting options during earnings periods and other dates

Tenor	5dp	10dp	25dp	50dc	25dc	10dc	5dc	Regime
10D	0.54	0.49	0.37	0.06	0.47	1.11	1.65	Earnings periods
1M	0.03	0.37	0.44	0.27	0.32	0.33	0.26	Earnings periods
3M	-0.21	0.00	0.47	0.39	0.26	0.34	0.35	Earnings periods
6M	-0.14	-0.11	0.01	0.13	0.02	0.12	0.02	Earnings periods
1Y	-0.09	-0.07	-0.14	-0.32	-0.42	-0.25	-0.15	Earnings periods
10D	0.08	0.12	0.13	0.22	0.40	0.31	0.45	Other dates
1M	0.05	0.11	0.12	0.18	0.50	0.63	0.64	Other dates
3M	-0.13	-0.05	0.16	0.25	0.23	0.08	0.05	Other dates
6M	-0.04	0.02	-0.04	-0.03	-0.06	0.06	0.03	Other dates
1Y	-0.07	-0.06	-0.14	-0.34	-0.44	-0.32	-0.19	Other dates

Source: J.P. Morgan, OPRA, Bloomberg Finance L.P.

“Safer Carry” Legs

The safest carry comes from shorting front-end upside wings, which provides high hit ratios and low drawdowns even before optimization. The base strategy is a short 10D 5-delta call, with overlays added in riskier regimes and increased participation during earnings:

1. When the term structure inverts, overlay long 10D daily delta-hedged 25-delta calls.
2. When the trend is Extreme Positive, overlay long 1M daily delta-hedged 10-delta calls.
3. During earnings periods, increase participation by an additional 30%.

These enhancements increased the Sharpe ratio from 0.91 to 1.23 and reduced the worst drawdown to -1.04%. The strategy delivered a Sharpe ratio of 2.29 in 2025.

Figure 73: Cumulative PnL for base carry strategy and with enhancements

Notional per trade \$1M, as of 8/31



* Assumes no transaction costs
Source: J.P. Morgan

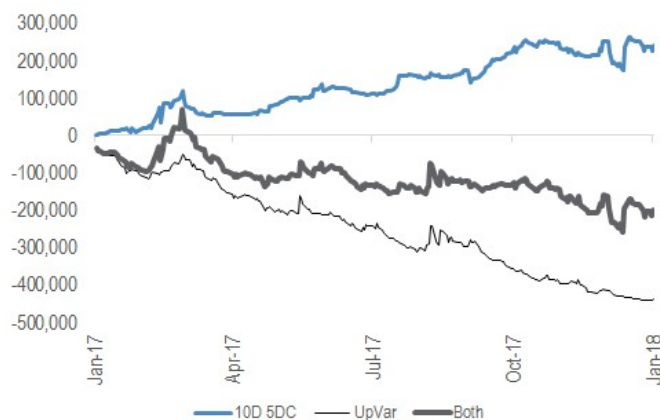
Diversifying strike concentration risk by spreading the strategy across more strikes improves the results. However, longer expiries increase path dependency, so unwinding the position early when spot is close to the strike significantly enhances the performance. We presented 1M 10-delta calls as an example in the [report](#), as this strategy provides the best Sharpe ratio for that tenor.

Overlay with UpVar Trade

The carry leg can be structured with various theta-paying positions. For example, we can overlay the enhanced short-dated call carry on 1Y SPX ATM UpVar ([link](#)). While 1Y ATM UpVar struggles during low-realized-volatility periods, adding a 10D 5-delta call carry overlay helps offset losses and reduce drawdowns. In favorable markets, such as after May 2024, this combined approach collects extra premium and handles corrections well, resulting in strong overall P&L.

Figure 74: Case study of a poor UpVar trade

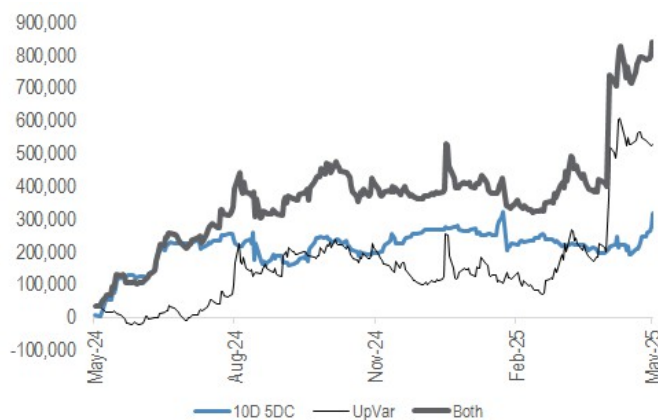
1Y ATM UpVar struck Jan 2017



Source: J.P. Morgan

Figure 75: Case study of a good UpVar trade

1Y ATM UpVar struck May 2024



Source: J.P. Morgan

3.1.7 Equity Volatility Carry with VIX Put Spreads

Short volatility conservatively using VIX spreads to **capture roll-down and volatility risk premium**, providing a good overlay to gamma dispersion strategies.

Puts on VIX can be a good alternative to harvest VRP in equities in a “safer” way, as the worst scenario is loss of the premium. We screened the most liquid tenors and strikes and found **3-month 50-25 delta put spreads** to be the most attractive structure because their PnL also benefits from the **VIX roll-down effect** (Figure 76). For additional carry harvesting, a **3-month 25-10 delta call spread** can be used, which also **caps the maximum loss**. However, this approach carries greater drawdown risk (Figure 78).

Figure 76: PnL and hit ratios for long VIX puts

2017-2025, zero transaction costs

Pnl Per Trade			
	10d P	25d P	50d P
1M	(0.05)	0.07	0.26
2M	0.00	0.09	0.69
3M	(0.01)	0.09	0.77

Hit Ratio per Trade			
	10d P	25d P	50d P
1M	19.0%	38.6%	52.0%
2M	22.6%	39.4%	58.6%
3M	20.9%	39.5%	62.4%

Source: J.P. Morgan, OPRA

Figure 77: PnL and hit ratios for short VIX calls

2017-2025, zero transaction costs

Pnl Per Trade			
	50d C (S)	25d (S)	10d C (S)
1M	0.35	0.25	0.11
2M	0.92	0.47	0.12
3M	1.37	0.60	0.08

Hit Ratio per Trade			
	50d C	25d C	10d C
1M	86.5%	95.4%	98.8%
2M	89.6%	96.1%	98.3%
3M	89.6%	96.1%	98.3%

Source: J.P. Morgan, OPRA

As we will discuss in 3.2.1 Relative Value in Equities — the Volatility Dispersion Trade, VIX put spreads can be used as a safer way to **increase the short index vega** leg in vega dispersion trades following a volatility event to **sell correlation** at an elevated level.

For owners of **gamma dispersion** trades, this strategy can serve as an effective overlay. During periods of market distress, gains from the gamma dispersion should **offset losses from the short VIX call spread**. Conversely, when VIX remains rangebound, the short call spread on VIX will generate additional carry (Figure 79).

Figure 78: 3M put spread on VIX vs. long PS, short CS combined

1/60 notional per day on VIX leg



*Transaction costs included
Source: Bloomberg Finance L.P., J.P. Morgan

Figure 79: Gamma dispersion with 80% participation in VIX systematic trades

1/60 notional per day on VIX leg



*Transaction costs included
Source: Bloomberg Finance L.P., J.P. Morgan

For a carry mandate, short vol of vol in the “safer” regions and **allocate to tail options in a calendar format to minimize drawdowns**. For tail mandates, allocate one-third to carry strategies to reduce bleed while maintaining effective tail risk response.

3.1.8 Equity Vol of Vol Carry and Tail with VIX Options

In equities, we have the luxury of a liquid market on VIX options to use as an alternative source of VRP. One can think of VIX calls as similar to SPX puts in terms of their reaction function during tail events.

Figure 80: Sharpe ratios of delta-hedged VIX options by strike and tenor

Jan 2017 - Aug 2025, unwound 1W prior to expiry due to liquidity, no transaction costs

	10d_P	25d_P	50d_S	25d_C	10d_C
1M	0.48	0.67	1.10	0.87	0.38
2M	0.59	0.65	0.64	0.28	-0.06
3M	0.64	0.62	0.56	0.26	-0.07

Source: Bloomberg Finance L.P., J.P. Morgan, OPRA

The Sharpe ratio for delta-hedged VIX options (Figure 80) shares some similarities to the SPX 3M tail, where tail events can overshoot the elevated implied vol (Figure 25).

Figure 81: Average initial strike prices by strike and tenor

	10d_P	25d_P	50d_S	25d_C	10d_C
1M	15.3	17.5	21.2	28.8	43.1
2M	15.0	17.6	22.4	32.7	52.1
3M	14.8	17.6	23.0	34.4	55.9

Source: Bloomberg Finance L.P., J.P. Morgan, OPRA

Figure 82: Hit ratios of delta-hedged VIX options by strike and tenor

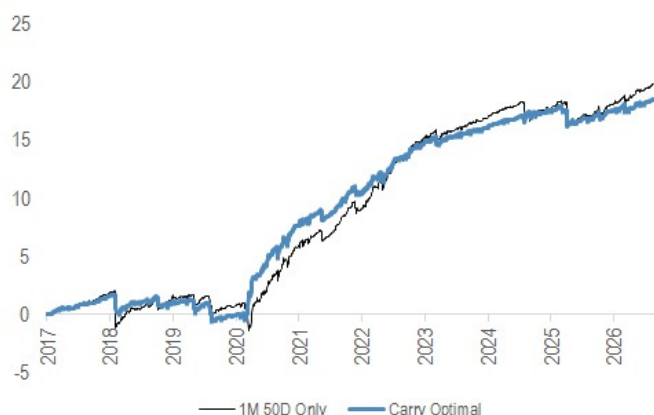
	10d_P	25d_P	50d_S	25d_C	10d_C
1M	61%	62%	78%	85%	89%
2M	64%	69%	76%	80%	82%
3M	67%	68%	74%	80%	81%

Source: Bloomberg Finance L.P., J.P. Morgan, OPRA

We find that the **best Sharpe ratios for VIX vol carry are for the front months ATM to downside strikes** (Figure 80) despite having lower hit ratios than the upside wings, where large drawdowns wipe out the high hit ratios from the carry (Figure 82).

Figure 83: Short 1M 50d straddle long 1x 3M 10dC on VIX

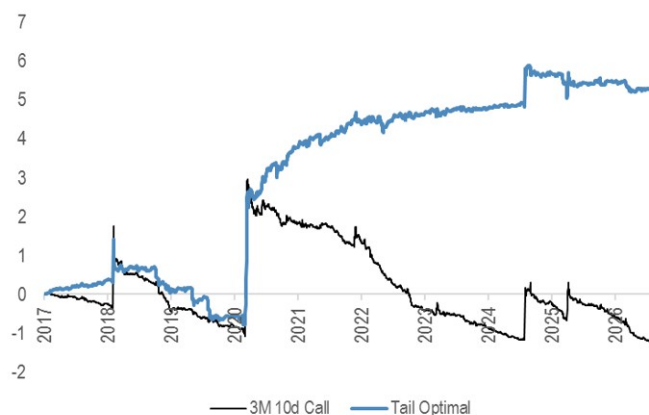
Delta-hedged daily, both legs unwound 1W prior to expiry



* Assumes no transaction costs
Source: Bloomberg Finance L.P., J.P. Morgan, OPRA

Figure 84: Long 3M 10dC short 33% 1M 50d straddle on VIX

Delta-hedged daily, both legs unwound 1W prior to expiry



* Assumes no transaction costs
Source: Bloomberg Finance L.P., J.P. Morgan, OPRA

We can use **3M 10dC as the tail leg** and **1M 50dC as the carry leg** base strategies. For PMs with a **carry-focused mandate**, 1x participation in the tail hedge has been sufficient to reduce drawdowns (Figure 83). For those who **focus on tail mandates**, allocating 33% to carry would reduce the bleed while keeping the a tail-like profile during large market corrections (Figure 84).

3.1.9 Commodity Brent Volatility Carry and Tail Strategy

For carry and tail mandates, investors can **adjust the sizing of the two legs of the calendar** strategy as needed. In commodities, directional tail risk tends to persist for longer periods.

For options on Brent, we use a similar framework to equities to uncover “safer” regions for volatility carry and the most leveraged solutions for tail ([link](#)). Unlike equities, where the trend has been on the upside, Brent had a sustained downside risk prior to 2021, whereas post 2022, the tail events have been on the upside instead (Figure 86).

Figure 85: Sharpe ratios for shorting options delta hedged at different tenors and delta strikes

2011-2026

Tenor	5dp	10dp	25dp	50dc	25dc	10dc	5dc
2M	0.28	0.42	0.62	0.72	0.59	0.46	0.51
3M	0.08	0.29	0.62	0.70	0.65	0.37	0.31
6M	-0.03	0.19	0.59	0.74	0.45	0.13	0.06
12M	-0.05	0.13	0.48	0.52	0.24	0.01	0.04

Source: J.P. Morgan, Bloomberg Finance L.P.

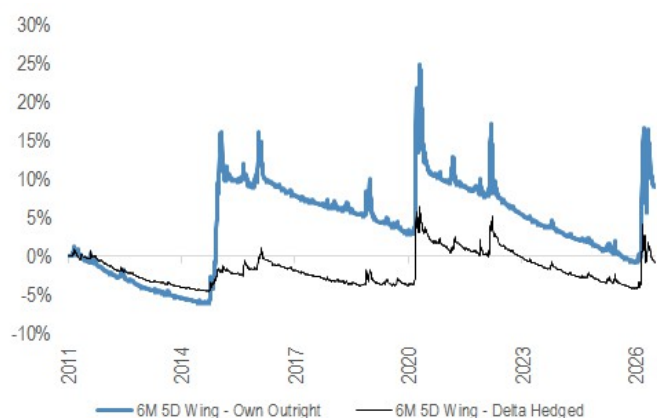
Figure 86: Average Sharpe ratios before and after 2022

Window	Tenor	5dp	10dp	25dp	50dp	25dc	10dc	5dc
2011-2021	2M	0.07	0.21	0.46	0.61	0.67	0.62	0.65
2011-2021	3M	-0.14	0.05	0.45	0.63	0.92	0.89	0.91
2011-2021	6M	-0.24	-0.07	0.37	0.73	0.80	0.77	0.66
2011-2021	12M	-0.22	-0.11	0.25	0.47	0.44	0.25	0.38
2022-2026	2M	1.98	1.47	1.18	1.02	0.43	0.21	0.38
2022-2026	3M	1.99	1.60	1.30	0.94	0.18	-0.18	-0.12
2022-2026	6M	2.04	1.87	1.45	0.77	-0.16	-0.42	-0.30
2022-2026	12M	1.70	1.72	1.37	0.68	-0.16	-0.31	-0.18

Source: J.P. Morgan, Bloomberg Finance L.P.

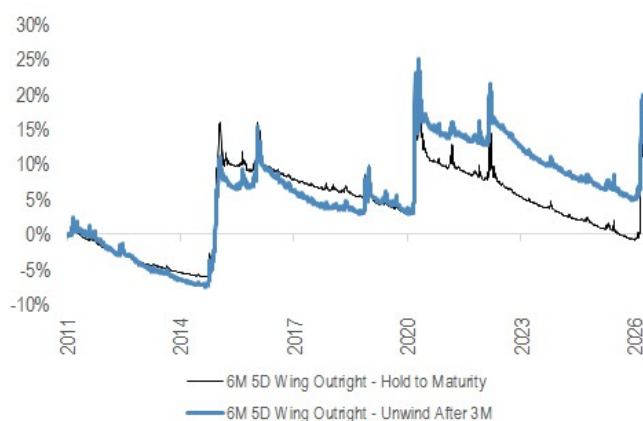
Regardless of the regime, the most “reliable” carry is around ATM strikes in front months, as VRP quickly becomes too conservative, sharing similarities with our findings on options in VIX. For the tail leg, the **longer-dated 5-delta wings represent an all-weather solution**.

Figure 87: Outright 5-delta wings outperform delta-hedged ones



Source: J.P. Morgan, Bloomberg Finance L.P.

Figure 88: PnL for outright 5-delta wings 6M maturity unwound after 3M vs. HTM



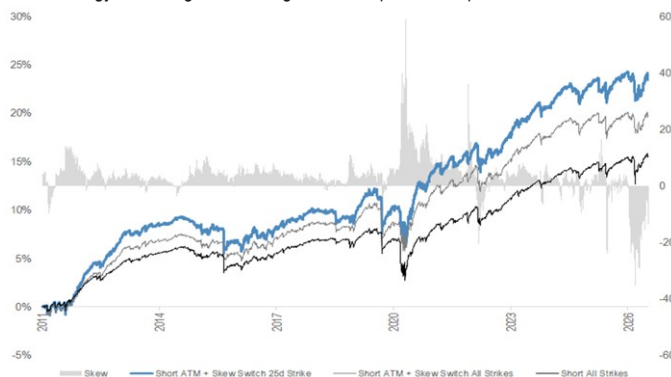
Source: J.P. Morgan, Bloomberg Finance L.P.

Unlike equities, where a “buy-the-dip” bias has tended to reverse the trend in corrections in recent years, in commodities, the directional moves are heavily driven by CTAs, which makes the **momentum accelerate in the same direction for much longer** once a shift starts to occur. Therefore, **owning the wings outright significantly outperforms the delta-hedged version in tail periods** (Figure 87). Unwinding early after around 3M also helps to reduce the drag post tail events (Figure 88).

For the carry leg, we can diversify the strike pin risks by including slightly out-of-the-money ones, with improvements obtained by using a **skew signal** to help identify the bias to avoid the riskier side (Figure 89). A further improvement can be achieved if one **avoids seeking carry** when implied volatility is too high or too low (Figure 90).

Figure 89: Using a skew signal to switch call or put strike to short

Left - strategy return, right - skew signal 3M 25dp - 25dc imp vol



Source: J.P. Morgan, Bloomberg Finance L.P.

Figure 90: Short 2M ATM carry vs. with 1.5x implied volatility z-score stop-loss

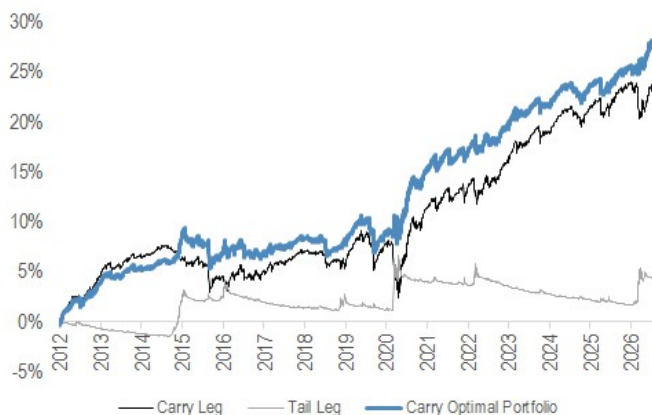


Source: J.P. Morgan, Bloomberg Finance L.P.

While the above shows potential improvements for the tail and carry legs individually, we prefer a solution **using both carry and tail to cover the drawbacks of each other** by assigning different weights depending on the mandate.

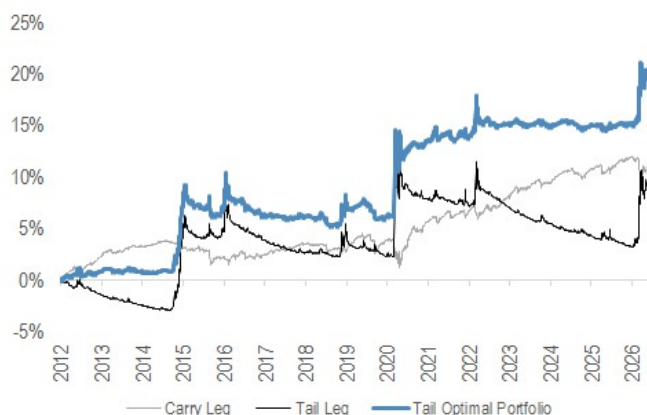
For optimizing carry, we can add 50% participation from the tail leg to achieve a Sharpe ratio of 0.89 with a worst DD of -3.67% (Figure 91). For tail mandates, we can add 50% participation to the carry leg to reduce drag, while maintaining robustness towards corrections and minimal bleed. The Sharpe ratio of the tail portfolio was able to reach 0.53, with a max DD of -4.66% (Figure 92).

Figure 91: Carry optimal portfolio



* Assumes no transaction costs
Source: J.P. Morgan, Bloomberg Finance L.P.

Figure 92: Tail optimal portfolio



* Assumes no transaction costs
Source: J.P. Morgan, Bloomberg Finance L.P.

FX volatility carry is strongest in shorter-dated, higher-vol surface points. The pattern is robust across regimes, except during periods of low volatility, broad USD rallies, and elevated central bank activity.

3.1.10 FX Volatility Carry Strategy

FX volatility carry behaves somewhat differently from its equities counterpart. Over the past 15 years, G10 FX implied volatility has traded only modestly above realized volatility. To identify the most persistent sources of carry and the key drivers of drawdowns, we evaluated delta-hedged option-selling strategies across a VXY-weighted G10 basket, spanning 7 strikes, 5 tenors, and 4 market regimes (see [original note](#)). The results point to a clear hierarchy across the volatility surface, while also highlighting the importance of volatility regimes, term structure dynamics, USD trends, and central bank activity in determining carry outcomes.

Across strikes and tenors, ATM structures repeatedly struggle to generate attractive risk-adjusted returns, as premium collection is frequently offset by realized volatility shocks. In contrast, low-delta USD call structures consistently outperform, benefiting from FX skew pricing and the systematic underperformance of spot-volatility correlation. Across these factors, short-dated OTM USD calls exhibited the strongest and most persistent Sharpe ratios.

Volatility Regimes

The first regime classification considers the level of FX volatility. We divide VXY into high-, medium-, and low-volatility environments using ± 0.75 standard deviations from the long-run mean. Sharpe ratios vary materially across volatility regimes. ATM structures remain weak performers regardless of the volatility backdrop, while wingier call structures consistently outperform comparable put structures. This result is consistent with the asymmetry of FX smiles, which are often skewed toward USD calls and therefore embed larger volatility premia on the USD topside (see Figure 93 and Figure 94).

Figure 93: Sharpe ratio for selling delta-hedged options for various tenors and strike combinations in specific VXY regimes

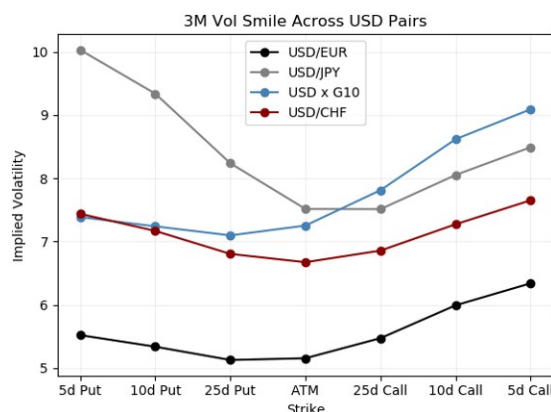
VXY weighted basket of options (puts designate USD puts and calls USD calls). Holding period: 2W and 1M until expiry, 3M for 2M, and 6M and 1Y for 3M

	5D Put	10D Put	25D Put	ATM	25D Call	10D Call	5D Call	Regime
2W	-0.05	-0.01	-0.1	-0.08	0.15	0.35	0.3	Low Volatility
1M	-0.19	-0.1	-0.12	-0.16	-0.01	0	-0.08	Low Volatility
3M	-0.14	-0.14	-0.11	-0.12	-0.11	-0.16	-0.19	Low Volatility
6M	-0.08	-0.08	-0.11	-0.14	-0.15	-0.17	-0.17	Low Volatility
12M	-0.06	-0.06	-0.11	-0.16	-0.16	-0.15	-0.13	Low Volatility
2W	0.38	0.6	0.7	0.62	1.12	1.36	1.29	Medium Volatility
1M	0.24	0.37	0.46	0.5	0.92	1.09	1.03	Medium Volatility
3M	0.33	0.43	0.45	0.5	0.88	1.06	0.98	Medium Volatility
6M	0.49	0.59	0.54	0.52	0.78	0.86	0.77	Medium Volatility
12M	0.71	0.75	0.57	0.48	0.66	0.77	0.77	Medium Volatility
2W	0.73	0.74	0.35	0.14	0.52	0.91	0.99	High Volatility
1M	0.35	0.29	0.15	0.24	0.7	0.91	0.83	High Volatility
3M	0.63	0.53	0.29	0.29	0.68	0.93	0.91	High Volatility
6M	0.69	0.61	0.36	0.34	0.65	0.85	0.84	High Volatility
12M	0.83	0.77	0.48	0.41	0.61	0.75	0.74	High Volatility

Source: J.P. Morgan

Figure 94: Most USD smiles price more premium towards the USD upside

Smiles are all in USD/FX notation



Source: J.P. Morgan

Call-selling performance is notably weak during low-volatility regimes. When volatility is already compressed, future outcomes become skewed toward mean reversion rather than further volatility compression, reducing the premium available to harvest.

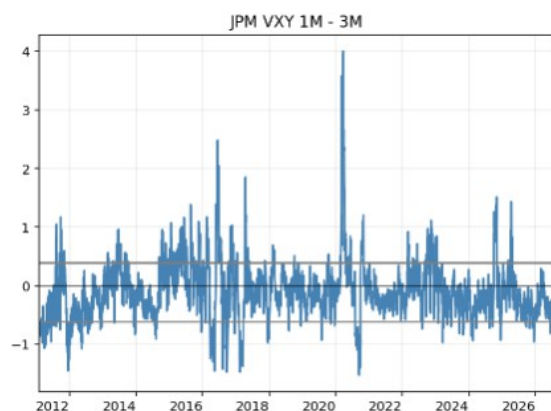
Volatility Term Structure

We next classify regimes according to the slope of the FX volatility curve, separating steep, normal, and inverted term structures using the relationship between 1M and 3M VXY (see Figure 95). The term structure acts as a proxy for how much stress is being priced into the market.

ATM performance remains weak regardless of curve shape. In contrast, short call structures perform best in the presence of steep and normal curves, where realized volatility remains contained and spot-volatility correlation appears overpriced. When curves invert, performance deteriorates, as inversion typically coincides with risk-off conditions, elevated event risk, and richer front-end volatility (see Figure 96).

Figure 95: VXY regimes can be distributed depending on the vol curve steepness

1M vs. 3M VXY, with +/- 1 std band around the long-term mean



Source: J.P. Morgan

Figure 96: Sharpe ratios for holding short delta-hedged options for different tenor and strike combinations in specific term structures

Hold 2W and 1M until expiry, 3M for 2M, and 6M and 1Y for 3M. Color coding: green = favorable vol shorts, red = poor vol shorts (favorable longs)

	5D Put	10D Put	25D Put	ATM	25D Call	10D Call	5D Call	Regime
2W	0.33	0.47	0.33	0.14	0.45	1.06	1.19	Steep Volatility Curve
1M	0.29	0.31	0.44	0.4	0.82	1.05	0.96	Steep Volatility Curve
3M	0.5	0.53	0.43	0.44	0.73	0.88	0.81	Steep Volatility Curve
6M	0.56	0.58	0.42	0.38	0.56	0.63	0.57	Steep Volatility Curve
12M	0.64	0.68	0.52	0.42	0.51	0.58	0.55	Steep Volatility Curve
2W	0.34	0.49	0.48	0.42	0.91	1.28	1.32	Normal Term Structure
1M	0.14	0.32	0.33	0.36	0.76	1	0.99	Normal Term Structure
3M	0.16	0.26	0.3	0.33	0.67	0.76	0.63	Normal Term Structure
6M	0.31	0.4	0.36	0.33	0.54	0.57	0.46	Normal Term Structure
12M	0.48	0.54	0.4	0.32	0.44	0.48	0.44	Normal Term Structure
2W	0.47	0.62	0.38	0.24	0.65	0.64	0.47	Inverted Volatility Curve
1M	0.1	0.11	0.03	0.12	0.49	0.43	0.23	Inverted Volatility Curve
3M	0.31	0.26	0.13	0.15	0.47	0.58	0.45	Inverted Volatility Curve
6M	0.41	0.36	0.2	0.22	0.54	0.67	0.55	Inverted Volatility Curve
12M	0.58	0.52	0.27	0.22	0.49	0.65	0.62	Inverted Volatility Curve

Source: J.P. Morgan

USD Trend Regimes

The third classification evaluates the impact of directional USD trends. Regimes range from strong USD weakness to strong USD strength and capture environments where directional spot moves persist for extended periods (see Figure 97).

Sharpe ratios remain close to zero for ATM structures regardless of trend direction. More broadly, USD call structures outperform USD put structures across most regimes, suggesting that USD trends affect option-selling performance primarily through changes in implied volatility and skew rather than through the spot direction itself (see Figure 98). USD put-selling performance is generally weaker, although longer-dated and wingier structures occasionally generate positive returns during moderate or rangebound USD environments. The USD/NOK experience between 2022 and 2023 provides one example where selling puts remained constructive despite a weakening USD backdrop.

Strong USD appreciation remains one of the more challenging environments for call-vol sellers. Rising implied volatility, increased demand for USD topside protection, and firmer call skew all reduce the attractiveness of short-vol positions, particularly in longer tenors, where vega exposure becomes more important.

Figure 97: There have been a few sparse periods in which USD trends continued for 6M+



Source: J.P. Morgan

Figure 98: Sharpe ratios for holding short delta-hedged options for different tenor and strike combinations during specific USD trends

Hold 2W and 1M until expiry, 3M for 2M, and 6M and 1Y for 3M

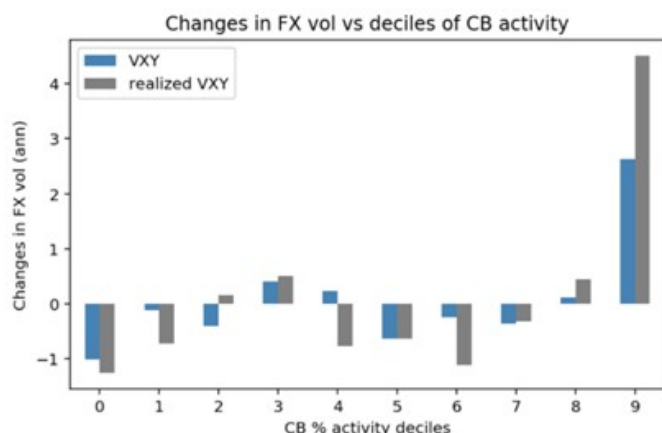
	5D Put	10D Put	25D Put	ATM	25D Call	10D Call	5D Call	Regime
2W	0.6	0.86	0.73	0.38	0.82	1.37	1.57	Strong USD Downside Trend
1M	0.46	0.58	0.5	0.46	0.87	1.22	1.21	Strong USD Downside Trend
3M	0.49	0.58	0.49	0.42	0.71	0.93	0.88	Strong USD Downside Trend
6M	0.42	0.51	0.4	0.31	0.49	0.56	0.46	Strong USD Downside Trend
12M	0.5	0.56	0.41	0.28	0.36	0.41	0.36	Strong USD Downside Trend
2W	0.61	0.53	0.21	0.08	0.55	1.04	1.08	Mild USD Downside Trend
1M	0.45	0.39	0.12	0.02	0.43	0.71	0.72	Mild USD Downside Trend
3M	0.36	0.36	0.24	0.32	0.79	0.94	0.73	Mild USD Downside Trend
6M	0.54	0.56	0.43	0.48	0.88	1.03	0.89	Mild USD Downside Trend
12M	0.66	0.67	0.52	0.5	0.8	0.98	0.95	Mild USD Downside Trend
2W	0.29	0.18	0.01	-0.05	0.31	0.51	0.49	Sideways USD Trend
1M	0.07	0.06	0.11	0.17	0.48	0.73	0.77	Sideways USD Trend
3M	0.15	0.2	0.19	0.29	0.69	0.89	0.78	Sideways USD Trend
6M	0.47	0.53	0.47	0.48	0.8	0.9	0.75	Sideways USD Trend
12M	0.68	0.68	0.52	0.49	0.72	0.8	0.73	Sideways USD Trend
2W	0.19	0.13	0.01	-0.08	0.27	0.68	0.64	Mild USD Upside Trend
1M	0.08	0.14	0.23	0.16	0.51	0.79	0.69	Mild USD Upside Trend
3M	0.25	0.29	0.39	0.46	0.75	0.8	0.7	Mild USD Upside Trend
6M	0.35	0.38	0.37	0.42	0.67	0.73	0.64	Mild USD Upside Trend
12M	0.61	0.62	0.46	0.44	0.63	0.71	0.69	Mild USD Upside Trend
2W	0.16	0.45	0.59	0.69	0.87	0.82	0.66	Strong USD Upside Trend
1M	-0.12	0.01	0.06	0.3	0.61	0.5	0.32	Strong USD Upside Trend
3M	0.05	0.1	0.08	0.1	0.34	0.4	0.3	Strong USD Upside Trend
6M	0.23	0.26	0.15	0.12	0.29	0.31	0.24	Strong USD Upside Trend
12M	0.42	0.43	0.23	0.16	0.27	0.32	0.3	Strong USD Upside Trend

Source: J.P. Morgan

Central Bank Activity

The final regime classification focuses on major developed-market central bank activity. Mapping policy-event concentration to FX volatility reveals only a modest relationship on average, with meaningful volatility impacts concentrated in periods of exceptionally low or exceptionally high activity (Figure 99). Option-selling performance generally deteriorates as central bank activity intensifies. Volatility frequently rises around major policy events, either due to regime uncertainty or the resolution of previously anticipated decisions. As a result, carry strategies become less reliable when macro catalysts cluster (see Figure 100 and Figure 101).

Figure 99: Volatility mostly affected only under extreme central bank activity levels (unusually low/high)



Source: J.P. Morgan

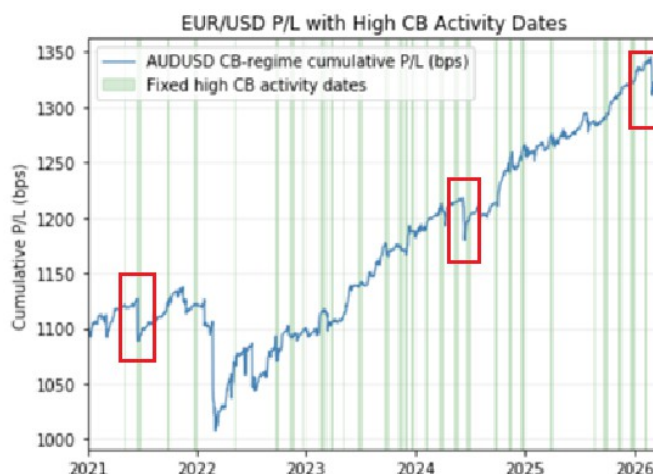
Figure 100: Sharpe ratios for holding short delta-hedged options for different tenor and strike combinations for specific central bank activity levels

Hold 2W and 1M until expiry, 3M for 2M, and 6M and 1Y for 3M

	5D Put	10D Put	25D Put	ATM	25D Call	10D Call	5D Call	Regime
2W	1.4	1.14	0.53	0.3	0.67	0.99	1.13	Very Low Central Bank Activity
1M	0.76	0.66	0.37	0.39	0.72	0.98	0.95	Very Low Central Bank Activity
3M	1.08	1	0.74	0.65	0.88	1.05	1.01	Very Low Central Bank Activity
6M	1.02	0.97	0.75	0.67	0.89	1.03	0.99	Very Low Central Bank Activity
12M	1.07	1.02	0.76	0.67	0.86	1.04	1.09	Very Low Central Bank Activity
2W	0.11	0.36	0.47	0.35	0.79	1.1	0.93	Low Central Bank Activity
1M	-0.1	0.09	0.32	0.38	0.67	0.55	0.33	Low Central Bank Activity
3M	0	0.09	0.17	0.3	0.61	0.55	0.33	Low Central Bank Activity
6M	0.24	0.35	0.35	0.4	0.61	0.54	0.38	Low Central Bank Activity
12M	0.39	0.49	0.44	0.42	0.51	0.51	0.42	Low Central Bank Activity
2W	0.09	0.16	0.13	0.22	0.58	0.96	1.15	Some Central Bank Activity
1M	-0.16	-0.1	0	0.14	0.49	0.84	0.95	Some Central Bank Activity
3M	0.04	0.05	0.01	0.06	0.31	0.53	0.56	Some Central Bank Activity
6M	0.1	0.11	0.04	0	0.17	0.32	0.3	Some Central Bank Activity
12M	0.19	0.18	0.03	-0.04	0.08	0.17	0.17	Some Central Bank Activity
2W	0.22	0.17	0.12	0.03	0.22	0.26	0.19	High Central Bank Activity
1M	0.3	0.22	0	-0.07	0.16	0.25	0.23	High Central Bank Activity
3M	0.22	0.16	0.03	-0.08	0.08	0.18	0.19	High Central Bank Activity
6M	0.18	0.14	-0.01	-0.09	0.02	0.07	0.06	High Central Bank Activity
12M	0.27	0.21	0.04	-0.06	0.03	0.07	0.09	High Central Bank Activity

Source: J.P. Morgan

Figure 101: EUR/USD P/L dips during some high-central-bank-activity periods



Source: J.P. Morgan

Low-delta USD calls continue to outperform across most policy regimes, although the magnitude of the advantage narrows when event risk is elevated and volatility repricing becomes more likely.

Managing Drawdowns and Enhancing Carry

While short-dated low-delta USD calls emerge as the strongest standalone carry structure, the earlier regime analysis highlights several environments where carry becomes more vulnerable, particularly during term structure inversion, concentrated central bank activity, and strong USD trends. The implication is less about identifying a single optimal strike and instead more about combining robust carry structures with disciplined regime management.

Hit ratios improve as structures move further out-of-the-money, with 5-to-10-delta USD calls delivering the strongest combination of return persistence and robustness (see Figure 97). Cross-currency performance broadly aligns with the aggregate findings. Although individual currencies exhibit varying sensitivities to volatility regimes, the strongest risk-adjusted performance continues to come from low-delta USD call-selling structures. Taken together, the evidence suggests that short 25-to-10 delta USD calls provide one of the cleanest and most persistent methods of monetizing FX volatility and skew premia.

Figure 102: Hit ratios for holding short delta-hedged options for different tenor and strike combinations

	5D Put	10D Put	25D Put	ATM	25D Call	10D Call	5D Call
2W	28.6%	28.3%	27.4%	27.2%	28.6%	30.0%	30.3%
1M	31.6%	31.2%	30.5%	30.1%	31.5%	33.0%	33.5%
3M	33.9%	33.4%	32.2%	32.2%	33.6%	35.6%	36.3%
6M	35.9%	35.4%	34.5%	34.2%	36.0%	37.6%	38.3%
1Y	35.7%	35.1%	34.2%	33.7%	35.1%	36.4%	37.2%

Source: J.P. Morgan

Figure 103: Worst drawdown for holding short delta-hedged options for different tenor and strike combinations

In bp	5D Put	10D Put	25D Put	ATM	25D Call	10D Call	5D Call
2W	-1129.4	-1510.9	-2417.3	-2499.1	-1601.3	-1422.8	-1217.3
1M	-941.9	-1374.6	-2811.1	-2551.4	-1638.2	-1577.8	-1402.8
3M	-723.2	-1122.6	-2206.9	-2441.8	-2063.1	-2076.7	-1921.9
6M	-713.6	-992.0	-1905.8	-2518.6	-2486.2	-2452.5	-2240.3
1Y	-818.5	-1151.8	-1846.0	-2751.0	-2946.0	-2777.0	-2447.8

Source: J.P. Morgan

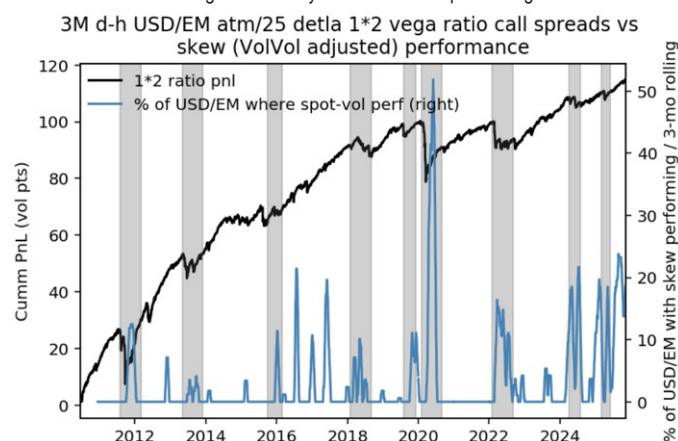
EM Skew Selling

In line with the above discussed findings for the example of a VXY-weighted portfolio, systematically selling USD/EM skew has delivered strong risk-adjusted returns. Skew performance (i.e., skew-selling underperformance) was concentrated in a limited number of carry-unwind episodes (see Figure 104). More recently, USD skew has continued to underperform across much of EMFX, leaving skew harvesting as one of the more attractive carry opportunities in the asset class. This is particularly relevant in the current environment, where outright vol levels remain relatively subdued, limiting the premium available from traditional short-vol strategies.

Conventional spot-vol correlation tends to overstate the attractiveness of long-skew positions, as it captures only the directional relationship between spot and realized vol while largely ignoring the magnitude and variability of volatility moves themselves. A VolVol-adjusted spot-vol correlation measure (see Figure 105) incorporates this additional dimension and has historically exhibited a much stronger relationship with realized skew P/Ls, making it a more reliable framework for identifying genuine skew risk premia and assessing the attractiveness of skew-harvesting opportunities.

Figure 104: Strong systematic performance from skew selling

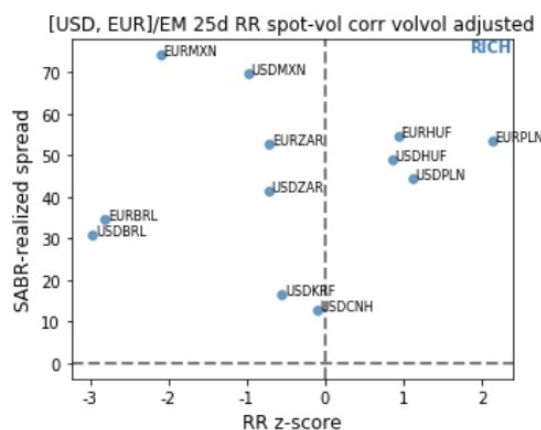
3M USD/EM ATM/25d 1*2 vega notional ratio call spreads. TC of 0.5vol accounted for. VolVol adjusted SABR rho (implied spot-vol corr) and trailing realized spot-vol corr are used in calculating the spot-vol implied-realized spread. That spread construes what we use in calculating the % of ccys whose skew is performing



Source: J.P. Morgan

Figure 105: Vol-of-vol adjusted spot-vol corr

Theoretical framework detailed [here](#)



Source: J.P. Morgan

Regarding the current backdrop, with a busier event calendar in Sep-Oct, these risks are more likely to generate intermittent gamma shocks than to trigger a sustained unwind of the broader carry regime and are thus not necessarily detrimental to skew-selling strategies. In this context, an implementation aspect to consider is that the prospect of episodic gamma noise argues for favoring delta-hedged ratio call spreads structures over outright short delta-hedged risk reversals.

3.1.11 FX Volatility Tail Strategy

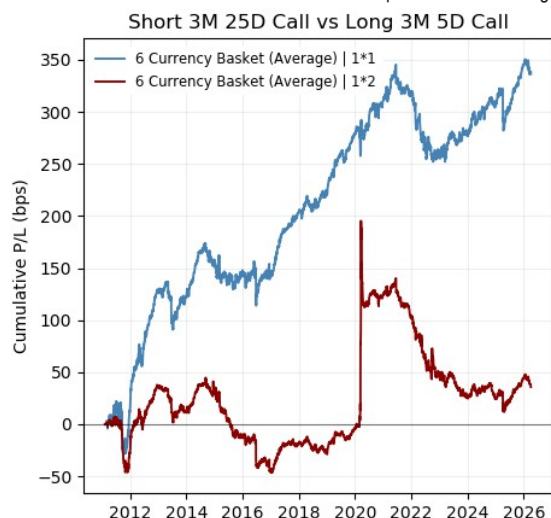
FX vol tails are valuable crash insurance, but their performance has historically been concentrated in major shock periods. The bleed experienced during the gamma-rich 2022-2023 regime highlights the importance of allocating alongside vol carry rather than relying on tails alone.

Tail exposure: Tail hedges are often criticized for persistent theta decay, but the cost of convexity is not uniform across the FX volatility surface. Historical pricing suggests that higher-strike volatility can partially fund exposure to deeper OTM tails, improving the long-run tradeoff between protection and carry. To evaluate this idea, we examine delta-hedged inverse call ratios that are short 25-delta USD calls and long 5-delta USD calls (1x2 notionals). This framework delivered its intended protection during the onset of COVID, when spot and volatility moves aligned with the structure's convexity exposure. Performance was notably weaker during the high-gamma environment of 2022-2023, when tail ownership was less effective in offsetting persistent gamma noise. Nevertheless, the results suggest that delta-hedged inverse ratios can provide a more efficient method of owning convexity than outright tail structures (Figure 106). Cumulative returns remained positive over the full sample, suggesting that the long-run bleed associated with convexity ownership was modest relative to the protection delivered during tail events.

Subsidized gamma: Tail hedges and long-gamma strategies address different market environments. While tail structures perform best during systemic shocks, subsidized gamma seeks to own realized volatility more efficiently during periods of elevated market movements that are not necessarily associated with crisis conditions. The framework finances long front-end ATM volatility through the sale of richer longer-dated skew, improving the economics of long-gamma ownership while retaining exposure to realized volatility and directional dislocations. The strongest post-COVID performance was in high-beta G10 currencies, particularly the Scandies and Antipodeans, where realized volatility increasingly exceeded implied volatility in an environment dominated by central-bank-driven market swings (Figure 107).

Figure 106: Convexity ownership via delta-hedged inverse ratios fared well through the COVID vol shock but gave back PnL during the gamma noise of 2022-2023

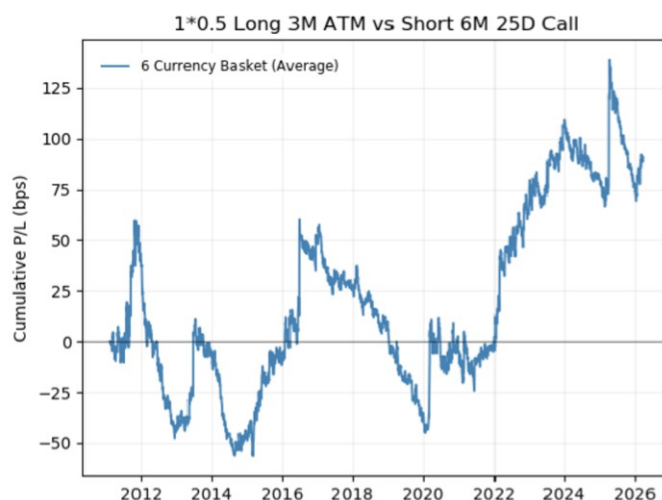
The currency basket includes USD call/G10 put options, excluding EUR, CHF and JPY. The 1*1 and 1*2 labels refer to the respective notional weightings.



Source: J.P. Morgan

Figure 107: Post-Covid, the high-beta Scandies and Antipodeans have exhibited robust subsidized gamma returns

The currency basket includes USD call/G10 put options, excluding EUR, CHF and JPY



Source: J.P. Morgan

Subsidized gamma and tail hedges should be viewed as complementary rather than competing approaches. Tail structures provide protection during crisis periods, while

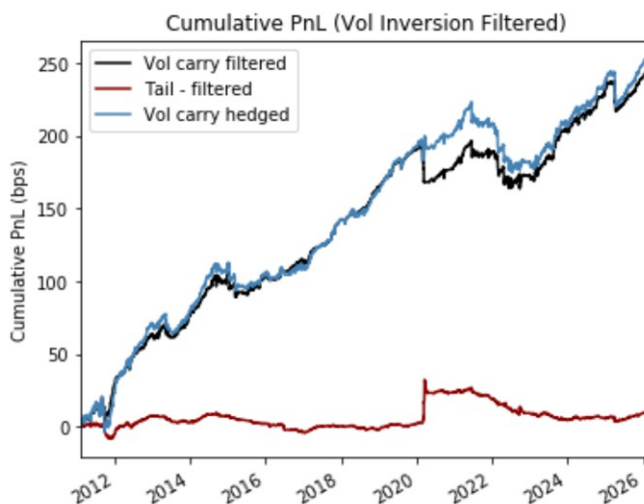
subsidized gamma performs better in sustained-realized-volatility environments. Combining the two addresses the key weakness observed in the tail-hedging analysis, namely periods where gamma noise dominates in the absence of a true tail event.

Optimal Carry Harvesting While Hedging Tail Exposure

The earlier inverse ratio spread offers a more sustainable way to own convexity. Overlaying this tail hedge (1x short 3M 25-delta calls vs. 2x long 3M 5-delta calls) on our preferred carry trade (short 1M 10-delta calls) improves carry outcomes while limiting losses during genuine stress episodes (Figure 108). We further enhance the portfolio by suspending additions to the portfolio when the volatility curve inverts, as inversion typically reflects acute demand for near-dated protection rather than durable roll-down, making structures that mostly rely on selling volatility mute during likely risk-off, high-vol periods. In the backtest, these parameters together materially reduced the COVID-era drawdown while preserving most of the underlying carry profile. Losses were modest and recovered quickly, versus it taking roughly six months for the unhedged portfolio to recover.

Figure 108: Tail hedge efficiently trimmed vol carry drawdown during the onset of COVID

Vol carry = short 1M 10-delta USD/FX call, delta-hedged. Tail = 3M USD/FX 25-delta / 5-delta 1*2 inverse ratio spread (i.e., short 25-delta / long 5-delta), delta-hedged



Source: J.P. Morgan

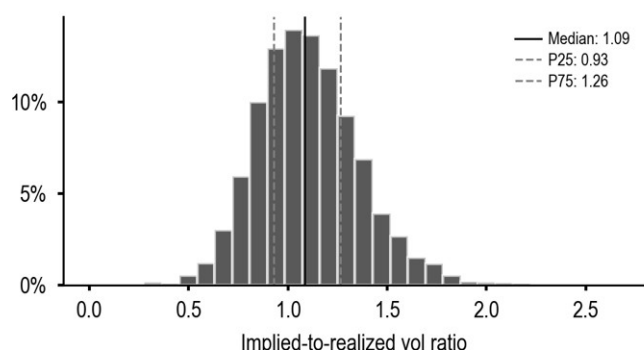
The upper right of the vol surface delivers the best carry in rates; allocate some long vega to minimize drawdowns.

3.1.12 Rates Volatility Carry Strategy

Selling rates vol (short gamma) has become a popular strategy for enhancing returns, in part because of the perception that rates vol tends to be overpriced due to high demand associated with mortgage convexity hedging, for example. Indeed, in the gamma sector, implied-to-realized vol ratios are biased towards values above one (Figure 109). Thus, in principle, **one can sell gamma to harvest this volatility risk premium**. However, due to the highly nonlinear nature of option returns, this ratio alone is insufficient to assess whether rates vol is systematically overpriced. **Jump risk is a primary concern:** rates vol tends to exhibit sharp spikes, often associated with event days or geopolitical developments, and a single-day spike in vol can erase multiple days worth of carry (e.g., [link](#)).

Figure 109: Implied rates vol tends to exceed delivered vol in the gamma sector

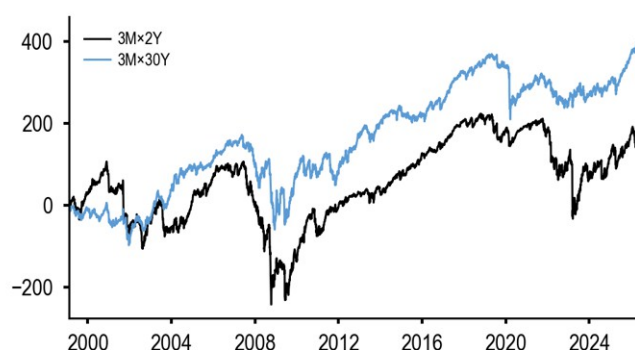
Distribution of daily implied-to-realized vol ratios* for 3Mx2Y swaptions, 1997 - present



* Implied vol divided by standard deviation of daily changes in the underlying LIBOR (through 12/31/2021) or SOFR (1/1/2022 - present) forward rate in the subsequent 21 business days
Source: J.P. Morgan

Figure 110: Over the past ~30Y, short straddles have been profitable in the shortest expiries

Cumulative total return (abp) from 3Mx2Y and 3Mx30Y short straddles in various swaption structures, rolled monthly and delta-hedged daily, 1997 - present

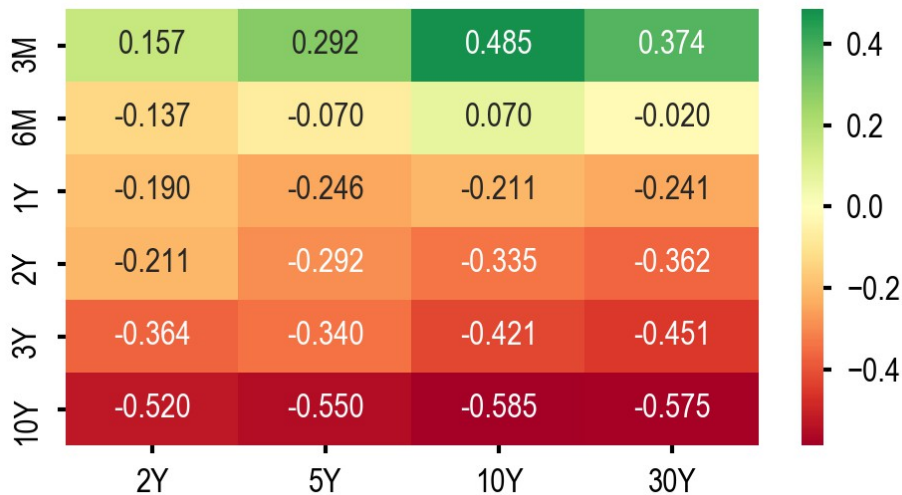


Source: J.P. Morgan

Instead, it is necessary to analyze the P/L for realistic short-gamma strategies. Our preferred expression for going short or long gamma is swaption straddles, usually with strikes reset monthly and daily delta hedging. Figure 110 shows the cumulative returns for short straddles, rolled monthly and delta-hedged daily, for 3Mx2Y and 3Mx10Y swaptions. We see that over the past ~3 decades, naively selling gamma continuously would have been profitable despite implied-to-delivered vol ratios being biased greater than unity. Figure 111 shows the Sharpe ratio for this short gamma strategy across the swaption vol surface. **The top right of the surface is the most attractive region for shorting gamma.**

Figure 111: The upper right of the swaption vol surface is best for harvesting carry

Sharpe ratios* for selling swaption straddles rolled monthly and delta-hedged daily, 1997 - present



* Sharpe ratio is mean daily return divided by standard deviation of daily returns
Source: J.P. Morgan

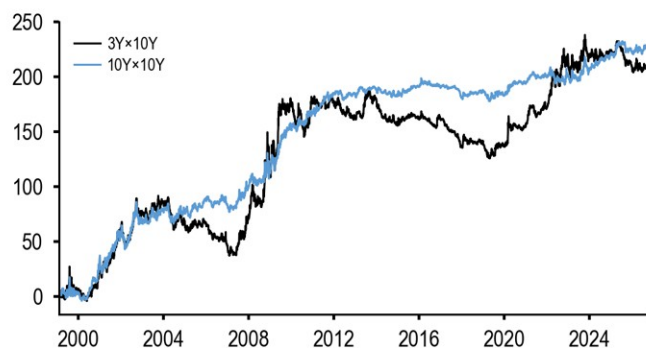
3.1.13 Rates Volatility Tail Strategy

In the vega sector, implieds have historically been cheap, resulting in long-vega strategies being profitable.

For tail, the vega sector minimizes bleed. Part of this is mechanical, as longer-expiry options simply decay at a slower pace; however, the rates volatility surface is also inverted in longer expiries, in part due to issuance of callable bonds. Due to this inverted term structure of vol, buyers of the deep vega part of the surface benefit from positive vol slide, which has led to the proliferation of systematic long vega strategies. While the positive vol slide is not quite large enough to overcome theta decay, buyers are able to maintain long vol exposure at a discounted cost of carry. **Over the past ~30 years, it has been profitable to buy vega** (Figure 112) because implied vols have been chronically (slightly) underpriced (Figure 113), in part because of the aforementioned callable bonds.

Figure 112: Buying vega has been profitable over the past ~30 years

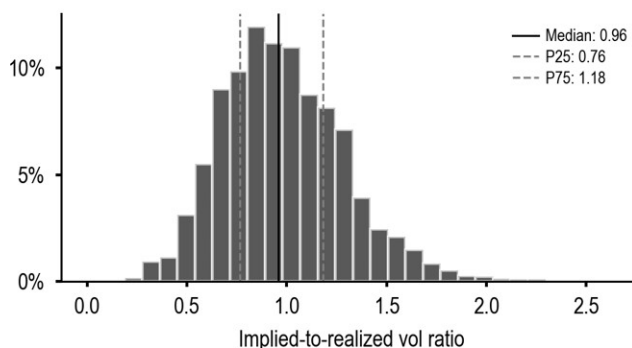
Cumulative PnL (abp) from buying swaption straddles, rolled monthly and delta-hedged daily, 1997 - present



Source: J.P. Morgan

Figure 113: Implieds tend to be underpriced in the vega sector

Distribution of daily implied-to-realized vol ratios* for 10Yx10Y swaptions, 1997 - present



* Implied vol divided by standard deviation of daily changes in the underlying LIBOR (through 12/31/2021) or SOFR (1/1/2022 - present) forward rate in the subsequent 21 business days
Source: J.P. Morgan

3.2 Relative Value

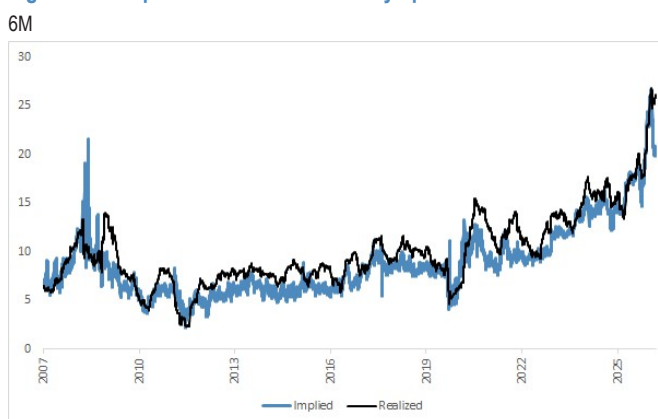
3.2.1 Relative Value in Equities — the Volatility Dispersion Trade

Dispersion is exhibiting a **dot-com replay**, characterized by **elevated stock volatility and low correlation**.

Dynamically sizing the index and stock legs to manage net short gamma can enhance alpha. Diversifying between Europe and the US helps mitigate correlation shocks, and stock selection is essential to improve carry.

In equities, the most popular trade in RV space is the **volatility dispersion trade between an index and single stocks**. Dealers receive single-stock vega from structured products clients shorting downside puts, which are less liquid to hedge. Dispersion trades allow the vega to be recycled to the index, which can be easily traded by dealers, and for investors to benefit from **realized volatility spread above implied** from the correlation risk premium.

Figure 114: Implied and realized volatility spread for SPXT75

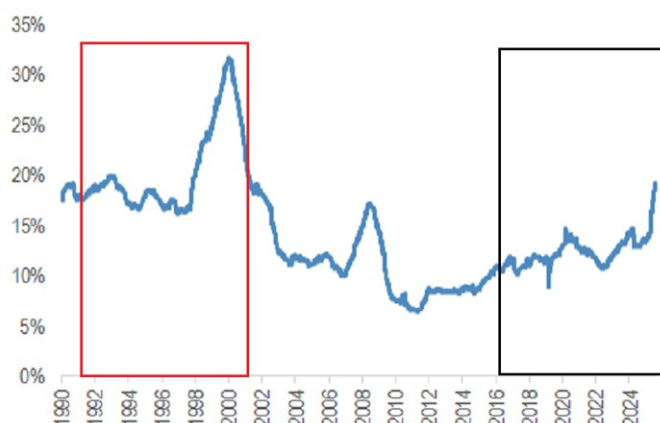


Source: J.P. Morgan, Bloomberg Finance L.P.

Vega neutral dispersion is the most common implementation. The trade has been very successful, especially since 2020, as correlation has kept resetting lower, increasing the volatility spread between stocks and the index.

Figure 115: Realized volatility spread for equal-weighted index, past 10Y vs. 10Y in the dot-com era

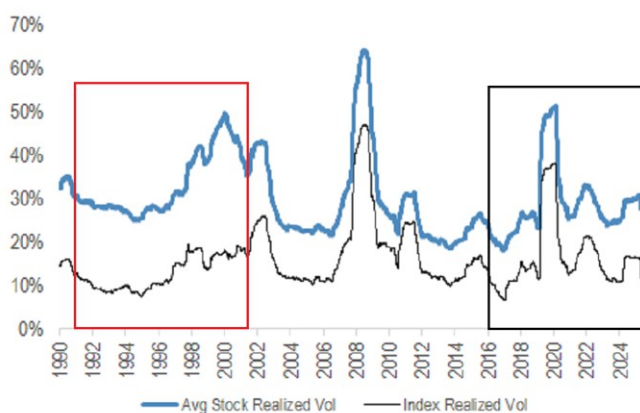
Equally weighted 200 US names with history from 1990 12M rolling



Source: J.P. Morgan, Bloomberg Finance L.P.

Figure 116: Realized avg stock vol and index vol 12M for equal-weighted index, past 10Y vs. 10Y in the dot-com era

Equally weighted 200 US names with history from 1990 12M rolling



Source: J.P. Morgan, Bloomberg Finance L.P.

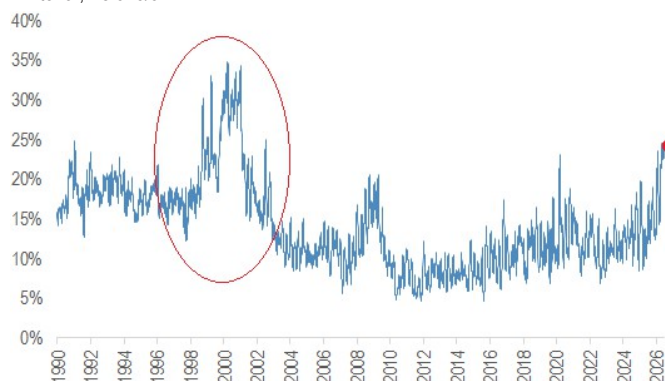
Some argue the increasing spread is due to higher-volatility names that are tech heavy growing in market cap and index weight. While this is true, when we examined an **equally weighted basket of a broader universe** extending the backtest window to **pre dot-com era**, not only did it also deliver well recently but also the **spread during the dot-com era was even higher** (Figure 115). The main reason for this was sustained high single-stock volatility with idiosyncratic moves and low correlation (Figure 116).

$$\sigma_{\text{index}} \approx \sigma_{\text{stock}} * \sqrt{\rho}$$

Similarly, the recent stellar returns in single-stock volatility, despite entering at elevated levels, are due to **unprecedented earnings uncertainty, which is driven by the AI revolution**. The magnitude of the reaction of stock prices to news echoes what we saw during the dot-com era, i.e., moves of 30% or more on news or earnings. As single stocks behave idiosyncratically, correlation remains low, as does index volatility, driving the vol spread to new highs.

Figure 117: Realized vol spread of stocks vs. a synthetic index compared to 1998-2000

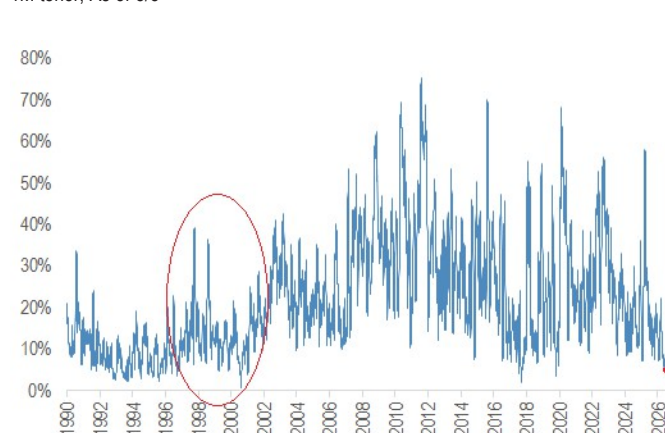
1M tenor, As of 8/5



Source: J.P. Morgan, Bloomberg Finance L.P.

Figure 118: Realized correlation compared to 1998-2000

1M tenor, As of 8/5



Source: J.P. Morgan, Bloomberg Finance L.P.

However, a challenge to vega-neutral dispersion at low implied correlation levels is the potential for **sharp drawdowns in correlation events**, such as what we witnessed around “Liberation Day” (Figure 119).

Figure 119: Dispersion vol spread scenario table for 75-stock basket

Rows - correlation level, columns - avg single-stock vol level

	22%	24%	26%	28%	30%	32%	34%	36%	38%	40%	42%	44%	46%	48%	50%
70%	3.2%	3.5%	3.9%	4.2%	4.5%	4.8%	5.2%	5.5%	5.8%	6.1%	6.4%	6.8%	7.1%	7.4%	7.7%
65%	3.8%	4.2%	4.6%	5.0%	5.3%	5.7%	6.1%	6.5%	6.9%	7.3%	7.6%	8.0%	8.4%	8.8%	9.2%
60%	4.4%	4.9%	5.3%	5.8%	6.2%	6.7%	7.1%	7.5%	8.0%	8.4%	8.9%	9.3%	9.8%	10.2%	10.7%
55%	5.1%	5.6%	6.1%	6.6%	7.1%	7.6%	8.1%	8.6%	9.2%	9.7%	10.2%	10.7%	11.2%	11.7%	12.2%
50%	5.8%	6.3%	6.9%	7.5%	8.1%	8.6%	9.2%	9.8%	10.4%	11.0%	11.5%	12.1%	12.7%	13.3%	13.8%
45%	6.5%	7.1%	7.8%	8.4%	9.1%	9.7%	10.4%	11.0%	11.7%	12.3%	12.9%	13.6%	14.2%	14.9%	15.5%
40%	7.2%	7.9%	8.7%	9.4%	10.1%	10.8%	11.6%	12.3%	13.0%	13.7%	14.5%	15.2%	15.9%	16.6%	17.3%
35%	8.0%	8.8%	9.6%	10.4%	11.2%	12.0%	12.8%	13.6%	14.4%	15.2%	16.0%	16.8%	17.6%	18.5%	19.3%
30%	8.9%	9.8%	10.7%	11.5%	12.4%	13.3%	14.2%	15.1%	16.0%	16.9%	17.8%	18.6%	19.5%	20.4%	21.3%
25%	9.8%	10.8%	11.8%	12.7%	13.7%	14.7%	15.7%	16.7%	17.6%	18.6%	19.6%	20.6%	21.6%	22.5%	23.5%
20%	10.8%	11.9%	13.0%	14.1%	15.1%	16.2%	17.3%	18.4%	19.5%	20.6%	21.6%	22.7%	23.8%	24.9%	26.0%
15%	12.0%	13.2%	14.4%	15.6%	16.8%	18.0%	19.1%	20.3%	21.5%	22.7%	23.9%	25.1%	26.3%	27.5%	28.7%
10%	13.3%	14.6%	16.0%	17.3%	18.6%	20.0%	21.3%	22.6%	24.0%	25.3%	26.6%	27.9%	29.3%	30.6%	31.9%

Black circle - average of single-stock vol and spreads for 1M prior to Feb 20th 2025, blue circle Apr 8th, pink circle Apr 30th

Source: J.P. Morgan, Bloomberg Finance L.P.

The reason is that being **vega-neutral** at low correlation results in high vol spread, which exposes the portfolio to a **large net short gamma** (Figure 120). In correlation shocks, stock vol and correlation both move up, but **correlation initially overtakes** the former. The best time to enter new trades is phase two, when **correlation remains elevated and stock vol settles**, but damage to existing trades is done ([link](#)).

Figure 120: Net short gamma for vega-neutral dispersion at different correlation levels, extra vega long from a gamma-neutral position, and PnL from the extra vega

6M tenor example

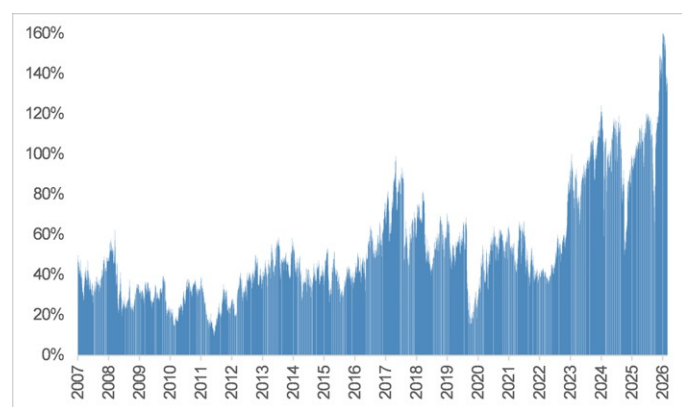
	Corr at 18 (Feb25)		Corr at 39 (Apr 8th)		Corr at 36 (Apr 30th)	
	Vol Level	Gamma	Vol Level	Gamma	Vol Level	Gamma
Index Vol	13.65%	-4.06%	27.13%	-2.06%	20.20%	-2.76%
SS Vol	29.00%	1.92%	41.50%	1.34%	32.60%	1.71%
Spread	15.35%	-2.13%	14.37%	-0.72%	12.40%	-1.05%
Extra Vega Long	112%					
Pnl From Extra Vega			14.06%		4.05%	

Source: J.P. Morgan, Bloomberg Finance L.P.

There are techniques to protect against this ([link](#)), such as intraday delta hedging on the index leg and **dynamically sizing up the index or stock vega** around the base position.

When both implied volatility and implied correlation are depressed, gamma-neutral dispersion is preferred over a reverse dispersion overlay. Historical backtests show that trading gamma-neutral dispersion when both volatility and correlation are below the 35th percentile of the 6M range captures most large correlation shocks with limited additional bleed (Figure 122).

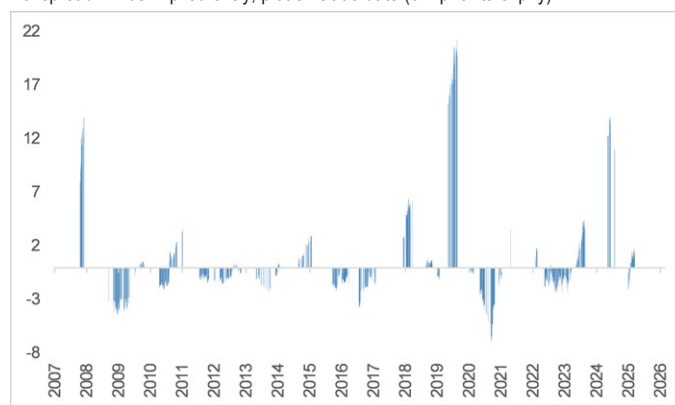
Figure 121: Extra single-stock long vega notional to make gamma-neutral



Source: J.P. Morgan

Figure 122: PnL estimate from the extra long-vega portion SPXT75 6M tenor when signal is on

Vol spread minus implied entry, plot on trade date (6M prior to expiry)



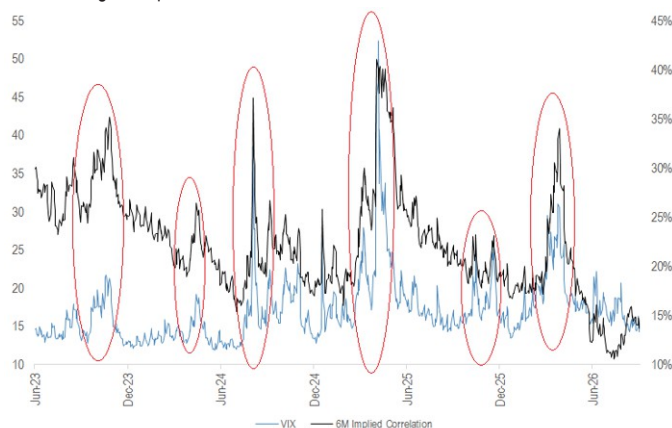
Source: J.P. Morgan

Switching to gamma-neutral dispersion in similar market conditions profits from **idiosyncratic risk and enables upside vega PnL during corrections**. Timely profit-taking on gamma-neutral trades is still essential to capture gains.

When correlations are low, long correlation exposure can be gained by increasing **index vega** positions, such as **VIX and UpVar**, or through reverse dispersion with careful implementation ([link](#)). A **call spread collar on VIX** acts like a low-bleed option on correlation. The best time to buy is usually towards the **end of the earnings period**, when correlation and index vol are typically low.

Figure 123: VIX and implied correlation of 6M SPXT75

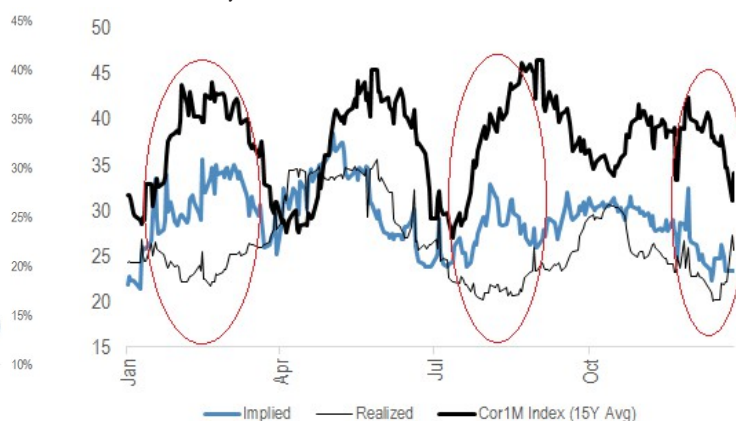
Left - VIX, right - implied correlation



Source: J.P. Morgan, Bloomberg Finance L.P.

Figure 124: SPXT75 correlation seasonality

2M tenor, 5Y history

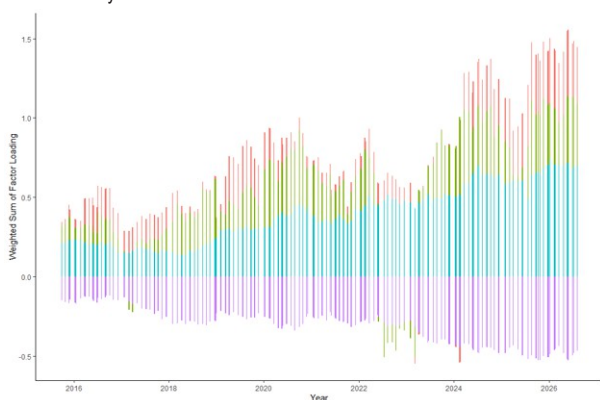


Source: J.P. Morgan, Bloomberg Finance L.P.

Diversification between SPX and SX5E could also be helpful due to their distinct characteristics. SPX typically exhibits lower correlation and index volatility than SX5E in normal market environments. However, during large sell-offs, SPX experiences higher realized volatility, driven by **concentration risk in factors and sectors**, particularly among tech stocks with strong growth and momentum attributes. On the other hand, SX5E is more diversified in terms of both sector and factor exposure, and its **correlation shocks tend to be driven more by geopolitical events**.

Figure 125: SPXT75 factor loading

Normalized by MSCI ACWI



Source: J.P. Morgan

Figure 126: EURO STOXX 50 index factor loading

Normalized by MSCI ACWI



Source: J.P. Morgan

Stock selection remains crucial for dispersion portfolios. Europe benefited from **lower entry levels on average**, and **energy** stocks delivered good carry as a result of the Middle East conflict. **Technology** stocks have been a key source of single-stock volatility PnL, as despite elevated implied volatility entry levels, the carry continues to outperform. **Industrials** and **healthcare** are next in line to experience the unprecedented earnings uncertainty associated with AI. **Financials** and **utilities** have underperformed overall, providing attractive entry levels for building long-dated vega positions.

Figure 127: EU names avg 3M realized vol — implied entry by sector

EU optionable universe >\$20bn mkt cap, average for 2026

Sector	Avg Carry	Hit Ratio	Avg Vol
Energy	5.0%	53%	29.0%
Information Technology	4.4%	56%	42.5%
Materials	2.8%	71%	33.7%
Industrials	2.4%	73%	31.1%
Consumer Staples	2.2%	46%	23.1%
Communication Services	1.7%	46%	26.5%
Consumer Discretionary	1.4%	61%	30.3%
Financials	0.3%	69%	27.0%
Health Care	0.1%	63%	30.3%
Utilities	-1.0%	44%	23.4%

Source: J.P. Morgan, Bloomberg Finance L.P.

Figure 128: US names avg 3M realized vol — implied entry by sector

US optionable universe >\$30bn mkt cap, average for 2026

Sector	Avg Carry	Hit Ratio	Avg Vol
Information Technology	7.3%	71%	59.4%
Industrials	1.1%	54%	37.7%
Materials	0.9%	49%	34.4%
Health Care	0.8%	46%	36.4%
Financials	0.4%	46%	33.2%
Communication Services	0.2%	45%	36.2%
Consumer Staples	-0.1%	45%	27.1%
Consumer Discretionary	-0.2%	45%	36.2%
Energy	-0.6%	42%	33.0%
Real Estate	-0.7%	41%	27.7%
Utilities	-1.4%	33%	28.7%

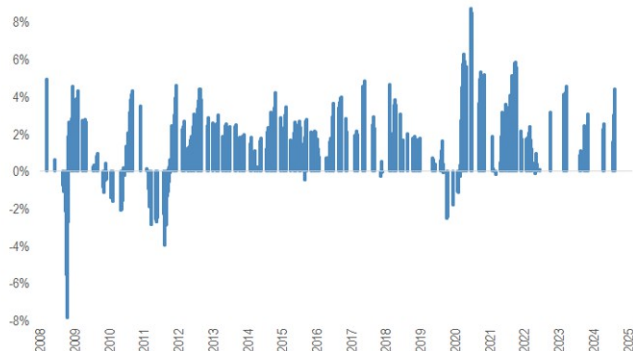
Source: J.P. Morgan, Bloomberg Finance L.P.

As we are in a regime similar to 1998-1999, one could adopt a “**buy the dip**” strategy to build a dispersion position after correlation shocks pump up index volatility ([link](#), [link](#)). We experimented with a signal of 3% correlation move, which roughly corresponds to a 1% vol spread increase. The trades entered have been largely successful (Figure 129).

Since the correlation risk premium is embedded within the index leg post shock, one can be more aggressive to bring the dispersion closer to being **theta-neutral** during such times. However, shorting extra index vega can be hurtful should momentum accelerate, such as in 2008 and 2020; therefore, we always prefer using **VIX put structures**, as discussed in 3.1.7 Equity Volatility Carry with VIX Put Spreads, for the implementation.

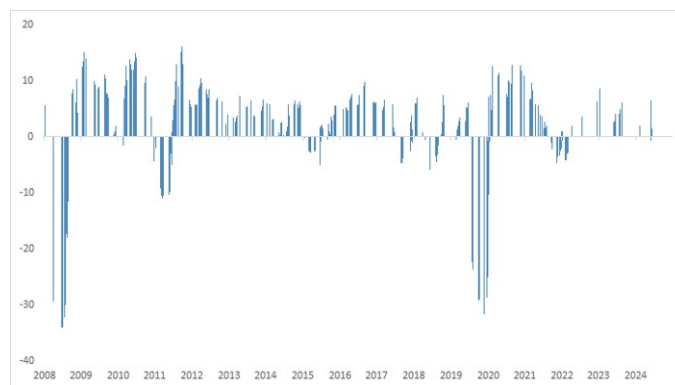
Figure 129: PnL for trades entered

Only trade when 6M implied corr pops by 3% above 1M average



Source: J.P. Morgan, Bloomberg Finance L.P.

Figure 130: Index implied minus subsequent 6M realized vol spread



Source: J.P. Morgan, Bloomberg Finance L.P.

The most profitable period for dispersion trading is **during earnings season**, with the Jan-Feb season being the best, followed closely by Jul-Aug, as companies deliver full-year and half-year updates during those times. We discuss further considerations in 3.4 Event Trading.

A multi-model framework combines high-frequency gamma signals, mean-reversion and vega-based approaches for longer expiries, and an autoregressive parametric vol surface model to identify vol RV opportunities across the curve.

3.2.2 Relative Value in FX Volatility and Model Framework

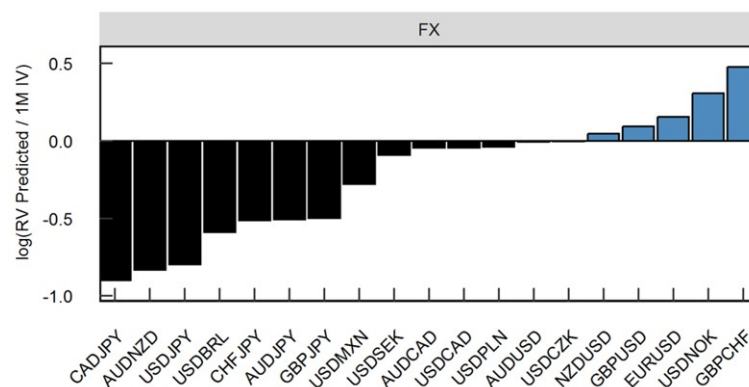
FX CARV Model

The main idea behind [our RV framework](#) is to identify vol RV opportunities by comparing current implied vol with the model forecast of “fair” realized vol. To do that, we use the first two moments of the return distribution, namely daily returns and daily realized power, to forecast realized vol. Daily realized power is defined as the sum of five-minute absolute returns across trading hours within a trading day. The resulting 120 features (2 features per day $\times$ 60 days) are passed through an attention layer with sigmoid activation. The attention mechanism incorporates a learnable similarity score to assign importance to previous time steps, and the weights are not required to follow a specific decay profile (i.e., recent observations are not necessarily assigned higher weights than more distant ones). A 20% dropout rate is applied to the hidden units to prevent overfitting.

Forecasting and historical performance: To evaluate performance, we rank FX vols on a daily basis based on the gap between implied vol and the model prediction. We then construct a portfolio with an equal number of long-vol and short-vol positions. Each position is scaled by the inverse of its vol level at entry and held for one month. As a result, each portfolio contains approximately $n \times 22$ overlapping long-vol and short-vol positions.

Figure 131: Cross Asset Relative Value (CARV) model ranking

Details given in the latest [Cross Asset Volatility](#) note



Source: J.P. Morgan

The model forms part of a Cross Asset Relative Value (CARV) framework. It performs best in equities and commodities (see 4.2 Cross-Asset Relative Value Volatility Trading for details). FX had a strong run prior to 2022; since then, the performance has flattened, as a high degree of correlation across FX has made it more difficult to identify profitable long/short divergences, resulting in longs and shorts offsetting. The cross-asset portfolio exhibits lower volatility and a higher Sharpe ratio than any individual asset class, suggesting that it benefits from diversification.

Relative Value Mean Reversion Model

A systematic framework (see the [original note](#) and related [follow-up](#)) identifies and trades relative value dislocations in FX volatility markets. By normalizing volatility data across currencies and constructing cross-market spreads, the model detects deviations from historical relationships that are expected to mean revert. The framework combines

statistical measures of dislocation with market-based filters to isolate robust opportunities and generate actionable volatility relative value trade recommendations.

Signal construction: The model evaluates all possible volatility spreads across a broad universe of currency pairs and tenors. Candidate trades are screened using three key statistical criteria: 1) mean reversion speed, measured through AR(1) half-life estimates; 2) stationarity of the underlying spread relationship; and 3) magnitude of dislocation, measured by standardized z-scores. Additional filters based on volatility risk premia and term structure dynamics are applied to improve signal quality. The final output is a binary matrix highlighting spreads that satisfy all requirements for implementation (Figure 132).

Figure 132: Number of filters activated for a spread supported by the mean reversion signal (USD/CCY 1Y ATM)

Rows represent short volatility, and columns represent long volatility. Details given in the latest [FXO Quant](#) note

Number of Active Filters	EUR	GBP	JPY	CHF	CAD	NOK	SEK	AUD	NZD	BRL	MXN	TRY	ZAR	PLN	HUF	SGD	KRW	CNH
EUR	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	2.0	0.0
GBP	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	2.0	0.0
JPY	0.0	0.0	0.0	3.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	4.0	0.0
CHF	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
CAD	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	3.0	0.0
NOK	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	3.0	0.0
SEK	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	3.0	0.0
AUD	1.0	2.0	0.0	1.0	2.0	0.0	0.0	0.0	1.0	0.0	0.0	0.0	0.0	0.0	2.0	0.0	1.0	0.0
NZD	0.0	3.0	0.0	1.0	2.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	3.0	0.0
BRL	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
MXN	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	1.0	0.0
TRY	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
ZAR	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	2.0	0.0
PLN	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	2.0	0.0
HUF	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
SGD	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	3.0	0.0
KRW	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
CNH	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	3.0	0.0

Source: J.P. Morgan

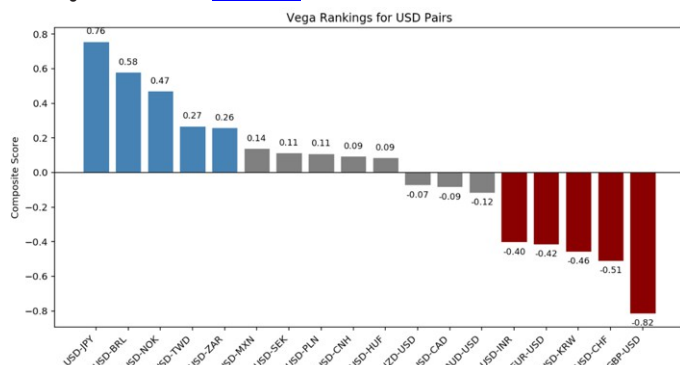
USD Vega RV Model

A machine learning framework (first discussed in our [2026 Year-Ahead Outlook](#)) forecasts relative USD vega opportunities across FX markets and systematically allocate volatility exposure. By combining cross-asset market signals with historical volatility per currency pair, the model predicts future vega performance and identifies the most attractive opportunities for long-volatility positioning.

Forecasting framework: The model uses Gradient Boosting Regressors trained on historical USD vega P/Ls, with a broad set of macro, rates, equities, and FX variables serving as inputs. Each day, forecasts are generated across the currency universe and ranked according to expected performance. Portfolio construction incorporates regime awareness, allowing the model to adapt between risk-on and risk-off environments, while a dynamic blending mechanism combines machine learning forecasts with a simpler baseline model when recent predictive performance deteriorates. To reduce turnover and improve stability, trade recommendations are based on an average of forecasts over the preceding eight trading days.

Figure 133: Vega model rankings

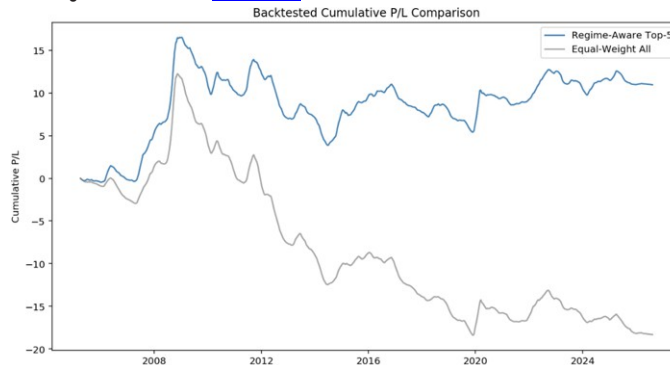
Details given in the latest [FXO Quant](#) note



Source: J.P. Morgan

Figure 134: Vega model backtest

Details given in the latest [FXO Quant](#) note



Source: J.P. Morgan

Volatility Surface RV Trading

We have developed a [framework](#) that allows us to forecast the evolution of the FX implied vol surface and quantify dislocations across its key dimensions. By modeling vol levels, skew, and term structure in a structured manner, we generate forecasts for the entire surface, identify rich and cheap segments of the market, and assess whether those signals can be systematically monetized through volatility trading strategies.

Vol surface specification: To model the implied vol surface, we fit a parametric specification to each currency pair using 35 surface points (7 expiries and 5 strikes). The model captures the key dimensions of the surface, including: 1) overall vol level, 2) call and put skew, 3) short- and medium-dated term structure, and 4) smile attenuation across maturities. The term structure decay parameter λ is estimated separately and then fixed, allowing the remaining coefficients to describe daily movements of the entire surface. The specification achieves an average adjusted R^2 above 89%, indicating a strong fit to observed FX vol surfaces.

$$\sigma_{i,t} = \beta_{0,t} + \beta_{1,t} 1_{\{\Delta_i > 0\}} \Delta_i^2 + \beta_{2,t} 1_{\{\Delta_i < 0\}} \Delta_i^2 + \beta_{3,t} \frac{1 - e^{-\lambda \tau_i}}{\lambda \tau_i} + \beta_{4,t} \left(\frac{1 - e^{-\lambda \tau_i}}{\lambda \tau_i} - e^{-\lambda \tau_i} \right) + \beta_{5,t} 1_{\{\Delta_i > 0\}} \Delta_i \tau_i + \beta_{6,t} 1_{\{\Delta_i < 0\}} \Delta_i \tau_i + \varepsilon_{i,t}$$

$\sigma_{i,t}$ denotes the implied volatility for pair i at time t

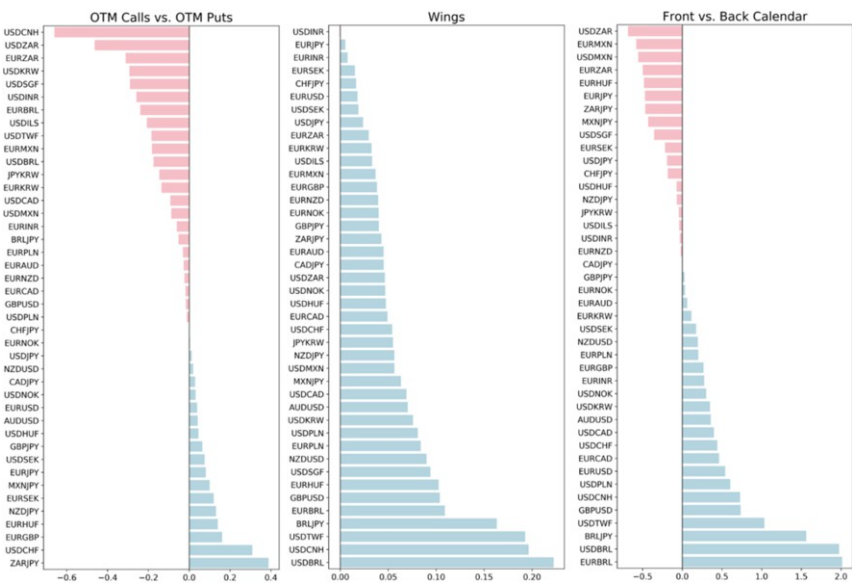
τ_i is the time to maturity measured in years

λ is taken as the median value by minimizing the daily sum of squared errors on the nonlinear estimation of all parameters including λ , then we fix λ and re-estimate the model for the β coefficients

Source: J.P. Morgan

The forecasting: The estimated surface coefficients exhibit significant persistence and cross-correlation, making them suitable for time series modeling. We use a VAR framework to forecast the evolution of the coefficients and reconstruct the entire implied vol surface out of sample. Trades are then selected based on the difference between forecasted vol and current market vol across the surface. EUR crosses and liquid USD/EM pairs exhibit the strongest predictability. JPY crosses and high-beta EM currencies (e.g., ZAR and TRY) turn out to be less predictable due to more idiosyncratic and episodic vol dynamics. USD/G10 pairs consistently generate the strongest results, with most individual strategies achieving Sharpe ratios above 1 before transaction costs. An equal-weighted portfolio aggregating all currency-specific trades (weighted based on dislocations, such as the ones illustrated in Figure 135) generates about 8.2 vols of annualized PnL.

Figure 135: Parametric vol surface model dislocations



Source: J.P. Morgan

Empirical fair value models and various rich/cheap measures can be employed to trade rates vol, both for outright swaption straddles and relative value trades such as expiry and tail switches.

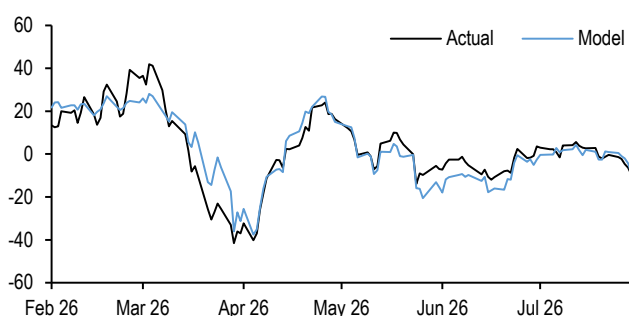
3.2.3 Relative Value in Rates Volatility and Model Framework

Many systematic rates vol strategies rely on active management of the positions, buying or selling vol opportunistically to reduce drawdowns and improve risk/reward. We frequently recommend discretionary vol trades in our research notes. To do so, we have developed **empirical fair value models for both implieds in various swaption structures and gamma returns**. Briefly, our fair value model for implieds regresses swaption implied volatilities on key economic drivers to determine where implieds should trade based on their relationship with those drivers. Our current model uses the underlying swap rate, inflation expectations, Fed hiking expectations, the size of the Fed balance sheet, and market depth as factors to obtain fair values for implieds.

Gamma returns depend on how future delivered vol compares with implied vol. With this in mind, our empirical model for gamma returns uses the current value of the ATMF implied vol for the structure under consideration plus one or more factors that can be used to forecast delivered vol. We tend to refresh this model every few months, as in our experience, the **primary drivers of volatility tend to change over time**. For example, prior to the Middle East conflict, our empirical gamma returns model included monthly S&P 500 returns, ex-ante 2Y inflation swap rate, and monthly change in market depth in addition to ATMF implied vol ([link](#)). However, since the start of the Middle East conflict, oil has become a key driver across asset classes ([link](#)). Consequently, we updated our gamma returns model to include only monthly changes in the Brent crude futures price and the 2Y inflation swap rate in addition to ATMF implied vol ([link](#)). As the conflict has continued, correlations with the price of oil have weakened somewhat, and we have found that Fed expectations are now playing a dominant role in driving volatility. **Our most recent model captures gamma returns well using only two drivers**, ATMF implieds and the 3Mx3M/1Yx1Y OIS curve, which serves as a proxy for medium-term Fed expectations (Figure 136, Figure 137) ([link](#)).

Figure 136: In recent months, gamma returns have been well described using a two-factor empirical model...

Rolling 21-day returns (bp/notional) from selling 6Mx2Y swaption straddles, actual (black) and modeled* (blue)



* Model line corresponds to the rolling 21-day returns forecast using our current model for gamma returns (see the next figure)
Source: J.P. Morgan

Figure 137: ...that uses only ATMF implieds and medium-term Fed expectations

Statistics from rolling 21-day returns* (bp/notional) from long straddles regressed** against ex-ante levels of ATMF implied volatility (bp/day) and monthly change in 3Mx3M/1Yx1Y OIS curve (bp)

	Coefficients				T-stats			
	6Mx2Y	6Mx5Y	6Mx10Y	6Mx30Y	6Mx2Y	6Mx5Y	6Mx10Y	6Mx30Y
Intercept	102.9	255.9	455.7	772.3	21.1	25.3	25.0	18.6
ex-ante ATMF vol	-18.2	-49.1	-97.5	-183.4	-21.9	-26.3	-26.0	-19.7
monthly change in 3Mx3M/1Yx1Y OIS	25.1	44.8	50.0	94.2	7.1	8.7	7.5	7.2
R-squared	85%	89%	88%	79%				
Standard error	6.8	10.1	13.2	26.5				

* Options are assumed to be delta-hedged daily and rolled each month. Assumes zero transaction costs

** Regression from Feb - Aug 2026
Source: J.P. Morgan

With over 40 ATMF tradable points on the swaptions surface alone (and skew adding a 3rd dimension), the opportunity set for relative value in rate vol is quite large. Common expressions include **tail switches** (two swaptions with the same expiry but different tails) and **expiry switches** (same tail but different expiries); more generally, any combination of two (or more) points can be traded. Skew and vol-of-vol can be traded via structures with different strikes but the same expiry and same tail, such as risk reversals or strangles vs. straddles. Cross-market vol expressions also exist; examples include swaption vol vs. exchange, cap/floor, or YCSO vol. Market participants use various rich/cheap measures, including vol ratios (Figure 138), regression analysis, PCA, fair value models, and other metrics, to find dislocations across the surface.

Figure 138: Implied-realized vol ratios across the swaption volatility surface

Implied volatility divided by 1M realized volatility* for various SOFR swaptions, 8/31

	1Y	2Y	5Y	10Y	30Y
1M	1.38	1.34	1.11	1.01	0.99
3M	1.37	1.34	1.14	1.06	1.03
6M	1.31	1.30	1.15	1.09	1.06
1Y	1.28	1.26	1.15	1.12	1.11
2Y	1.26	1.20	1.15	1.14	1.14
3Y	1.17	1.13	1.15	1.14	1.14
5Y	1.21	1.20	1.19	1.16	1.14
10Y	1.17	1.16	1.15	1.16	1.13

* 1M realized volatility is calculated as the 21-day standard deviation of daily changes in the underlying swap rate
Source: J.P. Morgan

Figure 139: PCA z-scores can help assess which parts of the volatility surface may be trading rich or cheap

3M z-score of 2-factor residuals* across the swaption implied volatility surface, 8/31

	1Y	2Y	5Y	10Y	30Y
1M	-0.21	-0.97	-1.61	-1.30	1.23
3M	-0.18	-0.82	-1.42	-1.31	1.89
6M	-0.48	-0.78	-1.29	-1.56	0.48
1Y	-0.93	-1.12	-1.50	-2.41	-0.19
2Y	-1.06	-1.16	-1.54	0.66	1.01
3Y	-1.29	-1.45	-1.28	0.95	1.07
5Y	-0.17	0.43	0.95	1.20	1.22
10Y	1.14	1.09	1.12	1.20	1.11

* To obtain the 2-factor residuals, on each day, we reconstruct the vol surface as a linear combination of the first two factor scores and eigenvectors (obtained via a 6M PCA) and then subtract each reconstructed vol from the actual vol
Source: J.P. Morgan

For example, one rich/cheap measure comes from **PCA residual z-scores**. One can perform a PCA on the vol surface and reconstruct the surface using the first few principal components (which explain most of the variance in the dataset). One can subtract the PCA-reconstructed surface from the actual vol surface on each day to obtain a time series of residuals. Those residuals can then be z-scored, and extreme values may identify good candidate buying/selling opportunities (Figure 139). One trade we recommended using this method was a short 5Yx2Y vs. long 5Yx5Y tail switch. The PCA z-score identified 5Yx5Y as cheap, and we preferred to buy it as a tail switch for additional relative value ([link](#)).

One can also use various metrics to assess where on the surface skew may be trading rich or cheap. A simple measure is to look at the percentile or z-score of, say, 25-delta receiver and payer skews across the surface (Figure 140). Another measure is to calculate the vol-adjusted 1x2 strike-gap breakeven z-score: in other words, one can calculate the strike gap required to make a 1x2 payer or receiver ratio spread costless, divide by the implied volatility, and compute the z-score (Figure 141). This method can also be applied to other distributions, such as ladders. Implied skew can also be compared to realized skew, analogous to implied/realized vol ratios. One can obtain the implied skew and vol-of-vol parameters by using a SABR model to fit the volatility skew and then compare to their realizations. While harvesting skew and vol-of-vol premiums is less straightforward compared to VRP, it can provide a source of uncorrelated return for portfolios ([link](#)).

Figure 140: 25-delta payer skew is elevated in intermediate expiries and longer tails

6M z-score of 25-delta payer swaption implied vol minus ATM swaption implied vol for various swaptions, 8/31

	1Y	2Y	5Y	10Y	30Y
1M	0.50	0.37	0.23	0.46	0.74
3M	0.71	0.63	0.68	0.47	0.37
6M	0.44	0.41	0.37	0.52	0.80
1Y	0.53	0.53	1.10	0.90	1.08
2Y	0.34	0.60	2.01	2.22	1.54
3Y	0.60	0.77	2.17	3.04	2.71
5Y	-0.63	-0.60	0.71	1.36	-1.67
10Y	-4.58	-4.33	-2.70	-2.97	-1.86

Source: J.P. Morgan

Figure 141: Payer ratio breakeven strike gaps flag payer skew in the bottom right of the surface as rich

6M z-scores of strike gap required to make 1x2 ratio spread costless adjusted by vol* for various swaptions, 8/27

	1Y	2Y	5Y	10Y	30Y
1M	0.90	0.91	0.42	0.25	-0.22
3M	0.75	0.75	0.42	0.20	-0.30
6M	0.55	0.63	0.45	0.48	-0.09
1Y	0.59	0.55	1.25	1.09	0.97
2Y	0.58	0.96	2.53	2.57	1.93
3Y	0.79	1.00	2.78	4.06	3.72
5Y	2.02	1.51	2.06	3.39	3.39
10Y	1.08	1.53	1.51	2.97	1.91

* On any given day, we solve for the approximate strike gap required to make a 1x2 payer ratio spread costless and then divide by the ATM swaption implied vol. We then z-score this time series as a rich/cheap measure for payer skew
Source: J.P. Morgan

One can use empirical models such as those discussed above to identify attractive rates vol RV opportunities. For example, in October of last year, our empirical gamma returns model indicated that the left side of the surface was likely to underperform the right side. Consequently, we recommended a tail switch, selling 6Mx2Y swaption straddles versus buying a gamma-neutral amount of 6Mx30Y swaption straddles, as a lower-beta proxy for short-gamma exposure ([link](#)). In December of last year, implieds appeared cheap relative to our fair value model, and the differential between the 6Mx10Y and 1Yx10Y points was near 2-year lows. As a result, we recommended an expiry switch, long 6Mx10Y swaption straddles versus selling a vega-neutral amount of 1Yx10Y straddles, where the vega weighting reduces sensitivity to parallel shifts in the vol surface and helps isolate RV ([link](#)).

Flow bias in GLD and GDX options contributes to improved entry points for relative value trades in the pair, based on realized volatility spreads.

3.2.4 Relative Value in Commodities — Gold and Gold Stocks

Volatility relative value opportunities in commodities are more constrained given the limited number of underlyings with liquid options markets. A popular play has been WTI vs. Brent, but the spread can be highly sensitive to geopolitical shocks.

We want to flag another angle using liquid markets in ETF options, where the **spread between GDX and GLD** is becoming interesting and offers good carry. GLD exhibits much **lower vol in terms of absolute value**, which makes the underlying the “go to” one for directional plays if momentum starts to build. At current vol levels, **call spreads** would likely be the preferred structure, as we have suggested. As a result, **higher strikes in GLD will likely result in more supply to dealers and reduce the GLD realized volatility if spot does get close**. The GDX flow suggests more **underwriters** and hence more potential volatility delivery for upside calls.

This is an ideal set up for the long GDX, short GLD vol spread, as GDX has a flatter skew; the best entry level is for near-term upside strikes (Figure 143).

Figure 142: Implied volatility surface of GLD

Strike % of forward

	1M	2M	3M	6M	1Y
120	29.9%	27.2%	26.1%	24.5%	23.7%
110	25.0%	24.3%	23.8%	23.2%	22.9%
100	22.3%	22.6%	22.5%	22.5%	22.5%
90	24.2%	23.3%	23.2%	22.8%	22.8%
80	32.8%	29.1%	26.9%	24.7%	23.9%

As of Aug 5th

Source: J.P. Morgan, Bloomberg Finance L.P.

Figure 143: GDX-GLD implied volatility spread

Strike % of forward

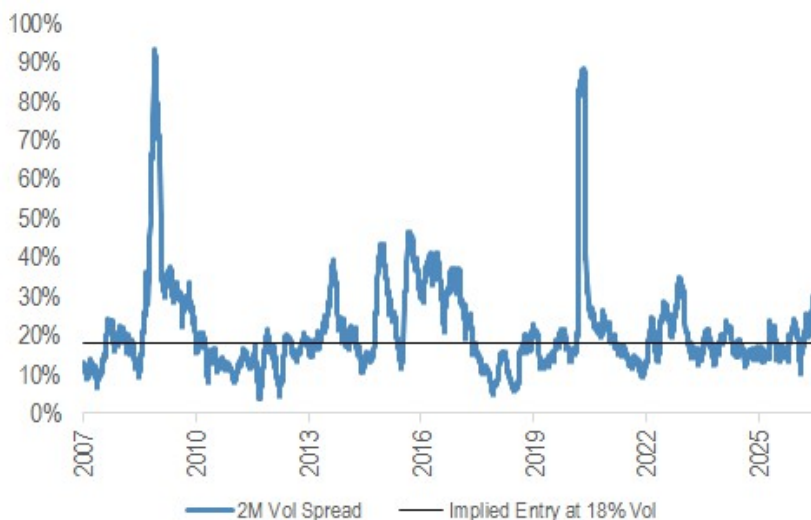
	1M	2M	3M	6M	1Y
120	13.9%	17.1%	17.8%	19.3%	19.8%
110	17.2%	18.8%	19.3%	20.0%	20.1%
100	18.9%	19.7%	20.1%	20.3%	20.1%
90	17.6%	19.3%	19.7%	20.0%	19.6%
80	12.1%	15.4%	17.2%	18.4%	18.4%

As of Aug 5th

Source: J.P. Morgan, Bloomberg Finance L.P.

We favor using **regions of 1-2M upside calls for 110% -115% strikes on GDX vs. GLD** to capture the potential realized volatility spread with delta hedged options.

Figure 144: 2M realized volatility spread of GDX- GLD vs. implied entry at 18% vol



As of Aug 5th

Source: J.P. Morgan, Bloomberg Finance L.P.

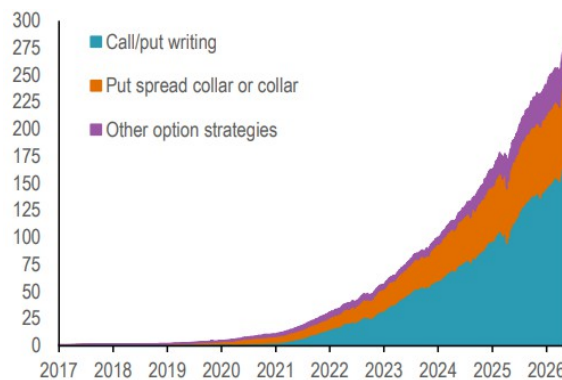
3.3 Beta with Options

3.3.1 Beta in Equities — Overwriting Improvement Strategy

While call overwriting generates attractive yield income, the total performance of buy-write strategies **can outperform instead of underperform benchmark equities if signals are used to enhance returns.**

The JPM ETF Handbook ([link](#)) highlights the rapid growth of income strategies in equities since 2022, with a significant portion from **call overwriting** (Figure 145). While the buy-write strategy can deliver superior returns compared to fixed income, it can underperform the equity benchmark when **spot keeps overshooting the short strikes.**

Figure 145: US option-based ETF assets, by strategy (\$bn)



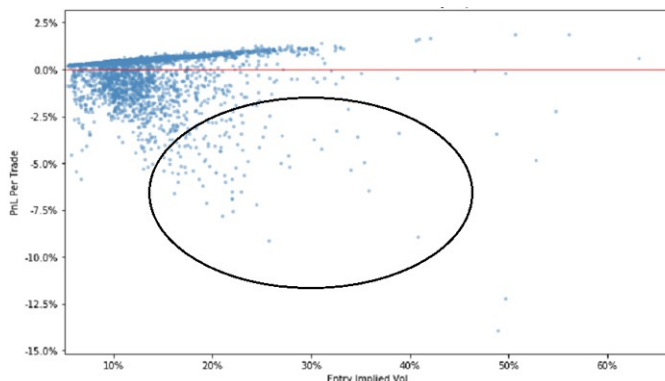
Source: J.P. Morgan, Bloomberg Finance L.P.

We conducted a [deep dive on the buy-write strategy \(link\)](#), aiming to improve the overwriting leg, and identified two periods where short calls underperform:

1. When markets are **extremely oversold** and a short squeeze drives aggressive buying.
2. When **upside momentum** is extremely strong and consistent, leading to a melt-up.

Figure 146: Short outright SPX1M 25D calls

PnL per trade vs. entry strike vol



Source: J.P. Morgan, Bloomberg Finance L.P.

Figure 147: Short outright SPX 1M 25D calls: comparison across stop-loss signals

PnL per trade vs. entry strike vol

	All Trades	3M-1M TS	VIX	Strike Vol
Cut off at	None	<0	>25	>18.25
Hit Ratio	68.21%	71.76%	68.69%	67.34%
Min	-13.07%	-8.36%	-6.02%	-6.10%
1st q	-0.42%	-0.16%	-0.36%	-0.40%
Median	0.389%	0.336%	0.350%	0.349%
Mean	-0.110%	-0.058%	-0.103%	-0.127%
3rd q	0.60%	0.51%	0.51%	0.51%
Max	3.07%	1.23%	1.03%	0.99%
%Trades cut at start		14%	13%	13%

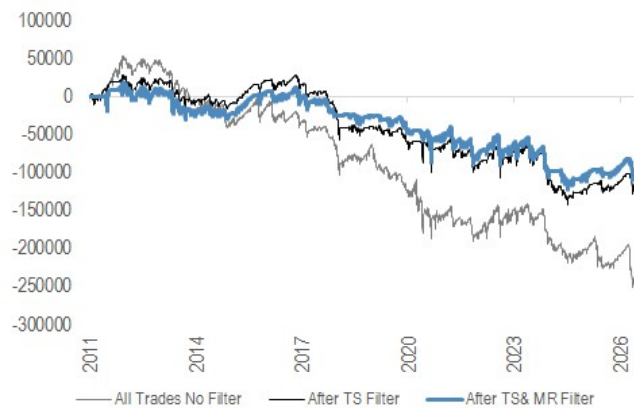
Source: J.P. Morgan

For case 1, short call drawdowns are associated with **high vol periods** in sell-offs (Figure 146). We tested various signals to identify what is “too oversold” using **strike vol, VIX cut off, or term structure signal** (Figure 147) and concluded that the TS

signal is the most robust and cleanest measure.

For case 2, **end-of-year seasonality and earnings periods** could see **equities overshoot on the upside**, but pulse trading during all of these periods is not realistic. Therefore, we tested using an **MR score**, which acts similarly to a short-term CTA signal, to identify periods in which **short-dated momentum is likely to continue**.

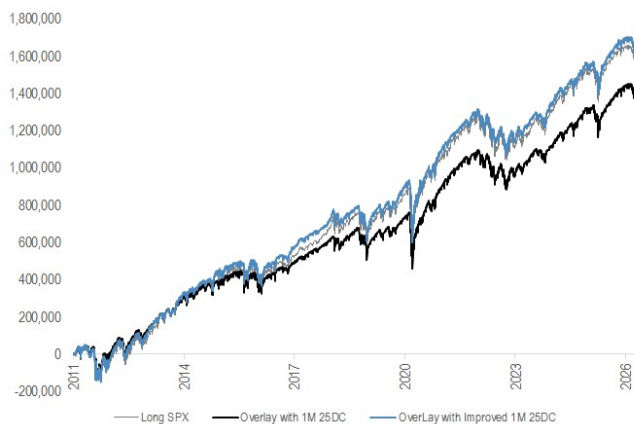
Figure 148: SPX short outright 1M 25D calls — improvement with TS & MR filters



Source: J.P. Morgan, Bloomberg Finance L.P.

We explored other points to consider, such as early unwind and treatment when implied volatility is too low to sell. The final buy-write strategy was able to achieve a **higher Sharpe ratio and better total return** compared to owning SPX outright.

Figure 149: SPX improved overlay vs. daily 25D and long-only update



Source: J.P. Morgan, Bloomberg Finance L.P.

Figure 150: SPX improved overlay versus daily 25D and long only

Sharpe is based on the full time series; return is average, not compounded

	Long SPX Only	Overlay with 1M 25DC	OverLay with Improved 1M 25DC
Sharpe	0.74	0.74	0.77
VaR95	-1.43%	-1.25%	-1.42%
Ann Rtn	11.21%	9.68%	11.43%

* Assumes no transaction costs

Source: J.P. Morgan, Bloomberg Finance L.P.

Replacing equity exposure with long call spreads can **significantly reduce drawdowns**. Enhancements to the strategy can outperform benchmark returns, delivering **higher Sharpe ratios, lower drawdowns, and improved overall performance**.

3.3.2 Beta in Equities — Long Equities with Options

Buy-write strategies provide extra income compared to owning equities but are still **exposed to drawdowns in market corrections**. A good alternative to **own equities without large drawdowns** is buying call spreads instead of long delta directly. In [What to Long, Short and When \(Part III\)](#), we conducted a deep dive into the preferred structure and techniques to improve it.

We ran a backtest on listed SPX options across all combinations of strikes and tenors, held to maturity. The best choice to own outright call options is 1M ATM, as longer tenors are penalized by higher cost and a greater chance of encountering the left tail.

Figure 151: Sharpe ratios for long calls

2017-2026

	50d	25d	10d	5d
10d	0.75	0.61	0.23	-0.14
1m	0.92	0.82	0.53	0.28
3m	0.74	0.54	0.25	0.01
6m	0.77	0.65	0.28	0.05
1y	0.73	0.67	0.48	0.22

Source: J.P. Morgan

Figure 152: Hit ratios for long calls

2017-2026

	50d	25d	10d	5d
10d	46%	29%	13%	6%
1m	53%	37%	19%	9%
3m	58%	37%	18%	8%
6m	64%	48%	24%	11%
1y	72%	58%	41%	18%

Source: J.P. Morgan

While 1M ATM calls are expensive to own, we use 10-delta calls to reduce the cost without giving up excess gains and retaining a defined loss profile. Our **baseline is 1M 50d/10d call spreads with 2x leverage**, entering every day with \$1M/22 notional and holding to maturity, to better match futures PnL. Compared with futures, the call spread replacement improves Sharpe to **0.97 vs. 0.73** and the **max drawdown to -13.6% vs. -26.1%**. The defined downside significantly improved performance during **COVID** and **“Liberation Day”**.

Figure 153: SPX baseline options strategy vs. futures

\$1M notional sized 1/22 per day; assumes zero transaction costs



Source: J.P. Morgan, Bloomberg Finance L.P.

Figure 154: Baseline strategy performance metrics

	Sharpe	Max DD	Return (p.a compounded)
Long 1M Futures	0.73	-26.1%	8.7%
Long 1M 50d/10d CS (2x)	0.97	-13.6%	9.3%

Source: J.P. Morgan, Bloomberg Finance L.P.

Trend Signal and Gamma Overlay

The first improvement is to find periods when the **market is likely to consolidate and reduce the exposure**. As discussed in [SPX Carry Strategy](#), with the trend signal from the [JPM Cross Asset Trend methodology](#), we note that in the extremely positive region (>0.75), the average PnL turns negative, and the posterior realized vol becomes higher than implied at entry, especially for the 3M tenor. Practically, we should **avoid entering the trade when the trend signal is beyond 0.75** and use a **delta-hedged 3M 10-delta call to capture the post-trend volatility**. We applied 25% notional on the gamma leg; this improved the return to 10.1% while keeping the Sharpe ratio and max drawdown superior to those of the base strategy.

Figure 155: Performance metrics with gamma overlay improvement

	Sharpe	Max DD	Return (p.a compounded)
Long 1M 50d/10d CS (2x)	0.97	-13.6%	9.3%
With Trend Filter + 3M Gamma Overlay	1.13	-12.7%	10.1%

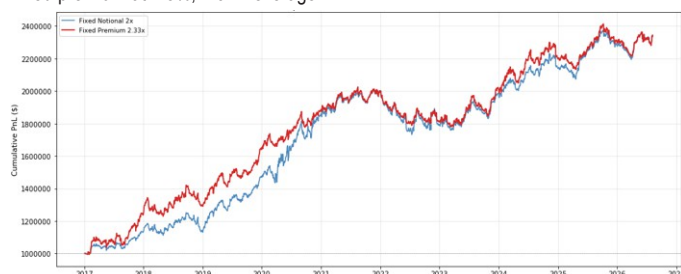
Source: J.P. Morgan, Bloomberg Finance L.P.

Fixed-Premium Approach

The 2022 drawdown reflects fixed notional, which can result in **overpaying when vol is high and call spreads are expensive**. A **fixed-premium** approach caps per-trade loss and helps in slow, choppy markets. We cap the call spread premium at 1.5% (with leverage to keep the terminal PnL comparable), lifting Sharpe to 1.01 and reducing maxDD to -12%.

Figure 156: Fixed notional vs. fixed premium

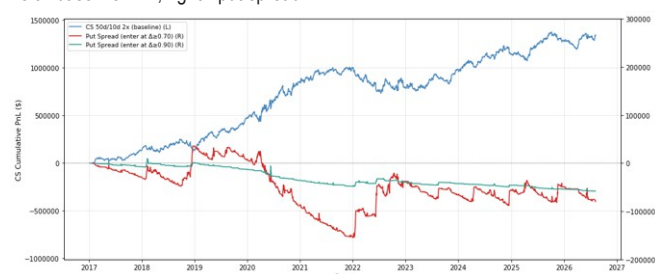
Fixed premium at 1.5%, 2.37x leverage



Source: J.P. Morgan, Bloomberg Finance L.P.

Figure 157: Call spread vs. put spread for profit-taking

Left - baseline PnL, right - put spread PnL



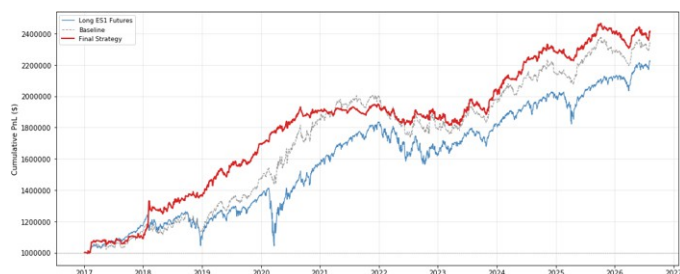
Source: J.P. Morgan, Bloomberg Finance L.P.

Profit-Taking With Put Spread Overlay

For profit-taking, we prefer adding a put spread rather than unwinding early, since early unwinds often forfeit more upside than they protect. When the **50d strike rolls to 70d**, a put spread **preserves upside with a cushion**, reducing maxDD with limited bleed and improving Sharpe.

Adding all the improvements, our final strategy is to long a 1M 50d/10d call spread with a 1.5% premium cap; when the trend signal is extremely positive, skip entry and overlay a delta-hedged 3M 10-delta call; when the 50d leg becomes 70d, long a 1M 50d/25d put spread for profit-taking. The strategy delivered a return comparable to holding futures, but with a significantly **higher Sharpe of 1.3 and a max drawdown of only -7%** since 2017.

Figure 158: Final strategy backtest PnL



*Assumes no transaction costs
Source: J.P. Morgan, Bloomberg Finance L.P.

Figure 159: Performance for options strategies vs. futures

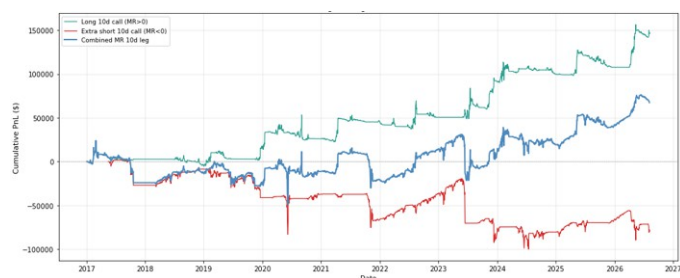
	Sharpe	Max DD	Return (p.a compounded)
Long 1M Futures	0.73	-26.1%	8.7%
Long CS Baseline	0.97	-13.6%	9.3%
Long CS Improved	1.30	-7.0%	9.6%

*Assumes no transaction costs
Source: J.P. Morgan, Bloomberg Finance L.P.

Return Improvement

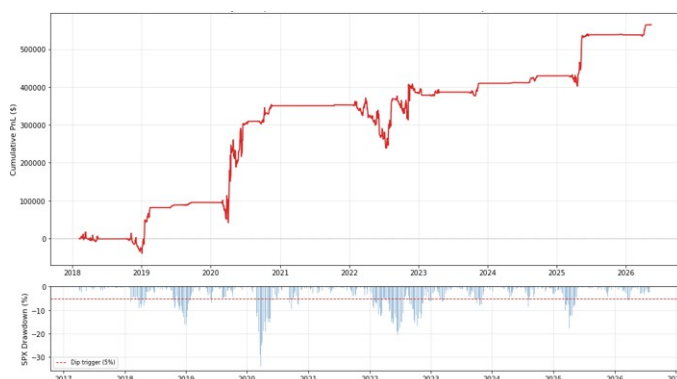
To further improve the returns, we proposed two extra enhancements: 1) using the [JPM Mean Reversion Score](#) to identify market dynamics, we remove the 10-delta cap when the market is strongly trending; for a rangebound market, we replace the call spread with a call ratio to source extra funding from the 10-delta leg. 2) We apply a buy-the-dip strategy by entering a 2w 25d/10d call spread and sizing it according to the magnitude of the drawdown (capped at 3x for a 15% drawdown).

Figure 160: Dynamically sizing the 10d leg according to MR score



Source: J.P. Morgan, Bloomberg Finance L.P.

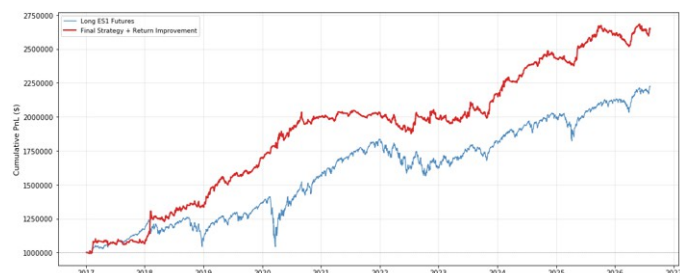
Figure 161: PnL for the buy-the-dip overlay



Source: J.P. Morgan, Bloomberg Finance L.P.

Overall, the enhanced strategy has delivered a **10.7% compounded annual return** and **1.34 Sharpe** since 2017, with materially **lower maxDD** versus delta-1 futures.

Figure 162: Call replacement strategy with improvement overlay compared to long SPX futures



*Assumes no transaction costs
Source: J.P. Morgan, Bloomberg Finance L.P.

Figure 163: Performance of return improvement strategy vs. futures

	Sharpe	Max DD	Return (p.a compounded)
Long 1M Futures	0.73	-26.1%	8.7%
Final Strategy + Return Improvement	1.34	-8.4%	10.7%

*Assumes no transaction costs
Source: J.P. Morgan, Bloomberg Finance L.P.

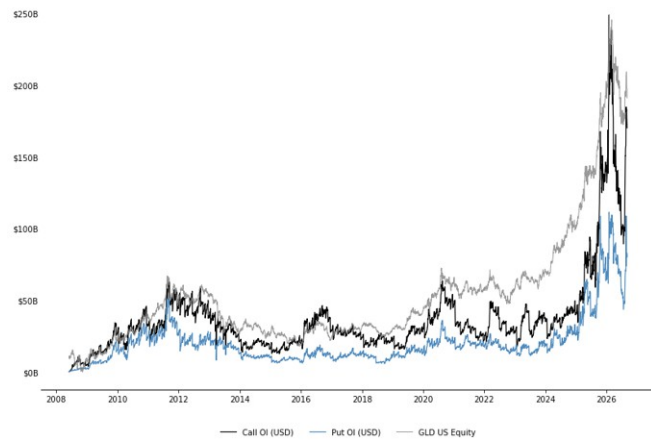
3.3.3 Beta in Commodities — Long Gold with Options

Systematic **call spread buying in gold** has been highly successful given the inverted skew and may continue to deliver strong results, supported by our long-term bullish outlook.

GLD option players had a huge impact on the gold rally since 2025, as we identified nearly **\$2bn short gamma** from October alone from clients lifting cheap outright calls, and investors switched the play to using call spreads instead after the volatility spike, pushing the momentum into Jan before Fed headwinds turned the corner ([link](#), [link](#)).

Figure 164: Open interest for GLD call and put options in USD

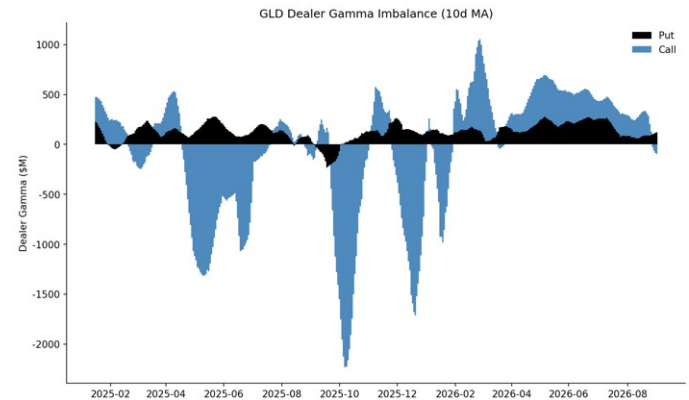
Open interest in calls and puts (left) and GLD level (right), past ~18 years, As of 9/1



Source: J.P. Morgan

Figure 165: GLD gamma imbalance for puts and calls

Dealer's perspective; negative means dealer short gamma. As of 9/1



Source: J.P. Morgan, Bloomberg Finance L.P., OPRA

Our commodity strategists continue to hold a long-term bullish view on gold. As **gold exhibits inverted skew**, systematically buying call spreads could act as a **cost-effective way to play upside on gold while minimizing drawdown risk**. We tested using a **1M 25%-10% delta call spread** on gold compared to **buying gold and holding for one month**, assuming we trade the same cash notional every day and hold to expiry. If we use a fixed-premium approach, the **call spread strategy** delivers a much **higher Sharpe ratio with a much smaller max DD**.

Figure 166: P&L for 1M 25-10 delta call spread on GLD vs. long 1M GLD spot

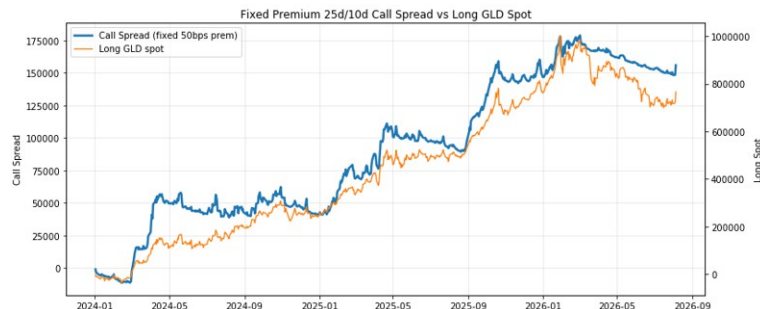
From 2024

	Call Spread	Long GLD Spot
Sharpe	1.6	1.36
Max DD (%)	-2.6%	-15.0%

Source: J.P. Morgan, Bloomberg Finance L.P.

Figure 167: Call spread replacement on GLD vs. long GLD delta-one

\$1M notional, sized 1/22 per day, adjusted by premium



*Assumes no transaction costs

Source: J.P. Morgan, Bloomberg Finance L.P.

Replacing spot FX carry with options can materially reduce drawdowns while preserving exposure to the underlying carry signal.

3.3.4 Beta in FX — FX Carry with Options

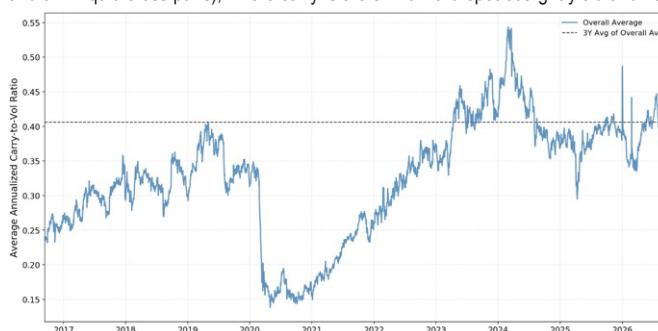
In our [Mid-Year Outlook](#), we highlighted carry as a key theme for H2, supported by resilient carry opportunities and subdued volatility despite concerns of escalation regarding Iran. The relative appeal of harvesting carry via options has only improved. Low starting volatility levels have reduced the attractiveness of pure vol carry, as compressed realized volatility and already-adjusted implieds leave less room for short gamma and vega strategies to monetize risk premia.

In contrast, carry-to-vol ratios remain elevated and have recently rebounded towards levels seen at the start of the year (Figure 168), reinforcing the case for directional carry implementations via options. We therefore continue to favor collecting linear carry through options over harvesting outright vol carry.

This is not to say that vol carry opportunities are absent; rather, they are increasingly concentrated in specific segments of the surface, including structures that monetize skew, as highlighted in our [recent work](#).

Figure 168: Carry-to-vol ratios in FX are near their levels at the start of the year

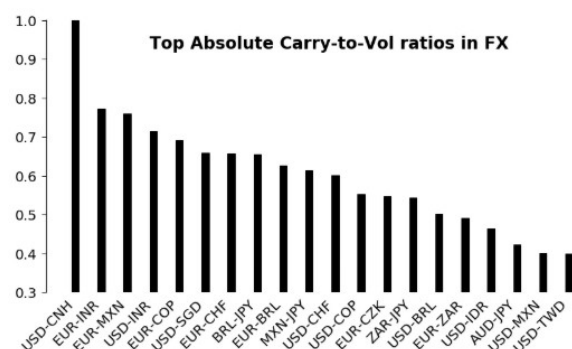
Daily level of the average annualized absolute carry-to-vol ratio across the Global FX cash carry portfolio (USD, EUR, and JPY liquid cross pairs), where carry is the 3M forward-spot outright yield and vol is 3M implied volatility



Source: J.P. Morgan

Figure 169 ranks the top 20 liquid carry-to-vol ratios (ex-TRY owing to liquidity concerns), containing USD pairs and EUR and JPY crosses.

Figure 169: Top carry-to-vol ratios in liquid FXO pairs/crosses



Source: J.P. Morgan

3.3.5 Beta in FX — Q-Score Model For One-Touch Calendar Spreads

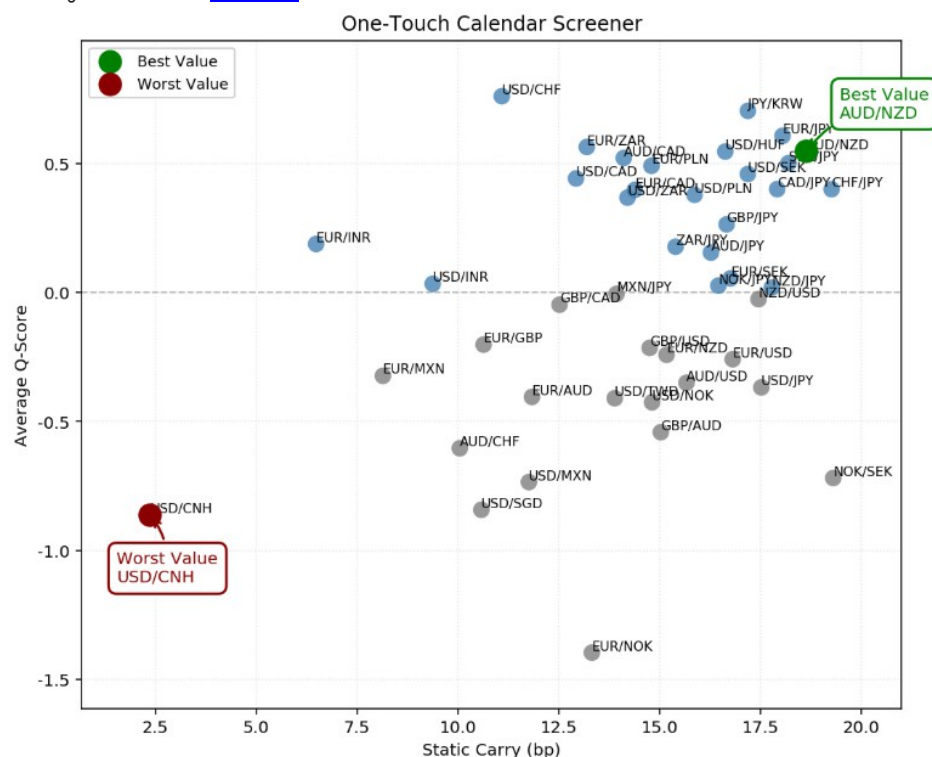
The Q-Score model identifies carry-collecting calendar spreads where carry and momentum conditions are aligned.

One-touch calendar spreads are primarily carry-harvesting structures that benefit from flat or inverted volatility curves, but spot dynamics are often the key determinant of realized outcomes. To account for this, our [screening framework](#) combines static carry with a composite Q-Score that incorporates normalized spot-barrier distance and trend strength, measured by ADX. The intuition is that attractive carry is most likely to be realized when spot remains rangebound and avoids persistent trends toward the barrier, while strong directional momentum can quickly overwhelm otherwise favorable carry characteristics. The model therefore seeks to identify structures where carry and momentum conditions are aligned.

Static carry has continued to move higher across much of the FX universe, with many currency pairs now offering carry levels near the upper end of the screen. However, directional momentum has become increasingly selective, as the majority of pairs cluster around neutral or negative Q-Scores despite elevated carry opportunities (Figure 170). Several LY/JPY opportunities remain attractive on the surface but are subject to intervention uncertainty that can distort the risk-reward profile of calendar structures. Excluding these event-driven themes, higher-ranking opportunities have become concentrated in a smaller set of SEK-, HUF-, CAD-, and AUD-related crosses, which continue to offer relatively strong combinations of carry and spot momentum. The screen continues to favor candidates for which trend persistence remains limited, allowing positive carry to dominate realized returns.

Figure 170: Carry and momentum screener for one-touch calendar spreads

Details given in the latest [FXO Quant](#) note



Source: J.P. Morgan

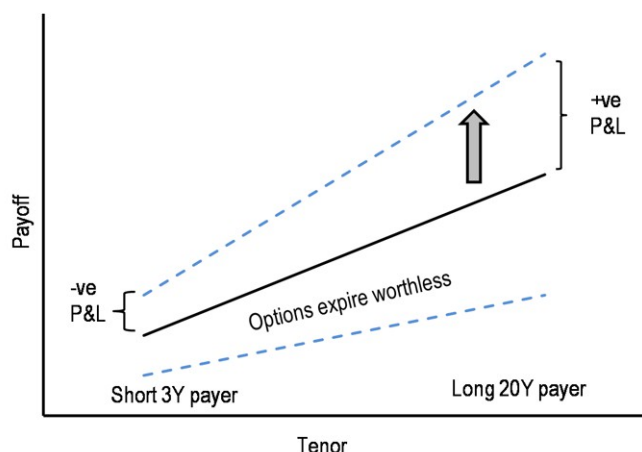
Swaptions can be used to construct conditional curve trades, which offer more flexible expressions of directional views compared with linear instruments.

3.3.6 Beta in Rates with Options

Swaptions can be used to express directional views more flexibly than with linear instruments. A popular type of trade in this space is **conditional curve trades**, which express a view on the yield curve conditional on the directionality between the steepness of the curve and the level of yields. For example, a 3s/20s **bear steepener** would be constructed by selling a 3Y payer and buying a 20Y payer (Figure 171). This trade would benefit if the 3s/20s curve steepened and yields sold off, while the principal risk would be a bear flattening. In either a bull steepening or bull flattening scenario, both swaptions would expire worthless. **It is desirable to construct these trades in a premium-neutral manner** (or at a premium intake), so conditional trades work well when the market is underpricing the directionality between the curve and yields. One can construct an “efficient frontier” of conditional curve trades by plotting the premium paid/received at trade entry vs. the correlation of the yield curve to the level of rates (Figure 172). Compelling candidates would display strong curve/yield directionality while being relatively cheap to enter (i.e., end up near the top right of the graph). Additional measures, such as risk-weighting the curve or striking one of the options away from ATMF, may be taken to reduce the premium outlay at inception ([link](#)). Swap spread trades or cross-market rates trades (e.g., USD vs. EUR yields) can also be done conditionally.

Figure 171: Conditional curve trades are one way that swaptions can be used to express directional views

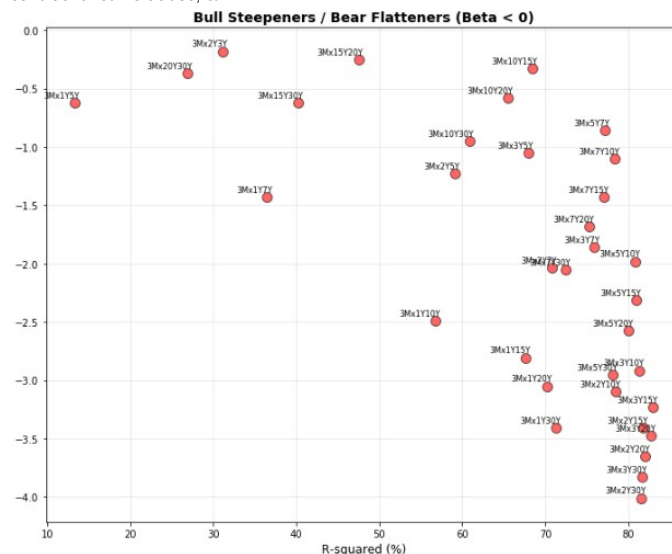
Schematic of a bear steepener conditional curve trade if the curve steepens



Adapted from [A primer on Conditional Trades](#), K. Gupta and F. Bassi, 5/26/2016
Source: J.P. Morgan

Figure 172: Currently, no ATMF bull steepeners/bear flatteners can be constructed in a premium-neutral manner

Premium received at entry vs. R² between curve and yield* for various 3M expiry conditional curve trades, 9/1



* For example, for a 3M 10s/30s bear flattener, it costs 1bp of yield to enter, and the R² of a 1Y regression of the 10s/30s curve onto the 10Y swap yield is around 60%
Source: J.P. Morgan

Other directional trades include outright options trades and distribution trades like 1x2s, ladders, butterflies, etc. These are not unique to rates options but have their applications in rates. For example, 1x2s in shorter tails (1y, 2y) can be used to take a view on the terminal rate during Fed hiking or cutting cycles, aiming to pin the OTM strike. Currently, we like 3Mx30Y 1x2s (initiated ATMF/ATMF+20bp on 7/31; [link](#)) as a positive-carry way to position for a moderate selloff in long-end yields.

3.4 Event Trading

3.4.1 Macro Event Trading with SPX Short-Dated Options

SPX short-dated options can be used to harvest event risk and protect against systematic strategies with a carry bias.

We conducted a deep dive into the PnL from owning SPX 2DTE options during macro events since 2017 and found increasing price reactions associated with FOMC meetings since 2022 ([link](#)).

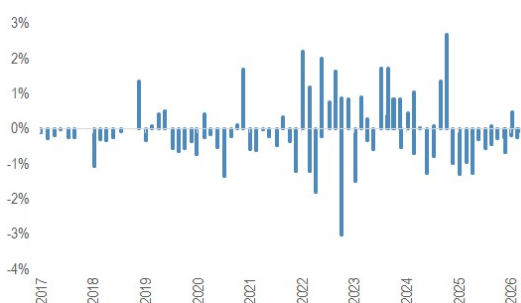
Figure 173: Buying 2DTE ATM straddles — avg premium and PnL for non-event (regular) and event days

2022-2026

	Premium	Payout	PnL	Hit Ratio
Regular	1.09%	1.1%	-0.01%	43%
FOMC	1.40%	1.4%	0.04%	47%
CPI	1.32%	1.2%	-0.08%	39%
NFP	1.22%	1.3%	0.11%	39%
ISMService	1.11%	1.2%	0.13%	46%
ISMManufacturing	1.12%	1.1%	-0.03%	41%

Source: J.P. Morgan, OPRA

Figure 174: PnL of long 2DTE ATM straddles on FOMC days



Source: J.P. Morgan, OPRA

There was a shift in focus on the **importance of indicators depending on the regime**. Regarding FOMC dates, in 2022, the Fed's policy regime changed, and SPX consequently became more reactive to FOMC event dates. CPI was also very relevant in 2022, whereas NFP event impacts were much more noticeable during 2018, when surprises due to unexpectedly weak prints caused market participants to question the Fed's hikes. In 2025 we saw more price reactions outside FOMC dates as Fed was more in a watch and see mode until the new chair started to induce volatility.

Figure 175: Buying 2DTE ATM straddles — avg PnL and hit ratio for positive PnL since 2017

	Regular		FOMC		CPI		NFP		ISM Service		ISM Manufacturing	
	PnL	Hit Ratio	PnL	Hit Ratio	PnL	Hit Ratio	PnL	Hit Ratio	PnL	Hit Ratio	PnL	Hit Ratio
2017	-0.04%	36%	-0.22%	0%	0.00%	31%	0.00%	35%	-0.11%	10%	0.05%	44%
2018	0.04%	42%	-0.29%	14%	0.03%	54%	0.40%	64%	0.53%	67%	-0.08%	23%
2019	-0.07%	41%	0.02%	50%	-0.27%	38%	0.28%	50%	0.49%	46%	0.22%	54%
2020	-0.31%	39%	-0.36%	22%	-0.33%	30%	0.38%	32%	0.15%	46%	0.08%	57%
2021	0.13%	50%	-0.04%	25%	-0.34%	9%	-0.30%	30%	-0.13%	38%	-0.06%	36%
2022	0.12%	50%	0.11%	56%	0.27%	44%	-0.09%	36%	0.02%	46%	-0.16%	35%
2023	-0.06%	39%	0.35%	69%	-0.29%	38%	0.14%	50%	0.17%	58%	0.00%	46%
2024	-0.04%	38%	0.35%	69%	-0.10%	33%	0.11%	38%	0.36%	63%	0.14%	54%
2025	-0.01%	46%	-0.55%	6%	-0.06%	50%	0.34%	36%	0.19%	29%	-0.16%	33%
2026	-0.09%	39%	-0.17%	30%	-0.31%	31%	0.10%	36%	-0.18%	25%	0.08%	38%

Source: J.P. Morgan, OPRA

2DTE enables us to compare trades dating back to 2017. With increased liquidity in 0DTE and 1DTE since 2022, we can isolate event-day moves more accurately. We also notice the **higher sensitivity on NFP dates**, a similar observation to the 2018 period.

Figure 176: Buying 1DTE ATM straddles — avg PnL and hit ratio for positive PnL since 2022

	Regular		FOMC		CPI		NFP		ISM Service		ISM Manufacturing	
	PnL	Hit Ratio	PnL	Hit Ratio	PnL	Hit Ratio	PnL	Hit Ratio	PnL	Hit Ratio	PnL	Hit Ratio
2022	0.08%	47%	0.14%	50%	0.54%	54%	-0.14%	25%	0.18%	50%	-0.36%	25%
2023	-0.03%	43%	-0.06%	50%	-0.43%	8%	0.13%	55%	0.04%	50%	-0.26%	17%
2024	-0.04%	37%	0.28%	63%	-0.03%	58%	0.02%	50%	0.19%	58%	0.12%	50%
2025	-0.04%	39%	-0.41%	13%	0.02%	50%	0.40%	64%	0.18%	33%	-0.02%	50%
2026	-0.05%	42%	0.14%	60%	-0.36%	13%	0.23%	71%	-0.13%	38%	-0.15%	25%

Source: J.P. Morgan, OPRA

0DTE delivered mixed results, as “Liberation Day” caused a sizable distortion, while 1DTE results provide clearer conclusions (Figure 176). With 1DTE, short straddles as a **systematic carry on “regular days” proved more profitable**, as holding long positions resulted in a net negative PnL and a low hit ratio. However, short positions should be **avoided on event dates**, when tail risk can induce large drawdowns.

Figure 177: Shorting 1DTE ATM straddles every day vs. avoiding shorting on event days



*Assumes no transaction costs
Source: J.P. Morgan, OPRA

Owning gamma is preferred on event dates, with **straddles typically delivering a better hit ratio than strangles**. For strangles, the spot must move beyond both the strike and the premium before PnL accrues.

Figure 178: Long 1DTE strangles and ATM straddles — average PnL and hit ratio

2022-2026

	25 Delta Strangle			ATM Straddle		
	Premium	PnL	Hit Ratio	Premium	PnL	Hit Ratio
Regular Days	0.29%	-0.01%	32%	0.76%	-0.01%	42%
FOMC	0.41%	0.04%	36%	1.09%	0.02%	46%
CPI	0.39%	0.00%	29%	1.06%	-0.02%	38%
NFP	0.36%	0.05%	35%	0.96%	0.09%	49%
ISM Service	0.30%	0.08%	45%	0.80%	0.11%	46%
ISM Manufacturing	0.30%	-0.09%	27%	0.80%	-0.13%	34%

Source: J.P. Morgan, OPRA

As with all events, **volatility risk premium generally decreases post the announcements**, which is part of the cost of owning event risk gamma. If gamma is expressed using longer-dated options, the PnL from implied volatility resets cannot be ignored. We witness a downside bias in general; hence, calendar positions should be considered if not using shorter-dated options on event dates.

Figure 179: Fixed-strike vol reset on event days and change in VIX level post event date

2022-2026

	1W Tenor		1M Tenor		VIX Level	
	Avg Chg	Hit Ratio	Avg Chg	Hit Ratio	Avg Chg	Hit Ratio
FOMC	-0.64%	70%	-0.03%	54%	-0.04	54%
CPI	-0.75%	69%	-0.18%	65%	-0.37	76%
NFP	-0.38%	75%	0.03%	60%	0.39	65%
ISM Service	0.41%	39%	0.13%	54%	0.22	57%
ISM Manufacturing	-0.11%	43%	0.08%	41%	0.14	57%

Source: J.P. Morgan

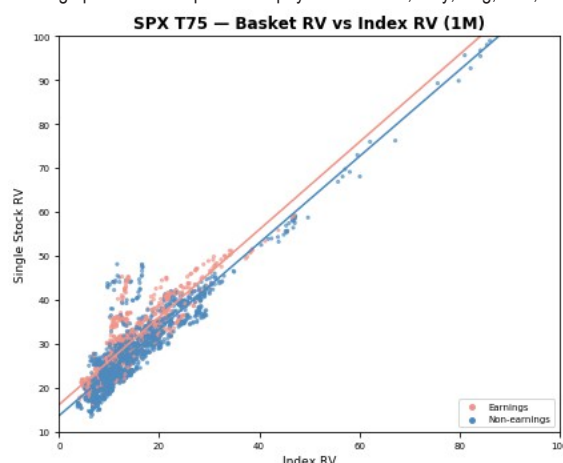
Earnings dispersion offers a compelling opportunity to capture company-specific events. Theta weighting should be considered, and calendar positions on stocks are essential to reduce bleed from volatility resets.

3.4.2 Company Earnings Event Trading

As discussed in 3.2.1 Relative Value in Equities — the Volatility Dispersion Trade, dispersion trades usually **deliver best during earnings periods**, when the spread between single-stock and index realized volatility can overshoot compared to non-earnings periods and versus implied entry ([link](#)).

Figure 180: Single stocks vs. index 1M realized volatility over earnings and non-earnings periods

Earnings periods correspond to expiry dates in Feb, May, Aug, Nov, 2017-2026

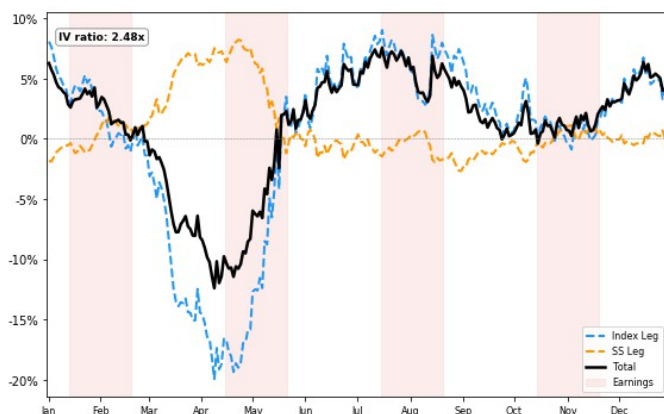


Source: J.P. Morgan

The seasonality is very strong: **Jan and July earnings usually deliver the best returns** as companies mark their progress for the full year and half year with outlooks. Realized correlations trough accordingly due to this phenomenon (Figure 124). The index leg contribution is more significant in **late summer and near the end of the year**.

Figure 181: SPX T75 theta-neutral dispersion vol spread contribution

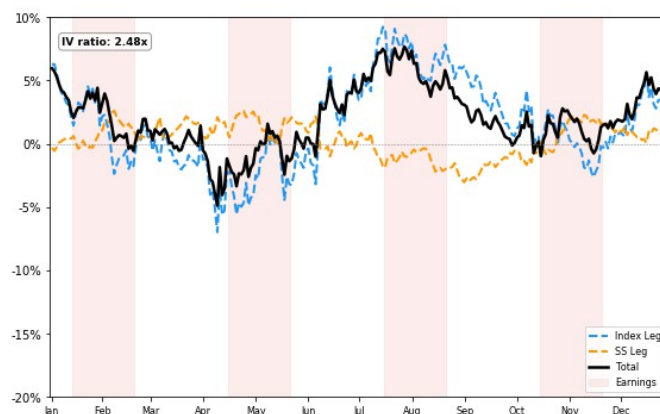
2M, 2017-2026



Source: J.P. Morgan

Figure 182: SPX T75 theta-neutral dispersion vol spread contribution

2M, 2017-2026 excluding 2020 and 2024 outliers shows similar pattern

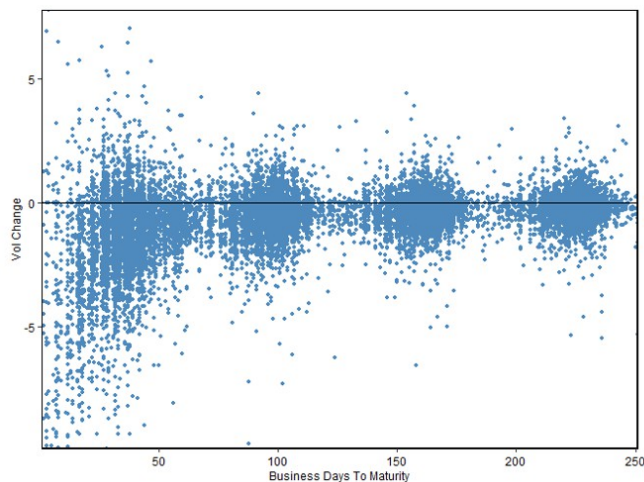


Source: J.P. Morgan

While one can size up the dispersion to be **theta-neutral**, particularly in **August and from the end of the year through Jan**, to maximize the carry from index options during earnings periods ([link](#)), on the single-stock leg, the **implied volatility reset** must be considered to balance the cost of gamma PnL ([link](#)).

Figure 183: Vol reset by option tenor

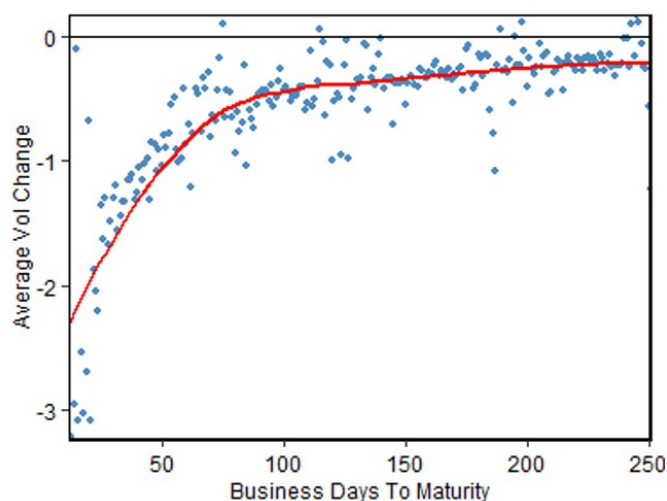
EU optionable stocks, 10Y history, fixed-strike ATM vol change on and post earnings days and post



Source: J.P. Morgan

Figure 184: Average vol reset by option tenor

EU optionable stocks, 10Y history, fixed-strike ATM vol change on and post earnings days



Source: J.P. Morgan

The fixed-strike vol reset is usually the highest **on the earnings date itself** and **across the curve**. While this could be a great opportunity to add new positions, particularly for non-earnings-sensitive names, for names that usually deliver a high earnings move, **calendar positions** are preferred in order to capture the vol reset while remaining long gamma.

Figure 185: Total PnL scenario for vol reset=1.25 in 2M and 0.46 in 5M

2M expiry, Vol reset ref =1.25%					
Vol \ Spot move	0.00%	2.00%	4.00%	6.00%	8.00%
22.00%	-0.48%	-0.30%	0.26%	1.18%	2.43%
24.00%	-0.49%	-0.32%	0.19%	1.04%	2.19%
26.00%	-0.50%	-0.34%	0.13%	0.91%	1.99%
28.00%	-0.51%	-0.36%	0.08%	0.80%	1.81%
30.00%	-0.51%	-0.38%	0.03%	0.71%	1.65%
32.00%	-0.52%	-0.40%	-0.02%	0.62%	1.51%
34.00%	-0.53%	-0.41%	-0.05%	0.55%	1.39%
36.00%	-0.54%	-0.43%	-0.09%	0.48%	1.27%
38.00%	-0.54%	-0.44%	-0.12%	0.42%	1.17%
5M expiry, Vol reset ref =0.46%					
22.00%	-0.28%	-0.17%	0.18%	0.77%	1.60%
24.00%	-0.29%	-0.18%	0.14%	0.68%	1.44%
26.00%	-0.29%	-0.20%	0.10%	0.60%	1.30%
28.00%	-0.30%	-0.21%	0.07%	0.53%	1.18%
30.00%	-0.30%	-0.22%	0.04%	0.47%	1.08%
32.00%	-0.31%	-0.23%	0.01%	0.42%	0.99%
34.00%	-0.31%	-0.24%	-0.01%	0.37%	0.91%
36.00%	-0.32%	-0.25%	-0.04%	0.32%	0.83%
38.00%	-0.32%	-0.26%	-0.06%	0.29%	0.77%
Long 2M short 5M					
	0.00%	2.00%	4.00%	6.00%	8.00%
22.00%	-0.20%	-0.13%	0.08%	0.41%	0.83%
24.00%	-0.20%	-0.14%	0.05%	0.36%	0.75%
26.00%	-0.21%	-0.15%	0.03%	0.31%	0.69%
28.00%	-0.21%	-0.15%	0.01%	0.27%	0.63%
30.00%	-0.21%	-0.16%	-0.01%	0.24%	0.57%
32.00%	-0.21%	-0.17%	-0.02%	0.21%	0.52%
34.00%	-0.22%	-0.17%	-0.04%	0.18%	0.48%
36.00%	-0.22%	-0.18%	-0.05%	0.16%	0.44%
38.00%	-0.22%	-0.18%	-0.06%	0.13%	0.40%

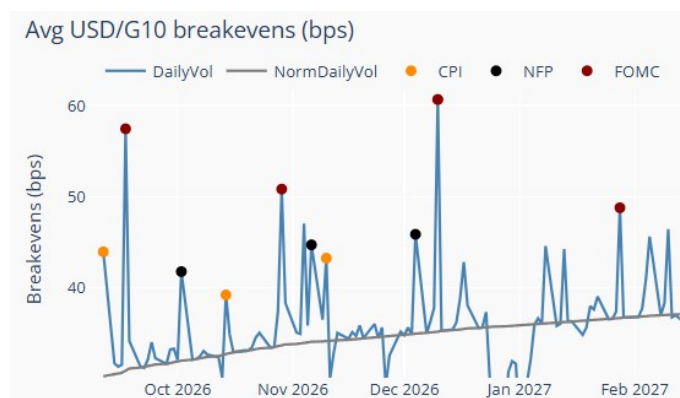
Source: J.P. Morgan

3.4.3 Macro Event Trading with FX Options

Subdued base vols and modest event premia tend to provide a supportive backdrop for long-gamma event trading across FX but still require a healthy dose of macro drivers.

Event trading remains attractive in FX. The key backdrop is that base volatility remains subdued, keeping event premia benign across most major releases (Figure 186). As a result, realized moves have frequently matched or exceeded implied pricing, particularly outside JPY, where volatility has remained elevated for idiosyncratic reasons (Figure 187).

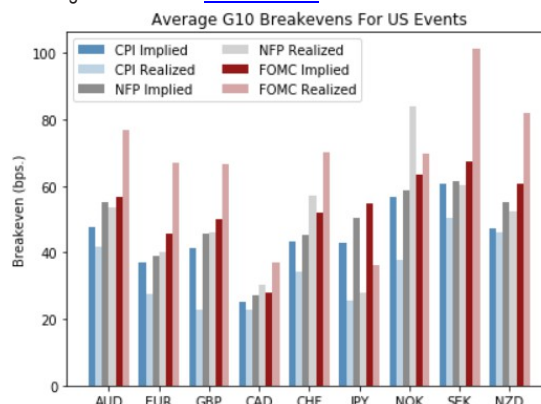
Figure 186: Low base volatility is keeping event premia contained across major FX events



Source: J.P. Morgan.

Figure 187: Realized volatility has generally exceeded implied event pricing across G10 FX

Details given in a recent [FX Derivatives](#) note.



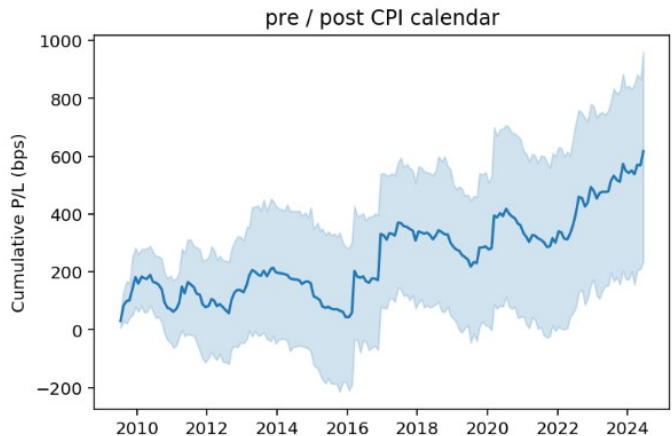
Source: J.P. Morgan.

This environment is supportive of long-gamma event strategies. As we highlighted in 2024, periods of modest event pricing can create opportunities for gamma scalping when realized volatility consistently exceeds breakevens. Figure 188 illustrates the cumulative profitability of systematically trading US CPI events, while Figure 189 shows that higher-beta and less-liquid G10 currencies, particularly NOK and SEK, generated the strongest event-trading returns. While the drivers have evolved, including Treasury-driven JPY volatility, UK political uncertainty, and renewed sensitivity of NOK and SEK to the oil and carry backdrop, these currencies remain among the most responsive to event risk.

Finally, event pricing should be assessed relative to the premium embedded in forward volatility. Our [2024 note](#) found that a forward-volatility-to-spot-volatility ratio of roughly 1.4x marked the threshold beyond which event risk premia became excessive and historically favored volatility sellers (Figure 190). The metric provides a useful gauge of whether event risk remains attractively priced or has become overly expensive relative to underlying spot volatility.

Figure 188: Event-trading US CPI releases generated alpha in 1H24 and is likely to do so again in 2026 if vols begin to tick up.

Details given in prior [FX Derivatives](#) note.



Source: J.P. Morgan.

Figure 189: Smaller/less-liquid high-beta G10 currencies, such as NOK and SEK, provided the highest returns in 2024

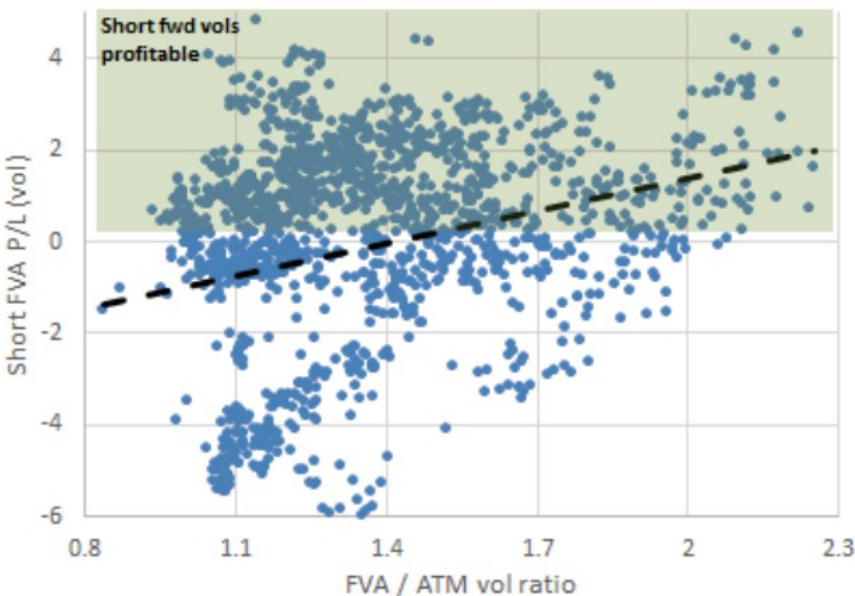
Details given in prior [FX Derivatives](#) note.

	2021	2022	2023	2024	Mean
AUD	-0.29	-0.46	1.80	1.90	0.73
EUR	-0.13	1.40	1.10	0.77	0.80
GBP	-1.60	2.50	-0.08	0.88	0.41
NZD	1.20	0.99	1.10	0.16	0.87
CAD	0.08	1.10	0.32	0.34	0.47
CHF	-1.10	-0.06	0.06	1.20	0.04
JPY	-0.79	4.10	-0.64	0.08	0.68
NOK	-0.27	2.20	0.57	3.10	1.40
SEK	-0.18	0.78	0.87	3.40	1.20

Source: J.P. Morgan.

Figure 190: A forward vol/spot vol ratio of 1.4x marks the threshold beyond which risk premium is too excessive

Details given in prior [FX Derivatives](#) note



Source: J.P. Morgan

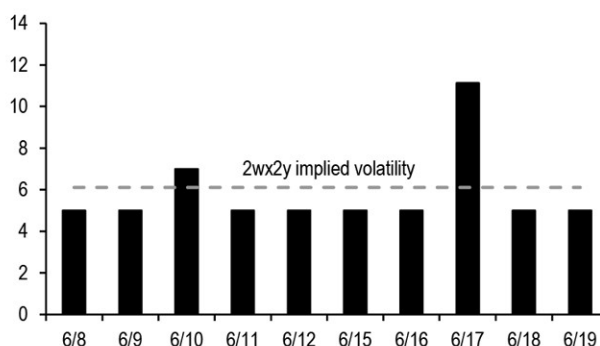
Swaption implied vols can be used to back out how much volatility markets are pricing in for event days such as FOMC meetings.

3.4.4 Macro Event Trading with Swaptions

For rates vol, some of the most significant recurring event days are FOMC meeting days and NFP and CPI prints. Elections can also bring significant event risk. These days tend to bring outsized delivered moves in yields compared to any “normal” day; hence, **rates options markets embed an event premium for these days. One can solve for how many normal days’ worth of variance markets are pricing for an event day by using a sum of variances formula, and market participants can take views on whether that vol is rich or cheap.** Figure 191 shows an example of this analysis for the June 2026 FOMC meeting. In a 10-day period containing 8 “regular” days, 1 CPI day, and 1 FOMC day, we assume “regular” days deliver 5bp/day of volatility and the CPI day delivers 7bp/day in the 2Y yield, estimated from historical averages. Then, by using the 2wx2y implied volatility and the sum of variances formula, we can approximate how much delivered volatility in the 2Y yield was priced into the July FOMC meeting by equating the delivered variance over the next 10 days with the implied volatility of the 2Wx2Y swaption.

Figure 191: One can back out how much vol is priced in for event days and take views on whether they are rich or cheap accordingly

Volatility priced into each day* around the June 2026 FOMC meeting (6/17/26); bp/day



* Under the assumption that over the 10 days shown, there are 8 “regular days” that will deliver 5 bp/day of volatility and 1 CPI day that delivers 7 bp/day, we solve for the delivered volatility priced in for the June FOMC day by equating the total delivered variance with the 2Wx2Y implied variance. Adapted from [9 to 5](#)
 Source: J.P. Morgan

4. Cross-Asset Opportunities

4.1 Diversification with Cross-Asset Portfolios

Portfolio optimization is a distinct topic that we will explore in detail in subsequent reports. Given the scope of this report, we aim to highlight the **diversification effect** achieved by implementing volatility strategies across multiple asset classes.

Diversified Volatility Carry Portfolio

For the **systematic volatility carry strategies with defensive improvements** discussed in 3. Opportunities Within Asset Classes, the time series exhibit **very low correlations between each other**.

A diversified portfolio of defensive carry across cross-asset volatility can generate **superior risk-return profiles** compared to single-asset solutions.

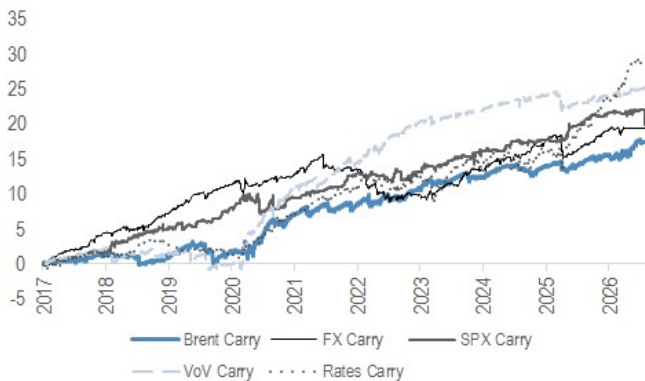
Figure 192: Correlations between systematic volatility carry strategies

	FX Carry	SPX Carry	VoV Carry	Brent Carry	Rates Carry
FX Carry					
SPX Carry	2%				
VoV Carry	11%	6%			
Brent Carry	13%	9%	13%		
Rates Carry	13%	1%	5%	6%	

Source: J.P. Morgan.

To construct a cross-asset volatility carry portfolio, we used a simple **inverse volatility scaling** to weight the different time series to illustrate the diversification effect.

Figure 193: PnL of volatility carry strategies after inverse vol scaling



Source: J.P. Morgan

Figure 194: Cross-asset volatility carry portfolio



Source: J.P. Morgan

As a result, we are able to achieve a much **higher Sharpe ratio for the cross-asset portfolio** than for single strategies (Figure 194). One can therefore increase leverage on the portfolio to enhance returns given the low drawdown.

Figure 195: Performance statistics for individual volatility carry strategies and cross-asset carry portfolio

Sized with inverse vol and leveraged up for the portfolio

	Ann Return	Ann Vol	Sharpe	Max DD
Brent Carry	1.7%	1.8%	0.93	-3.0%
FX Carry	1.8%	1.6%	1.14	-5.8%
SPX Carry	2.0%	1.7%	1.21	-2.4%
VoV Carry	2.4%	1.6%	1.45	-3.1%
Rates Carry	2.6%	2.3%	1.17	-4.6%
Portfolio	7.9%	3.4%	2.32	-4.6%

Annual returns are compounded. The rate strategy uses a 3M30Y-1Y30Y calendar as a defensive solution.

Source: J.P. Morgan

Even for equity tail hedging, a portfolio **including fixed income tail positions** can deliver robust results. **Post-crash profile management** is essential to enhance returns and reduce bleed.

Diversified Tail Hedge Portfolio

Investors with a tail mandate could face more challenges, as tail portfolios typically come with poor returns or bleed. At the same time, the down beta could fail to deliver in some scenarios, while profit capturing could be challenging.

However, if we **allocate across different tail strategies**, including a **profit-taking mechanism post crash**, the profile looks more appealing. We illustrate indicative results for blended hedges using strategies described in 3. Opportunities Within Asset Classes for equity portfolios and the **extra benefit of incorporating fixed income tail solutions**.

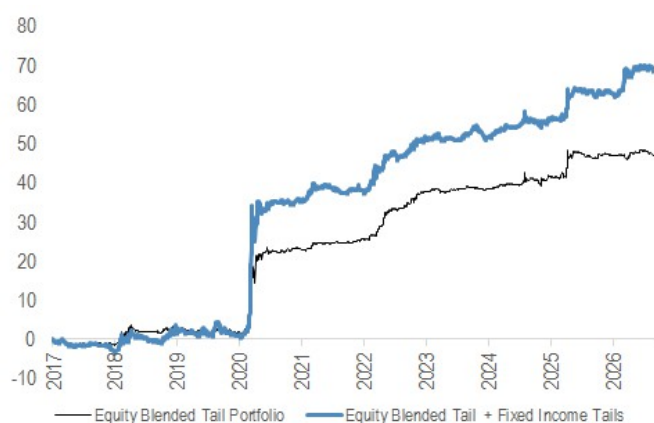
Figure 196: Correlations between systematic tail hedging strategies

	SPX Convexity	SPX Tail Post Crash	SPX Long Dated Vega	SS Vega	VIX Tail	SOXX Momentum	FX Tail	Enhanced Tail	Rates Tail
SPX Convexity									
SPX Tail Post Crash	48%								
SPX Long Dated Vega	23%	4%							
SS Vega	40%	-13%	47%						
VIX Tail	36%	15%	48%	43%					
SOXX Momentum	35%	20%	18%	29%	21%				
FX Tail	16%	2%	11%	22%	13%	16%			
Brent Enhanced Tail	14%	-12%	20%	35%	18%	6%	19%		
Rates Tail	2%	-3%	19%	16%	6%	-3%	11%	14%	

Source: J.P. Morgan

Unsurprisingly, while the equities-related tail strategies still diversify each other, they have higher correlations among themselves compared to using fixed income tails. For sizing a blended equities hedge, we included all three basic tail strategies (**long convexity short skew; own long-dated vega; and long single-stock downside vega**) with sizing adjusted according to maxDD. In addition, we added 30% to the 3-leg strategy for **tail trading post crash** and 10% to **intraday momentum**. In practice, we can safely allocate more to intraday momentum given the return profile.

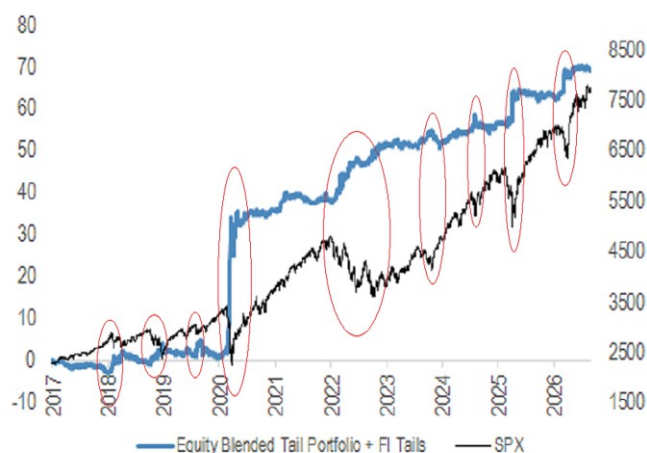
Figure 197: Equities blended tail portfolio vs. including Fixed Income tails strategies



Source: J.P. Morgan

Figure 198: Cross-asset tail hedge portfolio vs. SPX

Left - tail portfolio return, right - SPX



Source: J.P. Morgan

The equities blended tail hedge has delivered good returns with low drawdowns and bleed (Figure 197). However, there are some tail events with SPX drawdowns that were not efficiently captured with the equities blended tail hedge but that were picked up with **tail solutions in fixed income**, such as the **rates events** at the end of 2019 and 2023 and the **oil events** in 2022 and 2026 (Figure 197).

As we concluded above, fixed income tail strategies with low bleed can act as additional protection to equity corrections. If we also include allocations to these strategies on top of the tail hedges using only equity instruments, the combined **cross-asset portfolio delivers more comprehensive protection in almost all equities drawdown periods** (Figure 198).

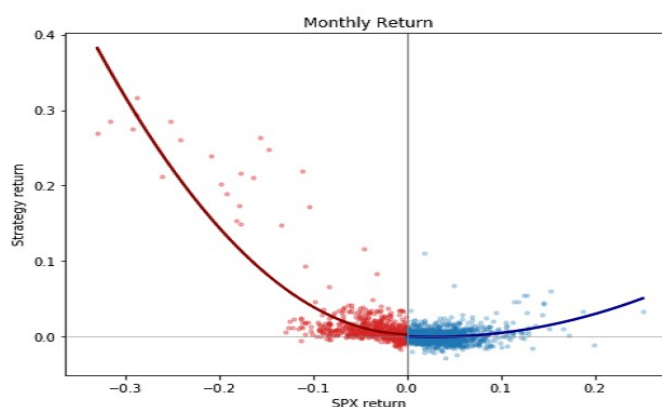
The Sharpe ratio is not the best measure for tail portfolios, but nevertheless, we note that the cross-asset solution is able to beat most single strategies and achieve a Sharpe of 1.05 with very robust downside beta and low bleed in “normal markets.”

Figure 199: Performance statistics for individual tail strategies and the cross-asset tail portfolio

	Annual Return	Sharpe	Max DD %
SOXX Momentum	10.14%	1.51	-9.7%
SPX Convexity	0.85%	0.65	-1.7%
SPX Tail Post Crash	4.52%	0.60	-8.9%
SPX Long Dated Vega	0.47%	0.56	-1.7%
Brent Enhanced Tail	1.40%	0.51	-3.7%
SS Vega	0.49%	0.27	-5.8%
Rates Tail	2.31%	0.22	-14.8%
FX Tail	0.01%	0.20	-0.3%
Cross Asset Tail Portfolio	5.60%	1.05	-6.8%

The rate strategy is constructed as a blend of long 10Y20Y and 20Y10Y.
Source: J.P. Morgan

Figure 200: Cross-asset tail portfolio vs. SPX return



Source: J.P. Morgan

Similarly, the benefits of owning positions across asset classes can also apply to a **rates-centered portfolio**, especially when market drivers that can be **distilled from other asset classes** are relevant. For example, for swaption trades, we can utilize our empirical model for gamma returns to motivate cross-asset trades. This can enable one to express a view on rates vol while reducing exposure to variables about which one doesn't have a view. For example, in April, our empirical model used in-sample monthly returns for the S&P 500 as one of the drivers. We recommended pairing longs in 6Mx5Y swaption straddles with longs in S&P 500 futures, weighted according to the partial beta in our empirical model, in order to reduce sensitivity to oil price-driven swings in the S&P 500 ([link](#)). Later in the conflict, when we included the Brent crude futures price as a driver, we recommended coupling shorts in 6Mx2Y swaption straddles with longs in Brent futures for similar reasons ([link](#)).

We have only used single-asset strategies for this discussion, but many relative value strategies also provide attractive carry and tail performance, such as dispersion trades. Overlaying such strategies can deliver further alpha to the portfolio.

Volatility relative value can be harvested across assets using ranking methods ranging from **basic VRP and EWMA to machine learning prediction** rankings. Broader performance can be achieved with a wider universe of cross-asset instruments, such as options on cross-asset ETFs.

4.2 Cross-Asset Relative Value Volatility Trading

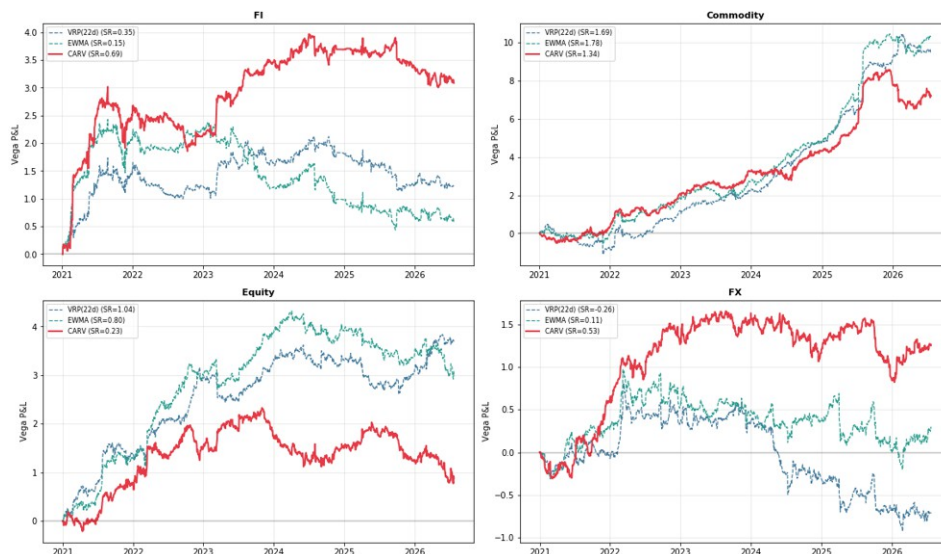
If we are evaluating a vol RV approach in a cross-asset context, the practical question is whether we can rank “rich” and “cheap” implied volatility across different underliers in a way that is stable and consistent enough to trade. That raises the question of **what constitutes a fair benchmark for implied volatility as a signal**.

A common starting point is the **volatility risk premium (VRP)** - the spread between implied volatility and realized volatility. In practice, however, realized volatility is time-varying and persistent: periods of high (low) volatility tend to be followed by high (low) volatility. This clustering motivates forecast-based benchmarks such as **EWMA volatility**, which places greater weight on recent price moves. A drawback of EWMA volatility is that excessively large moves are given a very high weight in the days immediately following the move. Therefore, a cap and a floor are usually applied to shrink the excessive move towards the median.

Alternatively, our **CARV model** ([link](#)) uses high-frequency intraday price information to improve the forecast of subsequent realized volatility and utilizes that forecast as the reference point for implied volatility relative value comparisons.

As an illustration, we evaluate signal quality by constructing a stylized cross-sectional relative value strategy on 1M variance swaps within each asset class (rates, commodities, equities, and FX). Each day, names are ranked by the signal (predicted RV minus current IV); the top basket is bought (long var), and the bottom basket is sold (short var), with basket sizes of 1, 2, 3, and 5 names for rates, commodities, equities, and FX, respectively. Each position is sized so that lower-vol names carry proportionally larger notional. The PnL blends two components weighted by time elapsed: realized vol accrual and implied vol mark-to-market. At expiry, the entire PnL is reduced to realized-vs-implied, mimicking varswap settlement. Note that this is not a tradable PnL: it abstracts away the path dependency of delta hedging, higher-order risks, and the practical frictions of replicating variance exposure with listed options.

Figure 201: Cross-asset vol RV backtest with VRP, EWMA, and CARV scores as signals for ranking

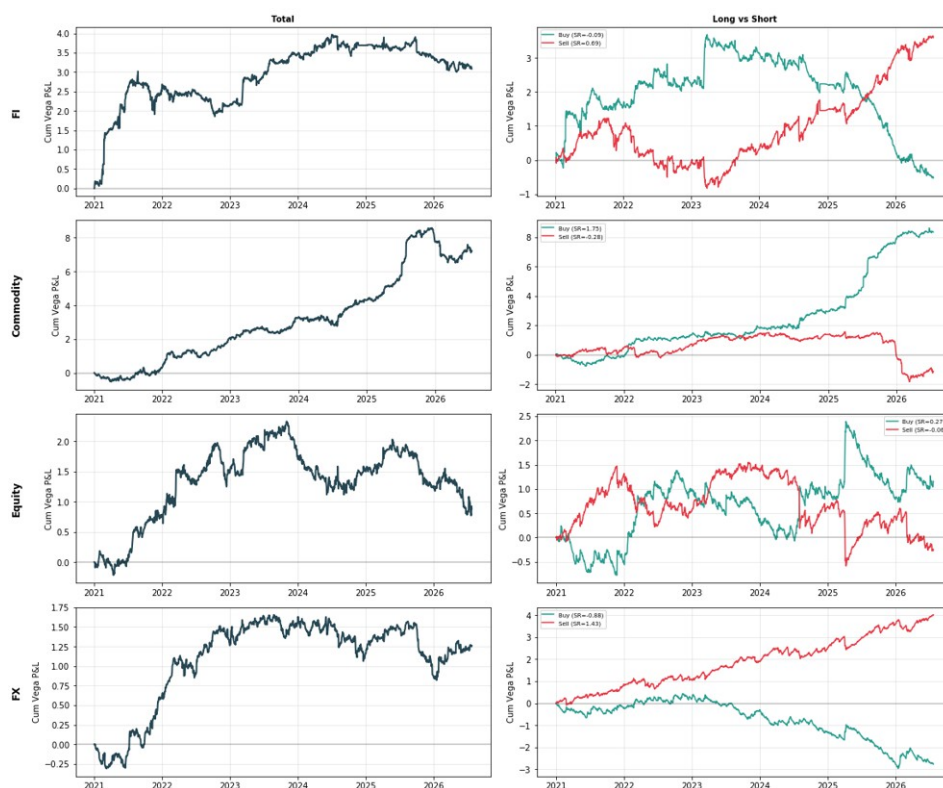


Source: J.P. Morgan, Bloomberg Finance L.P.

The CARV model delivers strong outperformance in rates and FX, demonstrating clear value in asset classes in which volatility exhibits stable mean-reverting dynamics. Rates benefits from a reliable cross-sectional structure, while FX benefits from the model's 2Y rolling-window estimation, which adapts well to evolving regimes. In rates and FX, the **PnL is mainly driven by the short leg**, while in commodities, the **PnL is driven by the long leg**, and equities is a mixed picture.

In equities and commodities, CARV underperforms relative to **simpler signals such as EWMA**, though this is largely driven by a handful of concentrated positions during exceptional regime transitions, such as persistent shorts in KOSPI and RTY in equities and in silver (commodities), where these names shifted abruptly to sustained high-vol regimes with limited historical precedent. Notably, both EWMA and CARV suffer during trending vol periods (e.g., KOSPI in 2026), whereas VRP's shorter lookback allows it to respond more quickly. This is a known limitation of models trained on historical patterns: structural breaks by definition fall outside the training distribution.

Figure 202: CARV PnL split between legs

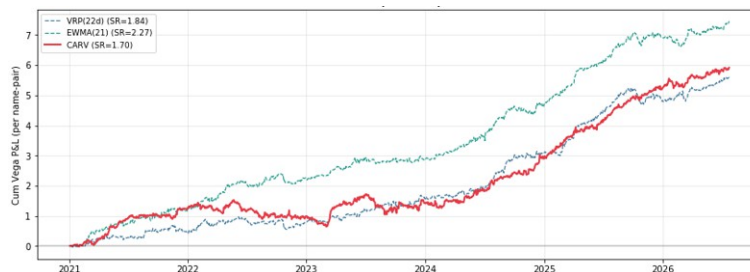


Source: J.P. Morgan, Bloomberg Finance L.P.

Separately from signal quality, cross-asset implementation introduces operational friction: options markets differ by asset class in terms of liquidity, contract conventions, and transaction cost profiles, and these differences can constrain both universe breadth and scalable execution. A practical way to expand breadth while keeping execution and monitoring more standardized is to express cross-asset exposures through ETF options. Our ETF RV Model ([link](#)) utilizes a similar concept, using high-frequency data to forecast the fair value of implied volatility.

Within the ETF space, we extend our universe to the 38 US ETFs ([link](#)) with the highest traded option volume and focus on the 1M tenor of the volatility curve to ensure the liquidity of our investment universe.

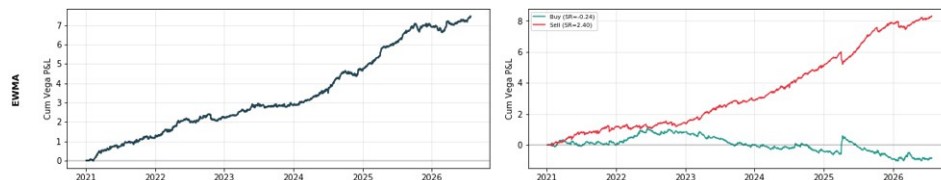
Figure 203: ETF RV backtesting



Source: J.P. Morgan, Bloomberg Finance L.P.

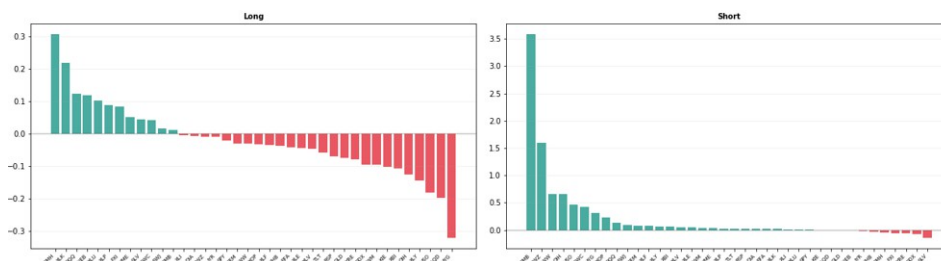
All three strategies deliver strong Sharpe ratios for the ETF universe in this period, with the **PnL overwhelmingly driven by the short leg**, and large weights on selling vol in EM bonds. The long leg functions as a hedge, where the signal targets the cheapest vol to reduce the bleed. **EWMA signal is the strongest performer**, partially due to the long position in SMH vol in 2022 (Figure 204, Figure 205). The return profile is typically strongest when volatility is stable/mean-reverting and can be poor during sharp volatility regime shifts.

Figure 204: EMWA strategy PnL split into long/short legs



Source: J.P. Morgan, Bloomberg Finance L.P.

Figure 205: EWMA strategy PnL split by underlying



Source: J.P. Morgan, Bloomberg Finance L.P.

Hybrid options serve as an effective tool to **leverage correlation inputs and reduce the cost** of expressing directional views across assets.

4.3 Bringing Cross-Asset Correlations to Equities

With ongoing monitoring of Cross-Asset Correlations with Equities and Volatility Parameters, hybrid options provide a tactical way to convert cross-asset views into a tradeable payoff whose valuation is driven by the **joint probability of multiple conditions**, making correlation a first-order risk driver ([Equity Hybrid Radar](#)).

Hybrid options are primarily used to **cheapen the expression** of a view versus assembling comparable vanilla exposure for each leg: either when there is conviction across both underlyings or when choosing a secondary asset to obtain a structural discount on primary views. They can also be used for scenario-specific hedging and profit-taking overlays.

Choice of Payoff

- **Dual digital:** Pays a fixed amount only if both underlyings meet their respective conditions at expiry (else zero), so its value is mainly driven by the joint probability and therefore implied correlation.
- **Contingent option:** Delivers a vanilla-like payoff on the primary asset only if the conditioning asset meets a trigger at expiry (otherwise worthless), which reduces premium but introduces “second-asset monetization risk.”
- **Worst-of call/best-of put:** The payoff is driven by the worst-performing constituent (or best-performing for best-of put), meaning that the laggard (or least-down asset) drives the payoff, making it a “synchronized move” expression that is framed as long correlation.
- **Best-of call/worst-of put:** The payoff is driven by the best-performing constituent (or worst-performing for worst-of put), so one can win with one strong mover, aligning with a dispersion view and framed as short correlation.

A Case Study on the SPX/GLD Hybrid Trade

We walk through a trade idea that uses hybrid options to play SPX/GLD upside. In December 2025, we recommended the idea based on a trend signal for upside ([link](#)) and a good entry level for low realized correlation (Figure 207). We backtest with a May 102.5% worst-of call, assuming that the trade was struck on Dec 9, 2025 (when we published the note).

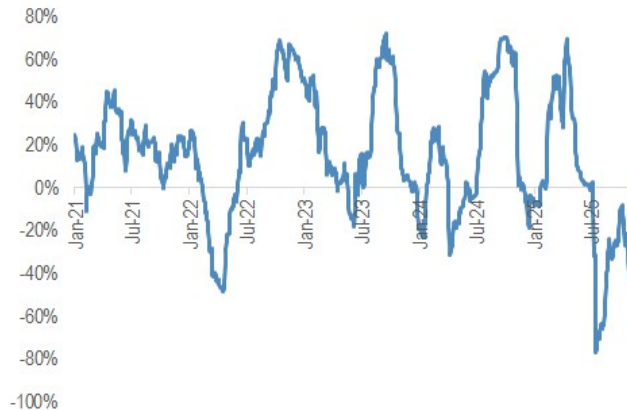
Figure 206: Worst-of call entry level and greek sensitivity

Worst of Call	1.45%
SPX Vanilla Call	3.59%
GLD Vanilla Call	4.69%
Discount to Average	65.10%
Delta	26.20%
Vega	0.09%
Phi	0.01%

Phi stands for correlation sensitivity for 1 correlation point change
 Source: J.P. Morgan, Bloomberg Finance L.P.

Figure 207: SPX/GLD realized correlation at lows at the time of trade

3M weekly return



Source: J.P. Morgan, Bloomberg Finance L.P.

At entry, the worst-of call is priced at 1.45% (indicative price), which is a **65% discount relative to the average of the vanillas on individual assets**, due to the low correlation. The vega and phi risks enabled attractive entry levels as the parameters were at the trough. Delta exposure from the structure provided efficient upside participation.

Figure 208: Worst-of call MTM vs. spot

Left - spot move since inception, Right - WoC value



Source: J.P. Morgan, Bloomberg Finance L.P.

Figure 209: SPX/GLD dual digital entry level

As of 12/9/2025

Tenor	GLD Barrier	SPX Barrier						GLD Digital
		100%	102%	104%	106%	108%	110%	
1M	100%	43.3%	31.7%	16.1%	5.1%	1.3%	0.3%	55.5%
1M	102%	32.0%	23.9%	12.5%	4.1%	1.0%	0.3%	39.7%
1M	104%	21.8%	16.6%	8.9%	3.0%	0.8%	0.2%	26.3%
1M	106%	13.8%	10.7%	5.9%	2.1%	0.6%	0.2%	16.4%
1M	108%	8.2%	6.4%	3.7%	1.3%	0.4%	0.1%	9.5%
1M	110%	4.6%	3.7%	2.2%	0.8%	0.2%	0.1%	5.3%
3M	100%	40.7%	35.5%	28.3%	19.9%	12.2%	6.7%	51.9%
3M	102%	34.2%	30.0%	24.1%	17.1%	10.6%	5.9%	42.8%
3M	104%	28.3%	25.0%	20.2%	14.5%	9.1%	5.1%	35.0%
3M	106%	23.0%	20.4%	16.6%	12.0%	7.6%	4.3%	28.0%
3M	108%	18.3%	16.3%	13.4%	9.7%	6.2%	3.5%	22.0%
3M	110%	14.4%	12.9%	10.6%	7.8%	5.1%	2.9%	17.2%
6M	100%	39.9%	36.9%	33.0%	28.1%	22.6%	17.2%	50.6%
6M	102%	35.4%	32.8%	29.5%	25.1%	20.3%	15.5%	44.2%
6M	104%	31.2%	29.0%	26.1%	22.4%	18.2%	13.9%	38.6%
6M	106%	27.2%	25.4%	22.9%	19.6%	16.0%	12.3%	33.3%
6M	108%	23.5%	21.9%	19.9%	17.1%	14.0%	10.9%	28.5%
6M	110%	20.2%	18.9%	17.2%	14.8%	12.2%	9.5%	24.4%

Source: J.P. Morgan, Bloomberg Finance L.P.

Figure 208 shows the MTM of the trade. Since entering the trade, the value increased along with GLD drifting upward while SPX hovered around the ATM level. When GLD peaked on 1/29 at up 28%, the worst-of was at around 3.2%, more than double the initial premium. However, due to the Middle East conflict, both SPX and GLD slid in March, and the trade was badly hit. Following the rebound in April and May, the trade fully recovered and expired at 5.7%.

Alternatively, one can choose dual digitals to play the upside on both assets. Figure 209 shows the price grid for different tenors and barrier levels, with a significant discount to GLD digitals. In general, 10x leverage is a good entry point for dual digitals to look interesting.

5. Comments and Conclusions

The insights from volatility trading opportunities across the four key areas reveal meaningful **parallels between asset classes**. For systematic carry and tail strategies, **calendar positions generally perform well** with delta-hedged options, though the regions of potential tail risk differ by asset class. It is important to **approach the implied volatility surface with humility**, as areas with elevated risk premia can overshoot unless the discount is driven by supply-demand dislocations or temporary overreactions. Similarly, surface inversions should prompt adjustments in positioning to align with the implied risk premium.

For systematic volatility carry strategies, it is essential to **incorporate built-in tail hedges**—either through calendar structures or wing options that avoid excessive bleed. Conversely, tail strategies should overlay “safer carry” components to minimize bleed while remaining responsive. A frequently overlooked trading opportunity is **after tail events**, when the market can offer substantial alpha through appropriate implementations; these can also serve as **profit-capturing mechanisms** for existing tail strategies.

Relative value positions can generate additional alpha through **dynamic trading** around core exposures. Event days warrant **special treatment**; tactical overlays should be added to systematic strategies where possible. Linear investors may consider yield enhancement or option replacement strategies to outperform benchmarks while maintaining **low-drawdown risk profiles**.

Building a **cross-asset volatility portfolio** significantly enhances diversification, increasing the Sharpe ratio for volatility carry strategies. Similarly, **cross-asset tail strategies** can help mitigate shocks originating from other asset classes, providing more comprehensive portfolio protection. Additional alpha can be generated from cross-asset relative value positions, and incorporating correlation inputs enables leveraged payoffs.

Ongoing innovation in quantitative investment strategies (QIS) and machine learning models will continue to refine volatility analytics and trading frameworks. We remain committed to deepening our coverage of volatility trading strategies across asset classes and to evolving portfolio construction techniques that drive superior alpha and risk-adjusted outcomes.

Appendix — Tradable QIS Indices

Asset Class	Style	Ticker	Instruments
Equity	Carry	JPMZVP5B Index	VIX Options
Equity	Carry	JPOSPSUT Index	1DTE Put Options on S&P 500
Equity	Defensive	AIJPVT1U Index	VIX & S&P 500 Futures
Equity	Defensive	JPOSDVU1 Index	S&P 500 Futures and Options
Equity	Defensive	JPOSDVUO Index	S&P 500 Futures and Options
Equity	Defensive	JPOSLCDU Index	S&P 500 Options
Equity	Defensive	JPOSLSU1 Index	S&P 500 Futures and Options
Equity	Defensive	JPOS LVM8 Index	MAG8 Single Stocks and Listed Options
Equity	Defensive	JPOS LVUS Index	Futures and Options on S&P 500
Equity	Defensive	JPOS PRU2 Index	S&P 500 options and e-mini S&P 500 futures
Equity	Defensive	JPUSVXCR Index	VIX Options
Equity	Defensive	JPRC85BU Index	S&P 500 Options
Equity	Intraday Momentum	JPOSHVSU Index	S&P 500 Futures
Equity	Intraday Momentum	JPOSIVSS Index	SOXX ETF
Equity	Intraday Momentum	JPOSIGN2 Index	Nasdaq Futures
Equity	Intraday Momentum	JPOSIOB5 Index	0-day-to-expiry options on S&P 500
Equity	Intraday Momentum	JPOSIVSU Index	S&P 500 Futures
Equity	Intraday Momentum	JPUSNQIM Index	Nasdaq Futures
Equity	Intraday Momentum	JPUSQEV2 Index	VIX Futures
Equity	Mean Reversion	JPOSBTDU Index	1-day to expiry S&P 500 options
Equity	Mean Reversion	JPOSMR5U Index	E-mini S&P 500 Futures
Equity	Mean Reversion	JPOSMR5N Index	Nasdaq-100 futures
Equity	Mean Reversion	JPOSMRHS Index	ICESEMIT Index and S&P 500 futures
Equity	Short Variance	JPOSC35Z Index	S&P 500 Futures & Options
Equity	Short Variance	JPOSRI5H Index	S&P 500 Futures and Options
Equity	Short Variance	JPOSSSNS Index	Nasdaq-100 options and futures
Equity	Short Variance	JPOSSK5H Index	S&P 500 Futures & Options
Equity	Short Variance	JPOSSVV1 Index	Futures and Options on VIX
Equity	Short Variance	JPOSSVV2 Index	Futures and Options on VIX
Equity	Short Variance	JPOSSVVE Index	Futures and Options on V2X
Equity	Short Variance	JPOSVVVU Index	1DTE Varswap and Options on S&P 500
Equity	Short Variance	JPOSVW2E Index	Varswap on SX5E
Equity	Short Variance	JPOSVU5H Index	Varswap on SPX
Equity	Volatility RV	JPOSCUV2 Index	Options on S&P 500 and on the top 75 S&P 500 names
Equity	Volatility RV	JPOSCUVS Index	Options on S&P 500 and on the top 75 S&P 500 names
Equity	Volatility RV	JPOSCEV2 Index	Options on EuroStoxx 50 and on its constituents
Equity	Volatility RV	JPOSCEVS Index	Options on EuroStoxx 50 and on its constituents
Equity	Volatility RV	JPOSCNV2 Index	Options on Nasdaq 100 and on the top 20 Nasdaq 100 names
Equity	Volatility RV	JPOSCNVS Index	Options on Nasdaq 100 and on the top 20 Nasdaq 100 names
Equity	Volatility RV	JPOSCUG2 Index	Options on S&P 500 and on the top 75 S&P 500 names
Equity	Volatility RV	JPOSCUGS Index	Options on S&P 500 and on the top 75 S&P 500 names
Equity	Volatility RV	JPOS DSV2 Index	Futures and Options on S&P 500 and on 11 S&P 500 Sector ETFs

Asset Class	Style	Ticker	Instruments
Equity	Volatility RV	JPOSDSVS Index	Futures and Options on S&P 500 and on 11 S&P 500 Sector ETFs
Equity	Volatility RV	JPOSDDVC Index	Options on S&P 500 and VIX
Equity	Volatility RV	JPOSDYVC Index	Options on S&P 500 and VIX
Equity	Volatility RV	JPOSRVUN Index	Futures and Options on S&P 500
Equity	Volatility RV	JPOSRVUS Index	Futures and Options on S&P 500
Equity	Volatility RV	JPUS2525 Index	S&P 500 Options
Commodities	Defensive	JPOSPRCL Index	Listed options and futures
Commodities	Flow	JPRCORHC Index	Commodity Futures
Commodities	Short Variance	JPOSSVGC Index	Gold Futures and Options
Commodities	Short Variance	JPOSCGLD Index	GLD US ETF Options
Commodities	Short Volatility	JPOSCOMV Index	Futures & Options on BCOM Universe
Commodities	Short Volatility	JPOSSVCO Index	Commodity options and futures
Commodities	Short Volatility	JPOSSVCR Index	Corn Futures and Options
Commodities	Volatility RV	JPOSDV1 Index	Listed options and futures
Commodities	Volatility RV	JPOSRVCO Index	Listed options and futures
Commodities	Volatility RV	JPOSRVC1 Index	Listed options and futures
Commodities	Volatility RV	JPOSRVN2 Index	Listed options and futures
Commodities	Volatility RV	JPOSRVSG Index	Gold and Silver Futures and Options
Rates	Carry	JCUBIXY3 Index	USD Swaptions and Swaps
Rates	Defensive	JCUBVUVT Index	USD Swaptions
Rates	Defensive	JCUBVETV Index	EUR Swaptions
Rates	Defensive	JCUBVJTT Index	JPY Swaptions
Rates	Defensive	JSFRMCGH Index	SOFR Options and Futures
Rates	Mean Reversion	JPOSMR5T Index	TLT US ETF
Rates	Short Variance	JPOSSFVW Index	Listed 5Y UST options and futures
Rates	Short Variance	JPOSSTYV Index	Listed 10Y UST options and futures
Rates	Short Volatility	JCUBGUCS Index	USD Swaptions and Swaps
Rates	Short Volatility	JCUBGUCT Index	USD Swaptions and Swaps
Rates	Volatility RV	JCUBGURV Index	USD Swaptions and Swaps
FX	Short Volatility	JPOSFES1 Index	FX OTC Options & Forwards
FX	Short Volatility	JPOSFVB7 Index	FX OTC Options & Forwards
FX	Short Volatility	JPOSFVE2 Index	FX OTC Options & Forwards
FX	Short Volatility	JPOSFVJ2 Index	FX OTC Options & Forwards
FX	Short Volatility	JPOSFSB1 Index	FX options and forwards
Credit	Carry	JCREOCH1 Index	Options on CDS Indices
Credit	Defensive	JCRDLR1 Index	Options on CDS Indices
Credit	Defensive	JCRDPCG2 Index	Options on CDS Indices
Credit	Defensive	JCREDCH1 Index	Options on CDS Indices
Credit	Defensive	JCREDPG2 Index	Options on CDS Indices
Credit	Defensive	JCRFDPI2 Index	Options on CDS Indices
Credit	Short Volatility	JCREVCG4 Index	CDS Indices and Options on CDS Indices
Credit	Short Volatility	JCREVCH4 Index	CDS Indices and Options on CDS Indices
Credit	Short Volatility	JPOSSVH2 Index	HYG US ETF and Options on HYG US ETF

Appendix — Related Publications

Equity Volatility Strategy

Date	Lead AC	Subtitle	Summary
8/31/2026	Yangyang Hou	Dispersion Update - Buy Some Dip on Singles Vega	Opportunities on single name volatility after implied reset, sector dispersion in Health Care and Staples, and timing for gamma dispersion strategies.
8/27/2026	Yangyang Hou	What to Long, Short and When (Part III)	Systematic call spread replacement for long SPX, enhanced risk-adjusted returns and reduced drawdowns through dynamic overlays and tactical participation.
8/5/2026	Yangyang Hou	Stellar PnL from Earnings Dispersion Echoes of 1998, How to Buy Correlation?	Historical parallels to dot-com, systematic and tactical strategies for trading equity vol and optimal ways to buy index correlation
7/24/2026	Yangyang Hou	Levered ETF Gamma, RV, Dispersion and FOMC	Levered ETF gamma effects, intraday momentum and hedging strategies
7/13/2026	Yangyang Hou	Intraday Momentum Deep Dive (Part I)	Deep dive into systematic intraday momentum strategies in equities, execution timing, and vol carry trades
6/15/2026	Yangyang Hou	Dispersion, Mag7 Volatility Opportunity And Inverted Skew	Opportunities with inverted skew, strategies for tail hedging and correlation trades
5/13/2026	Yangyang Hou	Dispersion At The Extremes - What Happens After a 100% 1M Return	Studies of posterior returns after the speed of the rally is too fast
5/6/2026	Yangyang Hou	Gamma Supply and Earnings Dispersion Update	Mag7 call overwriting, dispersion, sector vol trends, and RV opportunities in index and sector vol spreads
4/21/2026	Yangyang Hou	Have Parameters Normalized Too Much?	Hedging and RV trades on implied vol and skew reset, UpVar and index-EM volatility spreads
4/9/2026	Yangyang Hou	What To Long, Short And When (Part II)	Systematic strategies for tail hedging with short skew long convexity long gamma, minimize carry bleed and boost performance across market corrections
3/24/2026	Yangyang Hou	"Too Big to Move"? Dispersion Update and RV Opportunities	Mega-cap gamma supply effects, vol trends, and RV trades in index and sector dispersion
3/10/2026	Yangyang Hou	Lesson From The 2022 Playbook	Skew, vol-of-vol delivered well in 2022, compares current market dislocations to the 2022 volatility playbook, systematic vol -of-vol strategy
2/10/2026	Yangyang Hou	Vol RV opportunities amid the software correction	RV trades on IGV dispersion and SPX vol spreads, hedging for correlation risk
1/28/2026	Yangyang Hou	What to Long, Short, and When (Part I)	Systematic SPX carry strategies, optimal vol harvesting zones, and dynamic VIX-based positioning for safer carry
1/15/2026	Yangyang Hou	Why index vol is so low and MXWO dispersion	Why index volatility is very low, and the relationship with correlation and earnings seasonality, dispersion, and RV trades
11/18/2025	Yangyang Hou	Term-structure inversion, NVDA, Gamma dispersion and VIX	Gamma dispersion and systematic overlay with VIX structure, NVDA earnings impact
10/15/2025	Yangyang Hou	Dispersion Update, AI vs Dot-Com and Earnings	Dispersion update for 2025, comparison with dot-com, timing of gamma-neutral strategies and earnings seasonality, sector and single stock dispersion contributions
9/24/2025	Yangyang Hou	Take some chips off the table, replace with leveraged options	Palladium (spot dispersion) and long dated UpVar
8/20/2025	Yangyang Hou	Correlation and Volatility at the Lows: What to do	KO Var vs Var, single stock vega picks, gamma-neutral dispersion
7/29/2025	Yangyang Hou	Thinking Outside the Box for Tail Trading - Part 2	Extension of post tail trading to 2008, effect on delta hedging with different volatility levels, PnL outside signal days
7/23/2025	Yangyang Hou	Tactical Note: Ways to Long Cheap Volatility in Oil Stocks	Tactical long volatility trades in oil stocks with FVA structure, RV and dispersion strategies funded via index options
6/5/2025	Yangyang Hou	Thinking Outside the Box for Tail Trading	Three-pillar tail-trading framework for systematic trading post tail event
5/20/2025	Yangyang Hou	Start loading the bullets for longer dated VIX at 0 cost	VIX call spread collar, distribution post market correction
3/13/2025	Yangyang Hou	Trading US Macro Events with Short-Dated Options	Deep dive PnL on event days with different structures and hedging methods
1/7/2025	Yangyang Hou	The Case for Long Dated UpVar	Deep dive on Long-dated UpVar structures, optimal tenor/barrier selection, and why it is optimal to enter now

Date	Lead AC	Subtitle	Summary
4/29/2025	Yangyang Hou	Dispersion Update, Index Technique, Stock Leg and What Now	Dispersion: Deep dive into equity dispersion trades, index leg techniques, earnings volatility resets, and attractive entry levels
9/25/2024	Yangyang Hou	Systematic Dispersion Update and Trade Ideas	Dispersion: Analyzing index and single-stock vol drivers, recommending bespoke baskets based on value and revenue exposure
9/13/2023	Yangyang Hou	A Practical Dispersion Trading Handbook	Dispersion: Techniques to enhance PnL through earnings seasonality, index legging, and volatility overlays

Interest Rate Derivative Strategy

Date	Lead AC	Subtitle	Summary
8/7/2026	Ipek Ozil	August (NFP's version)	Most recent empirical model for gamma returns
7/31/2026	Ipek Ozil	Phish goes, the Fed doesn't	Example 1x2 payer spread to position for higher yields
6/29/2026	Ipek Ozil	Turkish (soccer) delights	Using PCA to identify vol trading opportunities
6/5/2026	Ipek Ozil	9 to 5	Backing out vol priced in for event days
6/2/2026	Ipek Ozil	Sell you twenty barrels worth	Empirical gamma returns model including the price of oil as a driver
5/8/2026	Ipek Ozil	Oil is the new principal component	Correlations between implieds and oil prices during the Middle East conflict
4/17/2026	Ipek Ozil	Abracadabra	Example empirical model for gamma returns
2/27/2026	Ipek Ozil	Second and goal	Effect of vol spikes on PnL of short gamma strategies
12/20/2025	Ipek Ozil	How the grinch stole volatility	Example swaption expiry switch RV trade
10/31/2025	Ipek Ozil	Trick and Treat	Example swaption tail switch as a lower-beta proxy for short gamma exposure
9/2/2022	Dobromir Tzotchev, Thomas Salopek	Options Skew Risk Premium: How to measure and harvest it	Primer on harvesting skew risk premium
5/26/2016	Khagendra Gupta, Fabio Bassi	A primer on Conditional Trades	Primer on using swaptions for conditional curve trades

FX Derivative Strategy

Date	Lead AC	Subtitle	Summary
7/17/2026	Ladislav Jankovic, Sanjana Shinde, Yangyang Hou	What to Long, Short, and When	Systematic carry and tail on FX Volatility
8/6/2026	Ladislav Jankovic, Sanjana Shinde	The Triennial Rebalancing of the VXY Indices: 2026 Edition	VXY Weights Unchanged. Introducing Our New Equal-Weighted FX Volatility Tickers. VXY Rebalancing 2026
8/13/2026	Sanjana Shinde, Ladislav Jankovic	FXO Quant: There's 104 Days of Summer, and Still No Vols	August 2026, FX Options Quant Monitor (a monthly refresh of our models and their signalling)
8/21/2026	Meera Chandan, Arindam Sandilya	Currencies amid long-end UST shenanigans	Most Recent FX Markets Weekly
8/21/2026	Ladislav Jankovic	Livin' On A Buyback	Most Recent FX Markets Weekly Derivatives Section
3/5/2019	Ladislav Jankovic	Timing FX short-vol strategies	Systematic timing of FX short-volatility
2/25/2020	Ladislav Jankovic, Twinkle Mehta	Total-return version of tactical filter allowing long Gamma trades	Tactical switching between gamma regimes
2/12/2020	Dobromir Tzotchev	Pair trading with FX vols: Should vol mean-reversion be your friend?	Volatility mean reversion pair trading
9/16/2020	Ladislav Jankovic	Market filters for FX Vol RV - what offers value now?	FX volatility relative value signals
11/24/2020	Arindam Sandilya	The gravitational pull of early cycle, zero rates	Trading Volatility Curves
12/8/2020	Ladislav Jankovic	Room to play steeper NOK and PLN vol curves	Continuing To Trade Volatility Curves
3/18/2021	Ladislav Jankovic, Arindam Sandilya	How effective are one-touch calendar spreads in fading flat & inverted vol curves?	One-touch calendar spreads for vol carry
9/25/2024	Dobromir Tzotchev, Thomas Salopek	Deploy premium and sleep well at night	FX carry strategies using options

Date	Lead AC	Subtitle	Summary
4/1/2020	Ladislav Jankovic	All weather vol ratio spreads excel when primed with risk on/off filter	Risk-filtered volatility ratio spread carry
7/18/2025	Ladislav Jankovic	A Forward-Looking Framework for FX Correlation Trading and Cross Asset Portfolio Integration	Systematic FX implied-correlation trading
9/5/2024	Ladislav Jankovic	Predicting and Trading the Volatility Surface	FX volatility surface forecasting and trading
8/3/2026	Sanjana Shinde, Ladislav Jankovic	Signals – August 3, 2026	Current FX hedge instrument allocation in portfolios
5/16/2019	Ladislav Jankovic, Arindam Sandilya	Pick your best FX hedges: A scorecard approach for selecting FX hedging instruments	Dynamic selection of FX hedging instruments
8/8/2025	Ladislav Jankovic	Scorecard hedging – a US hedger perspective	FX hedging scorecard for corporates
7/30/2025	Ladislav Jankovic, Arindam Sandilya	EM vol calendars - portfolio characteristics	EM volatility calendar spread carry
2/14/2025	Ladislav Jankovic	EM FX Vol Carry Portfolio - review and performance update	EM FX volatility carry performance
6/14/2024	Ladislav Jankovic	Term structure effects and timing of the trades	FX skew premium timing strategies
1/11/2024	Ladislav Jankovic	Deep Learning Model for Cross Asset Volatility Strategies	Deep learning for cross-asset volatility
10/23/2023	Ladislav Jankovic	Mahalanobis distance measure proves forward looking for FX vol	Mahalanobis distance for FX volatility

Cross Asset Volatility

Date	Lead AC	Subtitle	Summary
8/7/2026	Yangyang Hou, Ladislav Jankovic	Gold Waking Up and RV Model Update	Systematic RV and tactical trade with Gold GLD Options
7/17/2026	Yangyang Hou, Ladislav Jankovic	What to Long, Short, and When - (Commodity Part I)	Systematic carry and tail on Brent Options
4/28/2026	Yangyang Hou, Ladislav Jankovic	Trend, Oil and Gold Derivatives and ML Model Update	Trend and tactical vol strategies on equity, oil, and gold derivatives
3/17/2026	Yangyang Hou, Ladislav Jankovic	2022 Deep Dive, Oil and Gold Update, FOMC and ML Model Update	Cross asset behavior in 2022 deep dive, hedging strategies across equities, oil, and gold options and correlation-based structures
3/3/2026	Yangyang Hou, Ladislav Jankovic	Oil Derivatives and “Cheaper To Deliver” Macro Hedges	Oil related derivatives, systematic RV and macro hedging strategies on oil stocks and ETF
1/30/2026	Yangyang Hou, Ladislav Jankovic	Gold Through the Lens of Derivatives: Positioning and Trade	Deep dive on the derivative market impact to gold price/CTA dynamics, Systematic RV and tactical strategies on gold options and gold-linked ETFs
1/21/2026	Yangyang Hou, Ladislav Jankovic	Equity-USD, Buy the Dip, Japan and ML Model Update	Trend, ML model results update and tactical trades
12/9/2025	Yangyang Hou, Ladislav Jankovic	FOMC Implied Moves, AI Credit Hedge and ML Model Output Update	Gold spx worst of call to play trend, model results update, FOMC implied moves
11/14/2025	Yangyang Hou, Ladislav Jankovic	Gold ETF Option flow, FTSE vs Sterling and Machine Learning Model Output Update	Update on gold GLD option flow, Ftse GBP trade
10/22/2025	Yangyang Hou, Ladislav Jankovic	Record Short Gamma in Gold ETF and Machine Learning Based Trade Recommendations	Gold GLD option record short gamma, expect correction post expiry
9/30/2025	Yangyang Hou, Ladislav Jankovic	Machine Learning Based Trade Recommendations	Machine learning-driven trend and volatility strategies across equities, gold, and FX with tactical option structures
9/10/2025	Yangyang Hou, Ladislav Jankovic	Machine Learning Based Trade Recommendations	Machine learning-based trade recommendations on RV, event-driven strategies, on equities, fixed income, and FX
7/18/2025	Yangyang Hou, Ladislav Jankovic	A Forward-Looking Framework for FX Correlation Trading and Cross Asset Portfolio Integration	Systematic correlation and volatility strategies on FX options.
7/7/2025	Yangyang Hou, Ladislav Jankovic	2Q25 Update	Comprehensive analysis of cross-asset liquidity conditions and vol on equities, bonds, FX, and commodities

Risks of Common Option Strategies

Risks to Strategies: Not all option strategies are suitable for investors; certain strategies may expose investors to significant potential losses. We have summarized the risks of selected derivative strategies. For additional risk information, please call your sales representative for a copy of “Characteristics and Risks of Standardized Options.” We advise investors to consult their tax advisors and legal counsel about the tax implications of these strategies. Please also refer to option risk disclosure documents.

Put Sale: Investors who sell put options will own the underlying asset if the asset’s price falls below the strike price of the put option. Investors, therefore, will be exposed to any decline in the underlying asset’s price below the strike potentially to zero, and they will not participate in any price appreciation in the underlying asset if the option expires unexercised.

Call Sale: Investors who sell uncovered call options have exposure on the upside that is theoretically unlimited.

Call Overwrite or Buywrite: Investors who sell call options against a long position in the underlying asset give up any appreciation in the underlying asset’s price above the strike price of the call option, and they remain exposed to the downside of the underlying asset in the return for the receipt of the option premium.

Booster : In a sell-off, the maximum realized downside potential of a double-up booster is the net premium paid. In a rally, option losses are potentially unlimited as the investor is net short a call. When overlaid onto a long position in the underlying asset, upside losses are capped (as for a covered call), but downside losses are not.

Collar: Locks in the amount that can be realized at maturity to a range defined by the put and call strike. If the collar is not costless, investors risk losing 100% of the premium paid. Since investors are selling a call option, they give up any price appreciation in the underlying asset above the strike price of the call option.

Call Purchase: Options are a decaying asset, and investors risk losing 100% of the premium paid if the underlying asset’s price is below the strike price of the call option.

Put Purchase: Options are a decaying asset, and investors risk losing 100% of the premium paid if the underlying asset’s price is above the strike price of the put option.

Straddle or Strangle: The seller of a straddle or strangle is exposed to increases in the underlying asset’s price above the call strike and declines in the underlying asset’s price below the put strike. Since exposure on the upside is theoretically unlimited, investors who also own the underlying asset would have limited losses should the underlying asset rally. Covered writers are exposed to declines in the underlying asset position as well as any additional exposure should the underlying asset decline below the strike price of the put option. Having sold a covered call option, the investor gives up all appreciation in the underlying asset above the strike price of the call option.

Put Spread: The buyer of a put spread risks losing 100% of the premium paid. The buyer of higher-ratio put spread has unlimited downside below the lower strike (down to zero), dependent on the number of lower-struck puts sold. The maximum gain is limited to the spread between the two put strikes, when the underlying is at the lower strike. Investors who own the underlying asset will have downside protection between the higher-strike put and the lower-strike put. However, should the underlying asset’s price fall below the strike price of the lower-strike put, investors regain exposure to the underlying asset, and this exposure is multiplied by the number of puts sold.

Call Spread: The buyer risks losing 100% of the premium paid. The gain is limited to the spread between the two strike prices. The seller of a call spread risks losing an amount equal to the spread between the two call strikes less the net premium received. By selling a covered call spread, the investor remains exposed to the downside of the underlying asset and gives up the spread between the two call strikes should the underlying asset rally.

Butterfly Spread: A butterfly spread consists of two spreads established simultaneously – one a bull spread and the other a bear spread. The resulting position is neutral, that is, the investor will profit if the underlying is stable. Butterfly spreads are established at a net debit. The maximum profit will occur at the middle strike price; the maximum loss is the net debit.

Pricing Is Illustrative Only: Prices quoted in the above trade ideas are our estimate of current market levels, and are not indicative trading levels.

Correction: Tradeable QIS Indices table in Appendix is updated due to incomplete information.

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